

A General Theory of Capital Accumulation with Multi-Feature Dynamics and Functional Interactions

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Abstract

Background. Classical capital-circulation formulas such as $C - M - C$ and $M - C - M'$ capture the core distinction between commodity exchange and capital as self-expanding value, but their usual scalar form is too limited for analyzing heterogeneous, nonlinear, and dynamic accumulation processes.

Framework. This paper develops a general formula for capital accumulation by modeling capital motion as

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t) + u_t, \quad (1)$$

where K_t is a capital-state variable and $\mathbf{x}_t$ is a vector of accumulation-relevant features. The framework incorporates linear, quadratic, polynomial, nonparametric, constrained, dynamic, threshold, and regime-switching specifications.

Extension. The model is further extended to an infinite-dimensional Hilbert-space formulation,

$$K_{t+1} = K_t + a_0 + \langle b, x_t \rangle_H + \langle x_t, \mathcal{K}x_t \rangle_H + R(x_t), \quad (2)$$

allowing distributed features and kernel-based interactions to enter the accumulation law.

Empirical Illustration. A computational implementation is conducted using quarterly United States macroeconomic data. Because the available data do not directly measure Marxian capital stock, real GDP is used as an aggregate capital-state proxy, and the empirical target is the one-period variation $\Delta K_t = K_{t+1} - K_t$. A ridge-regularized finite-dimensional approximation with standardized features, lagged terms, and quadratic terms achieves a training R^2 of 0.2373, a test MAE of 121.1827, and a test RMSE of 143.1279. The predicted range of the capital-state variable is close to the observed range, differing by approximately 2.47% in relative amplitude. Prediction errors rise sharply during the 2007–2009 crisis period, where RMSE is approximately 1.84 times the normal-period RMSE, supporting the relevance of regime-dependent accumulation laws (Code link: <https://github.com/shjddjfi/PaperMKS.git>).

Contribution. The paper contributes a unified mathematical and econometric framework for capital accumulation. It preserves the classical theoretical intuition of capital as self-expanding value while generalizing it into a flexible model with multi-feature determinants, nonlinear coupling, dynamic feedback, constraints, thresholds, and infinite-dimensional interactions. The result is a bridge between Marxian capital-circulation theory, modern nonlinear economic modeling, and empirical estimation of capital-state dynamics.

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1 Classical Capital-Circulation Formula

1.1 Simple Commodity Circulation

The simplest point of departure is the classical circuit of commodity circulation. Let C_1 denote the commodity initially held by an economic agent, let M denote money as the universal equivalent, and let C_2 denote the commodity finally acquired through exchange. The circulation process may be written as

$$C_1 - M - C_2, \quad V(C_1) = M = V(C_2), \quad \Delta V = 0, \quad \Delta U \geq 0. \quad (3)$$

Here, $V(\cdot)$ denotes the exchange value of a commodity. The equality condition in Equation (3) expresses the defining feature of simple commodity circulation: commodities are exchanged through money at equivalent value. Money appears only as a mediating form. It does not initiate the process as self-expanding value, nor does it return as an enlarged magnitude. The purpose of the exchange is therefore not the augmentation of value, but the transformation of one use-value into another.

The preservation of value can be written as

$$\Delta V \equiv V(C_2) - V(C_1) = 0. \quad (4)$$

Equation (4) states that, under equivalent exchange, the value magnitude held at the beginning of the circuit is preserved at the end of the circuit. No surplus value is generated by the circulation form itself. This does not mean that the transaction is economically meaningless. The gain from simple commodity circulation lies in the domain of use-value. If $U_i(C)$ denotes the use-value of commodity C for agent i , then exchange may satisfy

$$\Delta U_i \equiv U_i(C_2) - U_i(C_1) \geq 0, \quad (5)$$

even when the exchange-value magnitude remains unchanged.

This distinction between value and use-value is central. In the circuit $C_1 - M - C_2$, money functions as a transient equivalent rather than as capital. The process begins with a commodity and ends with a qualitatively different commodity. Its economic rationality is consumption-oriented: the agent gives up a commodity that is less useful to her and obtains another commodity that better satisfies her needs. Hence the circuit can be summarized as

$$C_1 \longrightarrow M \longrightarrow C_2, \quad \Delta V = 0, \quad \Delta U_i \geq 0. \quad (6)$$

The formal implication is that simple commodity circulation cannot by itself explain capital accumulation. Since the value magnitude is conserved, repeated commodity circulation of this type yields

$$V(C_T) = V(C_0) \quad (7)$$

for any finite sequence of equivalent exchanges,

$$C_0 - M - C_1 - M - \cdots - M - C_T. \quad (8)$$

Thus, if accumulation is defined as a positive change in value,

$$\Delta K \equiv K_{t+1} - K_t > 0, \quad (9)$$

then simple commodity circulation is accumulation-neutral:

$$\Delta K = 0. \quad (10)$$

Simple commodity circulation therefore describes the redistribution and reallocation of commodities according to use-value, but not the production of surplus value. This limitation motivates the transition from the commodity circuit $C - M - C$ to the capital circuit $M - C - M'$, where money no longer serves merely as a medium of exchange but becomes the initial and terminal form of a self-expanding value process.

1.2 General Formula of Capital

The general formula of capital reverses the logic of simple commodity circulation. In simple circulation, the movement begins with a commodity and ends with another commodity. In capital circulation, by contrast, the process begins with money and ends with money. The circuit may be written as

$$M - C - M', \quad \Delta M_t \equiv M_{t+1} - M_t, \quad M_{t+1} = M_t + \Delta M_t. \quad (11)$$

Here, M_t denotes the initial money advanced at time t , C denotes the commodity form through which money is temporarily transformed, and M_{t+1} denotes the money recovered at the end of the circuit. The prime in M' indicates that the terminal money magnitude is not merely the original money restored, but money augmented by a surplus.

The defining condition of capital is therefore

$$M_{t+1} > M_t, \quad \Delta M_t > 0. \quad (12)$$

In this case, money is not used simply to mediate the acquisition of a use-value. It is advanced in order to return as a larger value. The object of the process is not the qualitative transformation of commodities, but the quantitative expansion of value. Capital is thus money in motion as self-expanding value.

The increment of money can be written as

$$\Delta M_t = M_{t+1} - M_t = M' - M. \quad (13)$$

This increment is the monetary expression of surplus value. If $\Delta M_t = 0$, the circuit degenerates into a value-preserving exchange and does not constitute capital accumulation. If $\Delta M_t < 0$, the circuit represents a loss, devaluation, or failed valorization. Hence the three possible regimes are

$$\Delta M_t \begin{cases} > 0, & \text{expanded reproduction or accumulation,} \\ = 0, & \text{simple reproduction or value preservation,} \\ < 0, & \text{devaluation or crisis.} \end{cases} \quad (14)$$

Unlike the circuit $C - M - C$, whose endpoint is a use-value, the circuit $M - C - M'$ has no natural terminal commodity. Its purpose is the expansion of abstract value. This gives the capital circuit a dynamic and recursive structure: the terminal money of one period can become the initial money of the next period. Formally,

$$M_{t+2} = M_{t+1} + \Delta M_{t+1}, \quad M_{t+h} = M_t + \sum_{s=0}^{h-1} \Delta M_{t+s}. \quad (15)$$

Thus, capital accumulation is not a single exchange but a repeated sequence of valorization cycles. The proportional rate of monetary expansion is given by

$$g_t^M \equiv \frac{M_{t+1} - M_t}{M_t} = \frac{\Delta M_t}{M_t}, \quad M_{t+1} = M_t(1 + g_t^M). \quad (16)$$

Iterating this expression yields

$$M_T = M_0 \prod_{t=0}^{T-1} (1 + g_t^M). \quad (17)$$

Equation (17) makes explicit the cumulative logic of capital: value is advanced not for consumption, but for repeated self-expansion across time.

The general formula of capital therefore provides the minimal abstract form of accumulation. It identifies the central object of analysis as the law of motion governing ΔM_t . The subsequent task is to explain where this increment comes from, how it is produced, and under what structural conditions it can be sustained. This requires moving beyond the surface form of exchange and introducing the production process, in which money is transformed into means of production and labor power before returning as an enlarged value.

1.3 Expanded Industrial-Capital Formula

The general formula $M - C - M'$ identifies the abstract form of capital as self-expanding value, but it does not yet specify the mechanism through which the monetary increment is generated. The industrial-capital circuit makes this mechanism explicit by inserting the production process between the purchase and sale of commodities. The expanded circuit is written as

$$M - C\{MP, LP\} \cdots P \cdots C' - M', \quad C = c + v, \quad C' = c + v + s. \quad (18)$$

Here, M is the money capital initially advanced, MP denotes means of production, LP denotes labor power, P denotes the production process, C' denotes the commodity output after production, and M' denotes the money realized after sale. The symbols c , v , and s denote constant capital, variable capital, and surplus value, respectively.

The first transformation in Equation (18) is the conversion of money into productive inputs:

$$M \longrightarrow C\{MP, LP\}. \quad (19)$$

This transaction does not by itself create surplus value. It is an exchange of money for commodities at their value. However, it changes the form of capital from money capital into productive capital. The purchased commodity bundle has two qualitatively distinct components:

$$C = c + v, \quad (20)$$

where c is the value advanced for means of production and v is the value advanced for labor power. The role of these two components differs fundamentally. Constant capital transfers its pre-existing value to the new commodity output, while variable capital purchases labor power, whose use in production can create new value. Thus, the production stage can be represented as

$$C\{MP, LP\} \cdots P \cdots C', \quad (21)$$

where the commodity output contains not only the value of the inputs but also surplus value:

$$C' = c + v + s. \quad (22)$$

The difference between C' and C is therefore

$$C' - C = s. \quad (23)$$

The final transformation realizes the commodity output as money:

$$C' - M', \quad M' = M + s, \quad (24)$$

assuming that commodities are sold at value and that the initial money advanced satisfies

$$M = C = c + v. \quad (25)$$

Under these conditions, the monetary increment in the capital circuit is identical to the surplus value generated in production:

$$\Delta M \equiv M' - M = s. \quad (26)$$

The expanded industrial-capital formula therefore separates two analytically distinct moments. The circulation stages,

$$M - C\{MP, LP\} \quad \text{and} \quad C' - M', \quad (27)$$

mediate the transformation of value between money and commodity forms. The production stage,

$$C\{MP, LP\} \cdots P \cdots C', \quad (28)$$

is the locus in which surplus value is generated. This distinction is essential because it prevents the monetary increase $M' - M$ from being misinterpreted as a mere consequence of exchange.

The rate at which the production process expands the advanced value can be expressed as

$$\frac{M' - M}{M} = \frac{s}{c + v}. \quad (29)$$

This expression anticipates the profit-rate formulation, while preserving the underlying value decomposition. It shows that the expansion of capital depends jointly on the amount of surplus value produced and on the total capital advanced.

Hence, the industrial-capital circuit gives the general formula of capital its material content. Capital is not merely money returning as more money; it is a sequential transformation of value through money, productive inputs, production, commodity output, and monetary realization:

$$M \longrightarrow C\{MP, LP\} \longrightarrow P \longrightarrow C' \longrightarrow M'. \quad (30)$$

This sequence provides the classical foundation for a dynamic theory of capital accumulation. It also establishes the basic accounting structure on which the later multi-feature and econometric formulations build: the increment of capital must ultimately be linked to the conditions governing production, realization, and reinvestment.

1.4 Value Composition

The industrial-capital circuit implies a corresponding decomposition of commodity value. Once production has taken place, the value of the commodity output can be written as the sum of three components:

$$W = c + v + s, \quad \Delta W = s, \quad W' = W + \Delta W. \quad (31)$$

Here, W denotes the total value embodied in the commodity output, c denotes constant capital, v denotes variable capital, and s denotes surplus value. The term ΔW denotes the increment of value generated through the production process.

The first component, constant capital, is the value of the means of production consumed or transferred during production:

$$c \equiv V(MP). \quad (32)$$

This component does not create new value by itself. Rather, its value is preserved and transferred to the new commodity. Machines, raw materials, energy, intermediate inputs, and other means of production reappear in the value of the final commodity to the extent that they are used up in the production process.

The second component, variable capital, is the value advanced for labor power:

$$v \equiv V(LP). \quad (33)$$

It is called variable because its use in production can generate a value greater than its own value. The wage payment represents the value of labor power, but the labor process may create new value exceeding this wage equivalent. This excess is surplus value.

The third component is surplus value:

$$s \equiv W - (c + v). \quad (34)$$

Equivalently, the new value created in production can be decomposed as

$$W - c = v + s. \quad (35)$$

Equation (35) separates the value transferred from means of production from the value newly created by labor. The latter is divided into the value equivalent of labor power, v , and the surplus appropriated as s .

The value increment of the production process is therefore

$$\Delta W \equiv W - (c + v) = s. \quad (36)$$

This expression gives a precise meaning to the claim that surplus value is not produced by the mere exchange of equivalents. It is the difference between the value of the final commodity and the value of the capital advanced in the form of means of production and labor power.

If the produced commodity is successfully realized in circulation, the value composition of commodity capital is converted into money form:

$$C' = c + v + s \implies M' = c + v + s. \quad (37)$$

Given the initial money advanced,

$$M = c + v, \quad (38)$$

the realized monetary surplus is

$$M' - M = (c + v + s) - (c + v) = s. \quad (39)$$

Thus, the value composition of the commodity output provides the accounting bridge between the production of surplus value and the monetary expansion represented by $M - C - M'$.

It is useful to distinguish the value initially advanced from the value finally realized. Let

$$W_0 \equiv c + v \quad (40)$$

denote the value advanced before production, and let

$$W_1 \equiv c + v + s \quad (41)$$

denote the value of the commodity output after production. Then

$$W_1 - W_0 = s. \quad (42)$$

In this notation, the expanded value can be written compactly as

$$W_1 = W_0 + \Delta W, \quad \Delta W = s. \quad (43)$$

The value-composition formula therefore provides the basic accounting identity for classical capital accumulation. It identifies surplus value as the increment that converts the preservation of advanced value into its expansion. In later sections, this scalar decomposition will be generalized into a multi-feature accumulation function, but the classical benchmark remains the same: accumulation requires a positive residual after the replacement of constant and variable capital,

$$s > 0 \quad \Longleftrightarrow \quad W > c + v. \quad (44)$$

1.5 Surplus-Value Rate

The decomposition of commodity value into constant capital, variable capital, and surplus value naturally leads to a measure of the intensity of surplus-value production. This measure is the surplus-value rate, defined as

$$e \equiv \frac{s}{v}, \quad s = ev, \quad e_t = \frac{s_t}{v_t}. \quad (45)$$

Here, e denotes the rate of surplus value, s denotes surplus value, and v denotes variable capital. In the time-indexed expression, e_t measures the rate of surplus value in period t .

The surplus-value rate compares surplus value not with the total capital advanced, but specifically with variable capital. This is because variable capital represents the value advanced for labor power, and labor power is the source of newly created value in the classical framework. Constant capital transfers pre-existing value to the final commodity, whereas variable capital enters a production process in which new value may exceed the value paid as wages. Therefore,

$$s = W - (c + v), \quad e = \frac{W - (c + v)}{v}. \quad (46)$$

Equation (46) shows that the rate of surplus value is a normalized measure of the excess value generated over and above the replacement of constant and variable capital. For a given level of variable capital, a higher value of e implies a larger surplus:

$$\frac{\partial s}{\partial e} = v > 0, \quad \frac{\partial s}{\partial v} = e. \quad (47)$$

Thus, surplus value may increase either because the exploitation intensity *rises*, as captured by e , or because the scale of variable capital employed in production expands.

The time-indexed formulation is especially useful for dynamic analysis:

$$e_t = \frac{s_t}{v_t}, \quad s_t = e_t v_t. \quad (48)$$

It allows the production of surplus value to vary across historical periods, firms, sectors, or economies. Changes in e_t capture shifts in the relation between surplus labor and necessary labor, while changes in v_t capture changes in the scale of labor power employed.

The growth of surplus value can be decomposed as

$$\Delta s_t \equiv s_{t+1} - s_t = e_{t+1} v_{t+1} - e_t v_t. \quad (49)$$

Adding and subtracting $e_t v_{t+1}$ yields

$$\Delta s_t = e_t \Delta v_t + v_{t+1} \Delta e_t, \quad (50)$$

where

$$\Delta v_t \equiv v_{t+1} - v_t, \quad \Delta e_t \equiv e_{t+1} - e_t. \quad (51)$$

This decomposition separates the scale effect of expanding variable capital from the intensity effect of a rising surplus-value rate.

In proportional terms, when $e_t > 0$ and $v_t > 0$, the gross growth of surplus value can be written as

$$\frac{s_{t+1}}{s_t} = \frac{e_{t+1}}{e_t} \cdot \frac{v_{t+1}}{v_t}. \quad (52)$$

Taking logarithms gives the additive decomposition

$$\Delta \log s_t = \Delta \log e_t + \Delta \log v_t. \quad (53)$$

This expression clarifies that the expansion of surplus value may come from a higher rate of exploitation, a larger wage-labor base, or both.

The surplus-value rate also provides a bridge between classical value analysis and later empirical specification. If s_{it} and v_{it} are observed for unit i in period t , one may define

$$e_{it} = \frac{s_{it}}{v_{it}}, \quad s_{it} = e_{it} v_{it}, \quad (54)$$

where i may index firms, sectors, regions, or national economies. This form makes the surplus-value rate a measurable object, subject to historical variation and econometric modeling.

The condition for positive surplus production is

$$e_t > 0 \iff s_t > 0 \text{ given } v_t > 0. \quad (55)$$

If $e_t = 0$, production reproduces the value of labor power without surplus. If $e_t < 0$, the production process fails to reproduce the value advanced as variable capital, indicating a loss or breakdown in valorization. Hence, the surplus-value rate is not merely an accounting ratio; it is the central classical measure of the intensity with which capital converts variable capital into surplus value.

1.6 Profit Rate

While the surplus-value rate measures surplus value relative to variable capital, the profit rate measures surplus value relative to the total capital advanced. It is defined as

$$r \equiv \frac{s}{c+v} = \frac{e}{\frac{c}{v}+1}, \quad r_t = \frac{s_t}{c_t+v_t}. \quad (56)$$

Here, r denotes the profit rate, s denotes surplus value, c denotes constant capital, and v denotes variable capital. The time-indexed expression r_t measures the realized profit rate in period t .

The profit rate differs conceptually from the surplus-value rate. The latter compares surplus value only with variable capital:

$$e = \frac{s}{v}. \quad (57)$$

The former compares surplus value with the whole capital advanced:

$$r = \frac{s}{c+v}. \quad (58)$$

Thus, the profit rate converts the production of surplus value into a measure of capital profitability. It asks not how much surplus value is produced per unit of variable capital, but how much surplus value is produced per unit of total capital committed to the production process.

Using the identity

$$s = ev, \quad (59)$$

the profit rate can be rewritten as

$$r = \frac{ev}{c+v} = \frac{e}{\frac{c}{v}+1}. \quad (60)$$

Equation (60) shows that the profit rate depends on two classical determinants: the surplus-value rate e and the ratio of constant to variable capital, c/v . The first captures the intensity of surplus-value production; the second captures the value composition of the capital advanced.

Let the organic composition of capital be denoted by

$$\omega \equiv \frac{c}{v}. \quad (61)$$

Then the profit rate can be written compactly as

$$r = \frac{e}{1+\omega}. \quad (62)$$

This expression makes explicit the basic comparative statics. Holding the organic composition fixed, a higher surplus-value rate raises the profit rate:

$$\frac{\partial r}{\partial e} = \frac{1}{1+\omega} > 0. \quad (63)$$

Holding the surplus-value rate fixed, a higher organic composition lowers the profit rate:

$$\frac{\partial r}{\partial \omega} = -\frac{e}{(1+\omega)^2} < 0 \quad \text{for } e > 0. \quad (64)$$

The time-indexed profit rate is

$$r_t = \frac{s_t}{c_t+v_t} = \frac{e_t v_t}{c_t+v_t}. \quad (65)$$

Defining

$$\omega_t \equiv \frac{c_t}{v_t}, \quad (66)$$

we obtain

$$r_t = \frac{e_t}{1 + \omega_t}. \quad (67)$$

This formulation separates the movement of the profit rate into an exploitation component, e_t , and a composition component, ω_t .

The exact change in the profit rate between two periods is given by

$$\Delta r_t \equiv r_{t+1} - r_t = \frac{e_{t+1}}{1 + \omega_{t+1}} - \frac{e_t}{1 + \omega_t}. \quad (68)$$

A first-order approximation around (e_t, ω_t) yields

$$\Delta r_t \approx \frac{1}{1 + \omega_t} \Delta e_t - \frac{e_t}{(1 + \omega_t)^2} \Delta \omega_t. \quad (69)$$

Equation (69) clarifies the classical tension: an increase in the surplus-value rate tends to raise profitability, whereas an increase in the organic composition tends to reduce it, all else equal.

In proportional terms, provided $e_t > 0$ and $1 + \omega_t > 0$, the gross movement of the profit rate can be expressed as

$$\frac{r_{t+1}}{r_t} = \frac{e_{t+1}}{e_t} \cdot \frac{1 + \omega_t}{1 + \omega_{t+1}}. \quad (70)$$

Taking logarithms gives

$$\Delta \log r_t = \Delta \log e_t - \Delta \log(1 + \omega_t). \quad (71)$$

The profit rate therefore rises when the growth of the surplus-value rate exceeds the growth of the capital-composition term.

The profit-rate formula also links the classical value categories to an empirical object. For an observational unit i in period t , one may write

$$r_{it} = \frac{s_{it}}{c_{it} + v_{it}} = \frac{e_{it}}{1 + \omega_{it}}, \quad \omega_{it} = \frac{c_{it}}{v_{it}}. \quad (72)$$

Here, i may denote a firm, sector, region, or national economy. This representation allows the profit rate to be studied as a dynamic outcome whose variation is associated with changes in surplus-value production, capital composition, technology, labor organization, and institutional conditions.

The condition for positive profitability is

$$r_t > 0 \iff s_t > 0 \text{ given } c_t + v_t > 0. \quad (73)$$

If $r_t = 0$, the circuit reproduces the advanced capital without profit. If $r_t < 0$, the realized value fails to cover the capital advanced, indicating devaluation or failed accumulation. Hence, the profit rate provides the central bridge between surplus-value production and the dynamic viability of capital accumulation.

1.7 Organic Composition of Capital

The relationship between surplus-value production and profitability depends not only on the rate of surplus value, but also on the composition of the capital advanced. The organic composition of capital is defined as the ratio of constant capital to variable capital:

$$\omega \equiv \frac{c}{v}, \quad r = \frac{e}{\omega + 1}, \quad \frac{\partial r}{\partial \omega} = -\frac{e}{(\omega + 1)^2} < 0. \quad (74)$$

Here, ω measures the relative weight of means of production in relation to labor power. A higher value of ω indicates that a larger share of the advanced capital is embodied in constant capital rather than variable capital.

The definition follows directly from the value decomposition of advanced capital:

$$K = c + v. \quad (75)$$

Dividing both sides by v gives

$$\frac{K}{v} = \frac{c}{v} + 1 = \omega + 1. \quad (76)$$

Since surplus value satisfies

$$s = ev, \quad (77)$$

the profit rate can be rewritten as

$$r = \frac{s}{c + v} = \frac{ev}{c + v} = \frac{e}{\frac{c}{v} + 1} = \frac{e}{\omega + 1}. \quad (78)$$

Equation (78) shows that the profit rate is jointly determined by the surplus-value rate and the organic composition of capital.

The comparative statics are immediate. Holding e fixed, the marginal effect of the organic composition on the profit rate is

$$\frac{\partial r}{\partial \omega} = -\frac{e}{(\omega + 1)^2}. \quad (79)$$

When $e > 0$, this derivative is strictly negative:

$$\frac{\partial r}{\partial \omega} < 0. \quad (80)$$

Thus, for a given surplus-value rate, an increase in the share of constant capital relative to variable capital lowers the profit rate.

The second derivative is

$$\frac{\partial^2 r}{\partial \omega^2} = \frac{2e}{(\omega + 1)^3}. \quad (81)$$

For $e > 0$, this implies

$$\frac{\partial^2 r}{\partial \omega^2} > 0. \quad (82)$$

The negative effect of ω on r is therefore convex: the profit rate declines as ω rises, but the absolute marginal decline becomes smaller at higher levels of ω .

The time-indexed form is

$$\omega_t \equiv \frac{c_t}{v_t}, \quad r_t = \frac{e_t}{1 + \omega_t}. \quad (83)$$

The exact change in the organic composition is

$$\Delta\omega_t \equiv \omega_{t+1} - \omega_t = \frac{c_{t+1}}{v_{t+1}} - \frac{c_t}{v_t}. \quad (84)$$

A first-order approximation gives

$$\Delta\omega_t \approx \frac{1}{v_t} \Delta c_t - \frac{c_t}{v_t^2} \Delta v_t = \omega_t \left(\frac{\Delta c_t}{c_t} - \frac{\Delta v_t}{v_t} \right). \quad (85)$$

Hence, the organic composition rises when constant capital grows faster than variable capital.

Substituting Equation (83) into the profit-rate change yields

$$\Delta r_t \equiv r_{t+1} - r_t = \frac{e_{t+1}}{1 + \omega_{t+1}} - \frac{e_t}{1 + \omega_t}. \quad (86)$$

A first-order approximation around (e_t, ω_t) is

$$\Delta r_t \approx \frac{1}{1 + \omega_t} \Delta e_t - \frac{e_t}{(1 + \omega_t)^2} \Delta \omega_t. \quad (87)$$

This expression formalizes the classical tension between the two forces. A rise in the surplus-value rate tends to increase profitability, while a rise in the organic composition tends to reduce it.

The condition under which the profit rate increases is therefore

$$\Delta r_t > 0 \iff \frac{1}{1 + \omega_t} \Delta e_t > \frac{e_t}{(1 + \omega_t)^2} \Delta \omega_t \quad (88)$$

under the first-order approximation. Equivalently,

$$\Delta e_t > \frac{e_t}{1 + \omega_t} \Delta \omega_t. \quad (89)$$

Thus, an increase in the surplus-value rate must be sufficiently large to offset the negative effect of a rising organic composition.

In empirical applications, the organic composition can be indexed by unit i and period t :

$$\omega_{it} = \frac{c_{it}}{v_{it}}, \quad r_{it} = \frac{e_{it}}{1 + \omega_{it}}. \quad (90)$$

Here, i may represent firms, industries, regions, or national economies. This formulation allows the organic composition to vary across production structures and historical periods.

The organic composition of capital therefore provides the classical link between technological or material transformation and profitability. When accumulation proceeds through a rising dependence on means of production relative to labor power, ω_t increases. Unless this increase is offset by a sufficiently large rise in e_t , the profit rate tends to decline:

$$\omega_t \uparrow \quad \text{and} \quad e_t \text{ constant} \implies r_t \downarrow. \quad (91)$$

This relation constitutes the final classical benchmark before the model is generalized into a multi-feature capital accumulation formula.

2 Finite-Dimensional Multi-Feature Capital Formula

2.1 Feature Vector

The classical formulas developed above express capital accumulation through a small set of value categories, such as constant capital, variable capital, surplus value, the surplus-value rate, and the organic composition of capital. These categories provide the conceptual foundation, but they are too restrictive for a general empirical theory of accumulation. Modern capital accumulation is shaped by a broader set of economic, institutional, technological, financial, ecological, and social conditions. To represent these conditions, we introduce a finite-dimensional feature vector:

$$\mathbf{x}_t = (x_{1t}, \dots, x_{dt})^\top \in \mathbb{R}^d, \quad \mathbf{z}_t = (1, \mathbf{x}_t^\top)^\top, \quad K_t \equiv M_t. \quad (92)$$

Here, $\mathbf{x}_t$ collects the observable or constructed state variables that affect the accumulation process at time t . The augmented vector $\mathbf{z}_t$ includes a constant term and is useful for linear and affine specifications. The notation $K_t \equiv M_t$ identifies the stock of capital with the money magnitude advanced or accumulated at time t .

The individual components of $\mathbf{x}_t$ may represent different dimensions of the accumulation environment:

$$x_{jt} \in \{\text{production, labor, technology, finance, demand, institutions, ecology}\}, \quad j = 1, \dots, d. \quad (93)$$

This notation does not require every feature to be directly observable in raw data. Some components may be measured variables, such as capital stock, employment, wages, debt, output, or energy use. Others may be constructed indices, such as technological intensity, market concentration, financial fragility, bargaining power, institutional quality, or ecological pressure.

The finite-dimensional representation generalizes the classical value categories by embedding them into a broader state vector. For example, one may include

$$\mathbf{x}_t = (c_t, v_t, s_t, e_t, \omega_t, \ell_t, q_t, \tau_t, \eta_t)^\top, \quad (94)$$

where c_t is constant capital, v_t is variable capital, s_t is surplus value, $e_t = s_t/v_t$ is the surplus-value rate, $\omega_t = c_t/v_t$ is the organic composition of capital, ℓ_t is a leverage measure, q_t is a demand or market-size variable, τ_t is a technology indicator, and η_t is an ecological or resource constraint indicator. The classical model is therefore not discarded; it appears as a lower-dimensional special case nested within the general feature space.

The augmented vector $\mathbf{z}_t$ is defined as

$$\mathbf{z}_t = \begin{pmatrix} 1 \\ \mathbf{x}_t \end{pmatrix} = \begin{pmatrix} 1 \\ x_{1t} \\ \vdots \\ x_{dt} \end{pmatrix} \in \mathbb{R}^{d+1}. \quad (95)$$

The leading constant allows accumulation functions to contain an intercept:

$$\alpha + \boldsymbol{\beta}^\top \mathbf{x}_t = \boldsymbol{\theta}^\top \mathbf{z}_t, \quad \boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}^\top)^\top. \quad (96)$$

This form is convenient for econometric estimation and provides the simplest finite-dimensional extension of the classical capital formula.

The identification

$$K_t \equiv M_t \quad (97)$$

should be understood as an analytical normalization. In the classical capital circuit, money is the initial and terminal expression of capital:

$$M_t - C_t - M_{t+1}. \quad (98)$$

The stock K_t denotes the value magnitude available for accumulation at the beginning of period t . When this magnitude is expressed in monetary form, it coincides with M_t . Therefore,

$$\Delta K_t \equiv K_{t+1} - K_t = M_{t+1} - M_t \equiv \Delta M_t. \quad (99)$$

This equivalence allows the subsequent model to use K_t as the general state variable while preserving the monetary logic of the classical formula.

For empirical applications, the feature vector may vary across units as well as time. If i indexes firms, sectors, regions, or national economies, then

$$\mathbf{x}_{it} = (x_{1it}, \dots, x_{dit})^\top, \quad \mathbf{z}_{it} = (1, \mathbf{x}_{it}^\top)^\top, \quad K_{it} \equiv M_{it}. \quad (100)$$

This panel representation allows capital accumulation to be modeled as a heterogeneous process across different productive units and historical contexts.

The feature-vector formulation therefore performs three functions. First, it translates the classical categories of capital circulation into a state-space language. Second, it permits the inclusion of additional determinants that are absent from the classical scalar formula but central to modern accumulation. Third, it prepares the ground for a general accumulation function in which the change in capital depends on both the existing capital stock and the multidimensional state of the economy:

$$\Delta K_t = A(K_t, \mathbf{x}_t; \theta). \quad (101)$$

This transition from a scalar circuit to a finite-dimensional state vector is the first step toward a general multi-feature formula of capital accumulation.

2.2 Capital Accumulation Function

Having represented the determinants of accumulation by the state vector $\mathbf{x}_t$, we now define the central object of the finite-dimensional framework: the capital accumulation function. Let K_t denote the capital stock, or value magnitude, available at the beginning of period t . The change in capital is defined as

$$\Delta K_t \equiv K_{t+1} - K_t, \quad \Delta K_t = \mathcal{A}(K_t, \mathbf{x}_t; \theta), \quad K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t; \theta), \quad \mathcal{A} : \mathbb{R}_+ \times \mathbb{R}^d \rightarrow \mathbb{R}. \quad (102)$$

Here, $\mathcal{A}$ maps the current capital stock and the vector of economic features into the net increment of capital. The parameter vector θ governs the structural relation between the accumulation environment and the resulting change in capital.

Equation (102) generalizes the classical formula $M - C - M'$ in dynamic form. Since $K_t \equiv M_t$, the monetary increment in the capital circuit can be written as

$$\Delta M_t = M_{t+1} - M_t = K_{t+1} - K_t = \Delta K_t. \quad (103)$$

The classical surplus-value relation is therefore nested as the special case

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) = s_t, \quad K_{t+1} = K_t + s_t. \quad (104)$$

In this benchmark case, the entire increment of capital is identified with surplus value. The multi-feature formulation relaxes this restriction by allowing the increment to depend on a broader set of state variables.

The sign of the accumulation function classifies the regime of capital motion:

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) \begin{cases} > 0, & \text{expanded accumulation,} \\ = 0, & \text{stationary reproduction,} \\ < 0, & \text{devaluation or contraction.} \end{cases} \quad (105)$$

Thus, $\mathcal{A}$ is not merely a prediction rule. It is the structural object that determines whether the capital circuit expands, reproduces itself without expansion, or fails to preserve the value advanced.

The corresponding gross growth rate of capital is

$$g_t \equiv \frac{K_{t+1} - K_t}{K_t} = \frac{\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{K_t}, \quad K_t > 0. \quad (106)$$

Hence,

$$K_{t+1} = K_t(1 + g_t). \quad (107)$$

The accumulation function can therefore be interpreted either in level form, as the net increment of capital, or in proportional form, as the determinant of the growth rate.

Iterating the law of motion gives

$$K_T = K_0 + \sum_{t=0}^{T-1} \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}). \quad (108)$$

Equivalently, when $K_t > 0$ for all t ,

$$K_T = K_0 \prod_{t=0}^{T-1} \left(1 + \frac{\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{K_t} \right). \quad (109)$$

These expressions show that long-run accumulation depends not only on the current value of $\mathcal{A}$, but also on the entire historical path of capital stocks and state variables.

The dependence of $\mathcal{A}$ on K_t allows for scale effects in accumulation. A larger capital stock may increase future accumulation through expanded productive capacity, market power, financial access, or technological complementarities. It may also reduce accumulation through overaccumulation, diminishing returns, debt burdens, or crisis tendencies. These possibilities can be expressed through the marginal effect

$$\frac{\partial \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial K_t}. \quad (110)$$

A positive derivative indicates self-reinforcing accumulation, whereas a negative derivative indicates capital-stock-induced deceleration.

The dependence of $\mathcal{A}$ on $\mathbf{x}_t$ captures the role of the multi-feature accumulation environment. For each feature x_{jt} , the marginal effect is

$$\frac{\partial \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial x_{jt}}, \quad j = 1, \dots, d. \quad (111)$$

These derivatives measure how changes in production conditions, labor relations, technological intensity, financial structure, market demand, institutional arrangements, or ecological constraints affect the net increment of capital.

A useful first-order local approximation around a reference point $(K_0, \mathbf{x}_0)$ is

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) \approx \mathcal{A}(K_0, \mathbf{x}_0; \boldsymbol{\theta}) + \mathcal{A}_K(K_0, \mathbf{x}_0; \boldsymbol{\theta})(K_t - K_0) + \nabla_{\mathbf{x}} \mathcal{A}(K_0, \mathbf{x}_0; \boldsymbol{\theta})^\top (\mathbf{x}_t - \mathbf{x}_0), \quad (112)$$

where

$$\mathcal{A}_K(K_0, \mathbf{x}_0; \boldsymbol{\theta}) \equiv \frac{\partial \mathcal{A}(K_0, \mathbf{x}_0; \boldsymbol{\theta})}{\partial K}, \quad \nabla_{\mathbf{x}} \mathcal{A}(K_0, \mathbf{x}_0; \boldsymbol{\theta}) \equiv \left(\frac{\partial \mathcal{A}}{\partial x_1}, \dots, \frac{\partial \mathcal{A}}{\partial x_d} \right)^\top. \quad (113)$$

This approximation provides the bridge from the general accumulation function to linear econometric specifications.

For empirical work, the accumulation equation may include an unobserved disturbance term:

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid K_t, \mathbf{x}_t] = 0. \quad (114)$$

In panel form, with observational unit i , this becomes

$$K_{i,t+1} = K_{it} + \mathcal{A}(K_{it}, \mathbf{x}_{it}; \boldsymbol{\theta}) + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (115)$$

This representation permits heterogeneity across firms, sectors, regions, or national economies while preserving the same structural interpretation of $\mathcal{A}$.

The accumulation function therefore transforms the classical capital circuit into a general law of motion. Instead of treating accumulation as a scalar identity, it defines accumulation as a mapping from the current capital stock and a multidimensional economic state into the next-period increment of capital. The central theoretical and empirical task of the paper is therefore to specify, estimate, and interpret the function

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}), \quad (116)$$

which governs the transition from classical capital circulation to a multi-feature theory of capital accumulation.

2.3 Linear Multi-Feature Formula

The simplest finite-dimensional specification is obtained by assuming that the accumulation function is affine in the feature vector. In this case, the law of motion for capital is

$$K_{t+1} = K_t + \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t, \quad \boldsymbol{\beta} = (\beta_1, \dots, \beta_d)^\top. \quad (117)$$

Here, α is a baseline accumulation term and $\boldsymbol{\beta}$ is the vector of marginal effects associated with the features in $\mathbf{x}_t$. Equivalently, using the augmented feature vector $\mathbf{z}_t = (1, \mathbf{x}_t^\top)^\top$, the model can be written as

$$K_{t+1} = K_t + \boldsymbol{\theta}^\top \mathbf{z}_t, \quad \boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}^\top)^\top. \quad (118)$$

Thus, the capital increment is

$$\Delta K_t = K_{t+1} - K_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t. \quad (119)$$

Equation (119) is the direct linear counterpart of the classical surplus-value expression. In the classical benchmark, the increment of capital is identified with surplus value:

$$\Delta K_t = s_t. \quad (120)$$

The linear multi-feature formula generalizes this by allowing the increment to be decomposed into contributions from multiple determinants:

$$\Delta K_t = \alpha + \sum_{j=1}^d \beta_j x_{jt}. \quad (121)$$

Each term $\beta_j x_{jt}$ represents the contribution of feature j to the net accumulation of capital in period t .

The coefficient β_j has a simple marginal interpretation:

$$\frac{\partial K_{t+1}}{\partial x_{jt}} = \frac{\partial \Delta K_t}{\partial x_{jt}} = \beta_j, \quad j = 1, \dots, d. \quad (122)$$

A positive coefficient indicates that an increase in x_{jt} raises the capital increment, holding other features fixed:

$$\beta_j > 0 \implies \frac{\partial \Delta K_t}{\partial x_{jt}} > 0. \quad (123)$$

A negative coefficient indicates that the feature depresses accumulation:

$$\beta_j < 0 \implies \frac{\partial \Delta K_t}{\partial x_{jt}} < 0. \quad (124)$$

In this sense, the linear formula provides the most transparent mapping from economic features to capital accumulation.

The baseline term α captures the component of accumulation not explained by the included features:

$$\Delta K_t = \alpha \quad \text{when} \quad \mathbf{x}_t = \mathbf{0}. \quad (125)$$

Depending on the empirical normalization of $\mathbf{x}_t$, α may be interpreted as autonomous accumulation, a mean drift term, or a residual institutional and historical component. If the features are centered around their sample means, then α measures the predicted capital increment at the average state:

$$\mathbb{E}[\mathbf{x}_t] = \mathbf{0} \implies \mathbb{E}[\Delta K_t] = \alpha. \quad (126)$$

The linear formula also yields a simple regime classification. Capital expands, reproduces itself, or contracts according to

$$\alpha + \boldsymbol{\beta}^\top \mathbf{x}_t \begin{cases} > 0, & \text{expanded accumulation,} \\ = 0, & \text{stationary reproduction,} \\ < 0, & \text{devaluation or contraction.} \end{cases} \quad (127)$$

The boundary between expansion and contraction is the hyperplane

$$\alpha + \boldsymbol{\beta}^\top \mathbf{x}_t = 0. \quad (128)$$

This hyperplane divides the feature space into an accumulation region and a devaluation region:

$$\mathcal{X}^+ = \{\mathbf{x} \in \mathbb{R}^d : \alpha + \boldsymbol{\beta}^\top \mathbf{x} > 0\}, \quad \mathcal{X}^- = \{\mathbf{x} \in \mathbb{R}^d : \alpha + \boldsymbol{\beta}^\top \mathbf{x} < 0\}. \quad (129)$$

The recursive dynamics implied by the linear formula are

$$K_T = K_0 + \sum_{t=0}^{T-1} (\alpha + \boldsymbol{\beta}^\top \mathbf{x}_t). \quad (130)$$

Thus, long-run capital accumulation depends on the accumulated history of the feature vector:

$$K_T - K_0 = T\alpha + \boldsymbol{\beta}^\top \sum_{t=0}^{T-1} \mathbf{x}_t. \quad (131)$$

If the feature vector is constant over time, $\mathbf{x}_t = \mathbf{x}$ for all t , then

$$K_T = K_0 + T(\alpha + \boldsymbol{\beta}^\top \mathbf{x}). \quad (132)$$

This makes clear that the linear specification generates arithmetic accumulation in levels unless additional dependence on K_t is introduced.

For econometric implementation, the stochastic version is

$$K_{t+1} - K_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t | \mathbf{x}_t] = 0. \quad (133)$$

In panel form, this becomes

$$K_{i,t+1} - K_{it} = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_{it} + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (134)$$

When persistent unit-level and time-level heterogeneity are important, the specification may be extended as

$$K_{i,t+1} - K_{it} = \alpha_i + \tau_t + \boldsymbol{\beta}^\top \mathbf{x}_{it} + \varepsilon_{it}, \quad (135)$$

where α_i captures unit-specific accumulation conditions and τ_t captures common historical shocks. The linear multi-feature formula is intentionally restrictive. It imposes constant marginal effects and excludes interactions, curvature, thresholds, and feedback from the capital stock unless these are explicitly included as features. Nevertheless, it provides the natural benchmark for the general theory. It is transparent, directly estimable, and interpretable as the first-order approximation to a more general accumulation function:

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) \approx \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t. \quad (136)$$

Subsequent nonlinear, log-linear, and nonparametric specifications can therefore be evaluated relative to this baseline.

2.4 Log-Linear Formula

The linear multi-feature formula models accumulation in levels. A natural alternative is to model capital accumulation in logarithms, so that the coefficients can be interpreted as elasticities. The log-linear specification is given by

$$\log K_{t+1} = \alpha + \rho \log K_t + \boldsymbol{\beta}^\top \log \mathbf{x}_t, \quad \log \mathbf{x}_t = (\log x_{1t}, \dots, \log x_{dt})^\top. \quad (137)$$

Here, $K_t > 0$ and $x_{jt} > 0$ for all $j = 1, \dots, d$, so that the logarithms are well defined. The parameter ρ captures the persistence of capital, while β_j measures the elasticity of next-period capital with respect to feature x_{jt} .

Equation (137) can be equivalently written in multiplicative form as

$$K_{t+1} = \exp(\alpha) K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (138)$$

This expression makes clear that the log-linear model is not additive in levels. Instead, each feature affects capital accumulation proportionally. The accumulation process depends on the product of capital persistence and feature-specific multiplicative contributions.

The elasticity interpretation follows directly from Equation (137). For each feature x_{jt} ,

$$\frac{\partial \log K_{t+1}}{\partial \log x_{jt}} = \beta_j. \quad (139)$$

Thus, a one-percent increase in x_{jt} is associated with an approximate β_j percent change in K_{t+1} , holding other features and current capital fixed. Similarly,

$$\frac{\partial \log K_{t+1}}{\partial \log K_t} = \rho, \quad (140)$$

so ρ measures the elasticity of future capital with respect to current capital.

The implied net increment of capital is

$$\Delta K_t \equiv K_{t+1} - K_t = \exp(\alpha) K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j} - K_t. \quad (141)$$

Hence, unlike the additive linear model, the log-linear formula implies that the absolute capital increment depends on the level of current capital and on the scale of the feature vector. The same proportional change in a feature generates a larger absolute change when the capital base is larger.

The gross growth factor is

$$\frac{K_{t+1}}{K_t} = \exp(\alpha) K_t^{\rho-1} \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (142)$$

Therefore, the log growth rate is

$$\log \frac{K_{t+1}}{K_t} = \alpha + (\rho - 1) \log K_t + \boldsymbol{\beta}^\top \log \mathbf{x}_t. \quad (143)$$

This representation is especially useful because it separates the role of capital persistence from the determinants of proportional accumulation.

The parameter ρ governs the scale dynamics of the model. If

$$\rho = 1, \quad (144)$$

then the growth rate is independent of the current capital stock, conditional on $\mathbf{x}_t$:

$$\log \frac{K_{t+1}}{K_t} = \alpha + \boldsymbol{\beta}^\top \log \mathbf{x}_t. \quad (145)$$

If $\rho < 1$, the model exhibits diminishing proportional accumulation with respect to the existing capital stock. If $\rho > 1$, current capital has an amplifying effect on future capital growth:

$$\rho \begin{cases} < 1, & \text{diminishing capital-scale effect,} \\ = 1, & \text{constant proportional scale effect,} \\ > 1, & \text{self-reinforcing capital-scale effect.} \end{cases} \quad (146)$$

The feature elasticities β_j classify how each dimension of the accumulation environment affects future capital:

$$\beta_j \begin{cases} > 0, & \text{feature } j \text{ raises future capital proportionally,} \\ = 0, & \text{feature } j \text{ is locally neutral,} \\ < 0, & \text{feature } j \text{ lowers future capital proportionally.} \end{cases} \quad (147)$$

Because the model is multiplicative in levels, the marginal effect in levels is

$$\frac{\partial K_{t+1}}{\partial x_{jt}} = \beta_j \frac{K_{t+1}}{x_{jt}}. \quad (148)$$

Thus, even when β_j is constant, the level effect varies with both K_{t+1} and x_{jt} .

The log-linear formula also provides a direct bridge to empirical growth regressions. Introducing an error term gives

$$\log K_{t+1} = \alpha + \rho \log K_t + \boldsymbol{\beta}^\top \log \mathbf{x}_t + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid K_t, \mathbf{x}_t] = 0. \quad (149)$$

Equivalently, in growth-rate form,

$$\log K_{t+1} - \log K_t = \alpha + (\rho - 1) \log K_t + \boldsymbol{\beta}^\top \log \mathbf{x}_t + \varepsilon_t. \quad (150)$$

This specification is convenient when the empirical object of interest is proportional accumulation rather than the absolute increment of capital.

In panel data, the corresponding model is

$$\log K_{i,t+1} = \alpha_i + \tau_t + \rho \log K_{it} + \boldsymbol{\beta}^\top \log \mathbf{x}_{it} + \varepsilon_{it}, \quad (151)$$

where α_i captures persistent unit-specific heterogeneity and τ_t captures common historical or macroeconomic shocks. This form is particularly suitable for comparisons across firms, sectors, regions, or national economies whose capital stocks differ substantially in scale.

The log-linear formula is therefore a multiplicative generalization of the finite-dimensional capital accumulation framework. It preserves the interpretability of a parametric model while allowing the effects of features to scale with the level of capital. Relative to the additive linear formula, it is better suited for contexts in which accumulation is proportional, elasticity-based, and path dependent:

$$K_{t+1} = \exp(\alpha) K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (152)$$

2.5 Cobb-Douglas-Type Capital Formula

A more structurally interpretable multiplicative specification is obtained by defining the accumulation increment itself as a Cobb-Douglas-type function of current capital and the feature vector. The accumulation function is given by

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) = A_0 K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j}, \quad K_{t+1} = K_t + A_0 K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (153)$$

Here, $A_0 > 0$ is a scale parameter, ρ measures the elasticity of the capital increment with respect to current capital, and β_j measures the elasticity of the capital increment with respect to feature x_{jt} . The parameter vector is

$$\boldsymbol{\theta} = (A_0, \rho, \beta_1, \dots, \beta_d)^\top. \quad (154)$$

The specification in Equation (153) differs from the log-linear formula in an important way. The log-linear model specifies the level of next-period capital, K_{t+1} , as a multiplicative function. The Cobb-Douglas-type formulation instead specifies the increment of capital, ΔK_t , as a multiplicative function:

$$\Delta K_t = K_{t+1} - K_t = A_0 K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (155)$$

Thus, the model directly generalizes the classical surplus-value term into a multi-feature accumulation function.

Taking logarithms of Equation (155) yields

$$\log \Delta K_t = \log A_0 + \rho \log K_t + \sum_{j=1}^d \beta_j \log x_{jt}, \quad \Delta K_t > 0. \quad (156)$$

Equivalently,

$$\log \Delta K_t = \alpha + \rho \log K_t + \boldsymbol{\beta}^\top \log \mathbf{x}_t, \quad \alpha \equiv \log A_0. \quad (157)$$

This log-linearized form provides a direct route to empirical estimation when the observed capital increment is positive.

The elasticity of the accumulation increment with respect to current capital is

$$\frac{\partial \log \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial \log K_t} = \rho. \quad (158)$$

For each feature x_{jt} , the corresponding elasticity is

$$\frac{\partial \log \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial \log x_{jt}} = \beta_j, \quad j = 1, \dots, d. \quad (159)$$

Hence, the coefficients have a proportional interpretation: a one-percent increase in x_{jt} changes the accumulation increment by approximately β_j percent, holding all other arguments fixed.

The marginal effect of feature x_{jt} in levels is

$$\frac{\partial \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial x_{jt}} = \beta_j \frac{\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{x_{jt}}. \quad (160)$$

Similarly, the marginal effect of current capital is

$$\frac{\partial \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial K_t} = \rho \frac{\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{K_t}. \quad (161)$$

These expressions imply that the absolute marginal effects depend on the scale of accumulation. Even when elasticities are constant, level effects are larger in states where the accumulation increment is larger.

The second-order properties are also transparent. For each feature,

$$\frac{\partial^2 \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial x_{jt}^2} = \beta_j(\beta_j - 1) \frac{\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{x_{jt}^2}. \quad (162)$$

Thus, for $\mathcal{A} > 0$ and $x_{jt} > 0$, the feature exhibits diminishing marginal effects when

$$0 < \beta_j < 1. \quad (163)$$

It exhibits increasing marginal effects when

$$\beta_j > 1. \quad (164)$$

The cross-partial derivative between two distinct features is

$$\frac{\partial^2 \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{\partial x_{jt} \partial x_{\ell t}} = \beta_j \beta_\ell \frac{\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})}{x_{jt} x_{\ell t}}, \quad j \neq \ell. \quad (165)$$

Therefore, two features are locally complementary when

$$\beta_j \beta_\ell > 0, \quad (166)$$

and locally substitutive in their marginal contribution to accumulation when

$$\beta_j \beta_\ell < 0. \quad (167)$$

The degree of returns to scale in the accumulation function is governed by the sum of exponents:

$$\chi \equiv \rho + \sum_{j=1}^d \beta_j. \quad (168)$$

For any scalar $\lambda > 0$,

$$\mathcal{A}(\lambda K_t, \lambda \mathbf{x}_t; \boldsymbol{\theta}) = \lambda^\chi \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}). \quad (169)$$

Hence,

$$\chi \begin{cases} < 1, & \text{decreasing returns to scale in accumulation,} \\ = 1, & \text{constant returns to scale in accumulation,} \\ > 1, & \text{increasing returns to scale in accumulation.} \end{cases} \quad (170)$$

The implied gross growth rate of capital is

$$g_t \equiv \frac{K_{t+1} - K_t}{K_t} = A_0 K_t^{\rho-1} \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (171)$$

Thus, the role of current capital in proportional accumulation depends on $\rho - 1$. If $\rho = 1$, the growth rate is independent of the scale of current capital, conditional on $\mathbf{x}_t$. If $\rho < 1$, the growth rate declines with the capital stock. If $\rho > 1$, the growth rate increases with the capital stock:

$$\rho \begin{cases} < 1, & \text{diminishing proportional accumulation,} \\ = 1, & \text{scale-neutral proportional accumulation,} \\ > 1, & \text{self-reinforcing proportional accumulation.} \end{cases} \quad (172)$$

The corresponding law of motion is

$$K_{t+1} = K_t \left[1 + A_0 K_t^{\rho-1} \prod_{j=1}^d x_{jt}^{\beta_j} \right]. \quad (173)$$

Iterating this expression gives

$$K_T = K_0 \prod_{t=0}^{T-1} \left[1 + A_0 K_t^{\rho-1} \prod_{j=1}^d x_{jt}^{\beta_j} \right]. \quad (174)$$

This representation highlights the path dependence of accumulation: future capital depends on the sequence of capital stocks and feature vectors, not only on their contemporaneous values.

The Cobb-Douglas-type formula also nests several useful special cases. If only variable capital enters the feature vector and $\beta_v = 1$, then

$$\mathcal{A}(K_t, v_t; \boldsymbol{\theta}) = A_0 v_t. \quad (175)$$

This corresponds to a constant surplus-value-rate interpretation with $A_0 = e$. If both the surplus-value rate and variable capital are included as features, then

$$\mathcal{A}(K_t, e_t, v_t; \boldsymbol{\theta}) = e_t v_t \quad (176)$$

is obtained by setting $A_0 = 1$, $\beta_e = 1$, and $\beta_v = 1$. The classical surplus-value identity is therefore a special case of the broader multiplicative accumulation formula.

For stochastic empirical implementation, one may write

$$\Delta K_t = A_0 K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j} \exp(\varepsilon_t), \quad \mathbb{E}[\varepsilon_t \mid K_t, \mathbf{x}_t] = 0, \quad (177)$$

which implies the log specification

$$\log \Delta K_t = \alpha + \rho \log K_t + \boldsymbol{\beta}^\top \log \mathbf{x}_t + \varepsilon_t. \quad (178)$$

In panel form, the model becomes

$$\log \Delta K_{it} = \alpha_i + \tau_t + \rho \log K_{it} + \boldsymbol{\beta}^\top \log \mathbf{x}_{it} + \varepsilon_{it}, \quad (179)$$

where α_i captures persistent unit-specific accumulation capacity and τ_t captures common historical shocks.

The Cobb-Douglas-type capital formula therefore provides a disciplined middle ground between the additive linear model and fully nonparametric accumulation functions. It preserves constant elasticity interpretation, permits scale analysis, nests classical surplus-value relations, and yields a tractable empirical specification. At the same time, it remains restrictive: the elasticities are constant, the functional form is globally multiplicative, and interactions are imposed through products rather than freely estimated cross-effects. These limitations motivate the subsequent introduction of more general nonlinear and nonparametric accumulation functions.

2.6 General Nonparametric Formula

The linear, log-linear, and Cobb-Douglas-type specifications impose strong functional-form restrictions on the accumulation process. They assume either additivity, constant elasticities, or globally multiplicative effects. A more general formulation allows the capital increment to be governed by an unknown smooth function:

$$K_{t+1} = K_t + g(K_t, \mathbf{x}_t), \quad g \in \mathcal{G} \subset C^2(\mathcal{X}) \cap L_2(\mathcal{X}), \quad g \in W^{2,2}(\mathcal{X}). \quad (180)$$

Here, g is the nonparametric accumulation function, $\mathcal{X}$ denotes the relevant state space for $(K_t, \mathbf{x}_t)$, and $\mathcal{G}$ is the function class over which the accumulation mapping is defined or estimated. The condition $g \in C^2(\mathcal{X})$ ensures twice continuous differentiability, while $g \in L_2(\mathcal{X})$ and $g \in W^{2,2}(\mathcal{X})$ impose square-integrability and Sobolev regularity.

The capital increment is now

$$\Delta K_t \equiv K_{t+1} - K_t = g(K_t, \mathbf{x}_t). \quad (181)$$

This expression nests the earlier parametric formulas as special cases. The linear model is obtained when

$$g(K_t, \mathbf{x}_t) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t. \quad (182)$$

The Cobb-Douglas-type model is obtained when

$$g(K_t, \mathbf{x}_t) = A_0 K_t^\rho \prod_{j=1}^d x_{jt}^{\beta_j}. \quad (183)$$

Thus, the nonparametric formulation does not reject the classical or parametric models; rather, it embeds them in a broader space of possible accumulation laws.

The function class $\mathcal{G}$ determines the admissible degree of flexibility. The smoothness condition

$$g \in C^2(\mathcal{X}) \quad (184)$$

allows first- and second-order marginal effects to be well defined. For a feature x_j , the first-order marginal effect is

$$ME_j(K, \mathbf{x}) \equiv \frac{\partial g(K, \mathbf{x})}{\partial x_j}. \quad (185)$$

The marginal effect of the capital stock is

$$ME_K(K, \mathbf{x}) \equiv \frac{\partial g(K, \mathbf{x})}{\partial K}. \quad (186)$$

Unlike the linear model, these marginal effects need not be constant. They may vary across the state space and may depend on the level of capital and on all other features.

Second-order effects are captured by the Hessian matrix:

$$\nabla^2 g(K, \mathbf{x}) = \left[\frac{\partial^2 g(K, \mathbf{x})}{\partial y_a \partial y_b} \right]_{a,b=0}^d, \quad \mathbf{y} \equiv (K, x_1, \dots, x_d)^\top. \quad (187)$$

The cross-partial derivative

$$\frac{\partial^2 g(K, \mathbf{x})}{\partial x_j \partial x_\ell} \quad (188)$$

measures whether features x_j and x_ℓ are complementary or substitutive in their effect on capital accumulation. In particular,

$$\frac{\partial^2 g(K, \mathbf{x})}{\partial x_j \partial x_\ell} > 0 \implies \text{local complementarity}, \quad (189)$$

whereas

$$\frac{\partial^2 g(K, \mathbf{x})}{\partial x_j \partial x_\ell} < 0 \implies \text{local substitution}. \quad (190)$$

The Sobolev condition

$$g \in W^{2,2}(\mathcal{X}) \quad (191)$$

means that g and its weak derivatives up to order two are square-integrable:

$$\sum_{|\nu| \leq 2} \int_{\mathcal{X}} |D^\nu g(\mathbf{y})|^2 d\mathbf{y} < \infty, \quad \mathbf{y} = (K, \mathbf{x}^\top)^\top. \quad (192)$$

This condition is useful because it rules out excessively irregular accumulation functions while still permitting nonlinearities, curvature, interaction effects, and heterogeneous marginal effects. It also provides a natural basis for regularized estimation.

A common nonparametric approximation represents g through a basis expansion:

$$g(K, \mathbf{x}) \approx g_M(K, \mathbf{x}) = \sum_{m=1}^M \theta_m \psi_m(K, \mathbf{x}), \quad (193)$$

where $\{\psi_m\}_{m=1}^M$ are basis functions and $\boldsymbol{\theta}_M = (\theta_1, \dots, \theta_M)^\top$ are coefficients. The basis may consist of polynomials, splines, wavelets, kernels, neural-network features, or other smooth approximating functions. Under suitable regularity conditions,

$$\|g_M - g\|_{L_2(\mathcal{X})} \rightarrow 0 \quad \text{as} \quad M \rightarrow \infty. \quad (194)$$

For estimation, suppose that observations are indexed by $i = 1, \dots, n$. Let

$$Y_i \equiv K_{i,t+1} - K_{it}, \quad \mathbf{y}_i \equiv (K_{it}, \mathbf{x}_{it}^\top)^\top. \quad (195)$$

The sieve estimator solves

$$\hat{\boldsymbol{\theta}}_M = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^M} \sum_{i=1}^n \left[Y_i - \sum_{m=1}^M \theta_m \psi_m(\mathbf{y}_i) \right]^2. \quad (196)$$

The estimated accumulation function is then

$$\hat{g}_M(K, \mathbf{x}) = \sum_{m=1}^M \hat{\theta}_m \psi_m(K, \mathbf{x}). \quad (197)$$

To control overfitting, the estimator may include a smoothness penalty:

$$\hat{g} = \arg \min_{g \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{i=1}^n [Y_i - g(\mathbf{y}_i)]^2 + \lambda \|g\|_{W^{2,2}(\mathcal{X})}^2 \right\}. \quad (198)$$

The penalty parameter $\lambda \geq 0$ governs the trade-off between empirical fit and functional smoothness. A larger λ produces a smoother accumulation function, while a smaller λ allows more local variation.

The stochastic version of the nonparametric accumulation equation is

$$K_{t+1} = K_t + g(K_t, \mathbf{x}_t) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid K_t, \mathbf{x}_t] = 0. \quad (199)$$

In panel form, this becomes

$$K_{i,t+1} = K_{it} + g(K_{it}, \mathbf{x}_{it}) + \alpha_i + \tau_t + \varepsilon_{it}, \quad (200)$$

where α_i captures persistent unit-specific heterogeneity and τ_t captures common time effects. If fixed effects are removed or controlled for, the nonparametric component g can be interpreted as the common accumulation law across units.

The nonparametric formula also provides a natural way to define local regimes of accumulation. The expansion region is

$$\mathcal{R}^+ = \{(K, \mathbf{x}) \in \mathcal{X} : g(K, \mathbf{x}) > 0\}, \quad (201)$$

the stationary reproduction boundary is

$$\mathcal{R}^0 = \{(K, \mathbf{x}) \in \mathcal{X} : g(K, \mathbf{x}) = 0\}, \quad (202)$$

and the devaluation region is

$$\mathcal{R}^- = \{(K, \mathbf{x}) \in \mathcal{X} : g(K, \mathbf{x}) < 0\}. \quad (203)$$

Unlike the linear specification, the boundary $\mathcal{R}^0$ need not be a hyperplane. It may be curved, disconnected, or state dependent.

The general nonparametric formula therefore provides the most flexible finite-dimensional representation of capital accumulation. It permits nonlinearities, threshold-like behavior, heterogeneous marginal effects, cross-feature interactions, and local instability without imposing a specific parametric structure. At the same time, the regularity conditions ensure that the model remains analytically tractable:

$$g \in \mathcal{G} \subset C^2(\mathcal{X}) \cap L_2(\mathcal{X}) \cap W^{2,2}(\mathcal{X}). \quad (204)$$

This formulation completes the finite-dimensional extension of the classical capital circuit and prepares the ground for the subsequent analysis of assumptions, theoretical validity, nonlinear coupling, and estimation.

3 Assumptions and Theoretical Validity

3.1 State Space and Bounds

The multi-feature accumulation formula requires a well-defined state space. We therefore impose boundedness conditions on both the capital stock and the feature vector. Let K_t denote the capital stock at time t , and let $\mathbf{x}_t = (x_{1t}, \dots, x_{dt})^\top$ denote the finite-dimensional vector of accumulation-relevant features. The admissible state space is specified as

$$K_t \in [\underline{K}, \overline{K}] \subset \mathbb{R}_+, \quad \mathbf{x}_t \in \mathcal{X} \subset \mathbb{R}^d, \quad \mathcal{X} \text{ compact}. \quad (205)$$

The capital stock is assumed to lie in a strictly positive and finite interval:

$$0 < \underline{K} < \overline{K} < \infty, \quad \underline{x}_j \leq x_{jt} \leq \overline{x}_j, \quad j = 1, \dots, d. \quad (206)$$

These restrictions define the economic domain over which the accumulation function is analyzed.

The lower bound on capital rules out degenerate states in which the capital stock vanishes:

$$K_t \geq \underline{K} > 0. \quad (207)$$

This condition is useful for two reasons. First, it ensures that growth rates and logarithmic transformations are well defined. Second, it preserves the interpretation of capital as an active value magnitude capable of entering the accumulation circuit. If $K_t = 0$, the formula $K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta})$

no longer represents the reproduction or expansion of an existing capital stock, but rather the emergence of capital from outside the model.

The upper bound on capital,

$$K_t \leq \bar{K} < \infty, \quad (208)$$

does not imply that capital cannot grow historically. Rather, it defines the local or sample-specific domain on which the theoretical analysis is carried out. In empirical applications, $\bar{K}$ may be interpreted as the largest economically relevant capital magnitude within the observed sample, industry, region, or historical period. This prevents the model from making claims about behavior arbitrarily far outside the support of the data.

The compactness of the feature space is similarly important. If

$$\mathcal{X} = \prod_{j=1}^d [\underline{x}_j, \bar{x}_j], \quad (209)$$

then each feature satisfies

$$\underline{x}_j \leq x_{jt} \leq \bar{x}_j, \quad j = 1, \dots, d. \quad (210)$$

The bounds may represent technological feasibility, labor-market limits, financial constraints, resource availability, ecological limits, institutional ranges, or measurement support. The compactness assumption ensures that the model is evaluated only over economically admissible states.

Combining the capital and feature restrictions, the full finite-dimensional state space can be written as

$$\mathcal{S} \equiv [\underline{K}, \bar{K}] \times \mathcal{X} \subset \mathbb{R}_+ \times \mathbb{R}^d. \quad (211)$$

A generic state is denoted by

$$\mathbf{y}_t \equiv (K_t, \mathbf{x}_t^\top)^\top \in \mathcal{S}. \quad (212)$$

The accumulation function is therefore evaluated on the domain

$$\mathcal{A} : \mathcal{S} \rightarrow \mathbb{R}. \quad (213)$$

The bounded state space also implies that continuous accumulation functions are bounded. If

$$\mathcal{A} \in C(\mathcal{S}), \quad (214)$$

then, by compactness of $\mathcal{S}$, there exists a finite constant $\bar{A} < \infty$ such that

$$\sup_{(K, \mathbf{x}) \in \mathcal{S}} |\mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta})| \leq \bar{A}. \quad (215)$$

This property is analytically useful because it rules out explosive one-period increments within the admissible domain.

The state transition induced by the accumulation function is

$$K_{t+1} = T_K(K_t, \mathbf{x}_t) \equiv K_t + \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}). \quad (216)$$

For the capital path to remain inside the admissible interval, the transition must satisfy

$$\underline{K} \leq K_t + \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) \leq \bar{K}, \quad (K_t, \mathbf{x}_t) \in \mathcal{S}. \quad (217)$$

Equivalently, the accumulation increment must obey the state-dependent bounds

$$\underline{K} - K_t \leq \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) \leq \overline{K} - K_t. \quad (218)$$

These inequalities ensure that the law of motion maps admissible states into admissible future capital stocks.

The feasible accumulation region is therefore

$$\mathcal{S}_A = \{(K, \mathbf{x}) \in \mathcal{S} : \underline{K} - K \leq \mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta}) \leq \overline{K} - K\}. \quad (219)$$

If $\mathcal{S}_A = \mathcal{S}$, the accumulation law is globally admissible on the state space. If $\mathcal{S}_A \subsetneq \mathcal{S}$, the model identifies states in which unconstrained accumulation would violate the economically admissible bounds.

For log-linear and Cobb-Douglas-type specifications, positivity of the features is additionally required. In that case, we impose

$$0 < \underline{x}_j \leq x_{jt} \leq \overline{x}_j < \infty, \quad j = 1, \dots, d. \quad (220)$$

This guarantees that

$$\log K_t, \quad \log x_{1t}, \dots, \log x_{dt} \quad (221)$$

are well defined for all admissible states.

The bounded-state assumption also supports empirical identification. If the observed data satisfy

$$(K_{it}, \mathbf{x}_{it}) \in \mathcal{S}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (222)$$

then the estimated accumulation function is interpreted on the support of the observed economic environment. Extrapolation beyond $\mathcal{S}$ is not part of the maintained theoretical claim. This is important because nonlinear and nonparametric accumulation functions may behave differently outside the observed or economically admissible domain.

Finally, compactness provides the foundation for the subsequent theoretical validity conditions. It allows one to define bounded derivatives, uniform approximations, maximum and minimum values, constrained optima, and stable local dynamics. The maintained state-space assumption may therefore be summarized as

$$(K_t, \mathbf{x}_t) \in \mathcal{S} = [\underline{K}, \overline{K}] \times \mathcal{X}, \quad \mathcal{S} \text{ compact}, \quad 0 < \underline{K} < \overline{K} < \infty. \quad (223)$$

Under this condition, the multi-feature capital accumulation formula is defined on a bounded and economically meaningful domain.

3.2 Smoothness and Boundedness

The state-space restrictions define the domain on which the accumulation function is evaluated. To ensure that the model has well-defined marginal effects, curvature, and local dynamics, we impose smoothness and boundedness on the accumulation function. Let

$$\mathcal{S} = [\underline{K}, \overline{K}] \times \mathcal{X}.$$

We assume that

$$\mathcal{A} \in C^2([\underline{K}, \overline{K}] \times \mathcal{X}), \quad \sup_{(K, \mathbf{x}) \in [\underline{K}, \overline{K}] \times \mathcal{X}} |\mathcal{A}(K, \mathbf{x})| \leq \overline{A} < \infty. \quad (224)$$

The first condition requires the accumulation function to be twice continuously differentiable on the admissible state space. The second condition requires the one-period capital increment to be uniformly bounded.

The smoothness condition implies that the first-order marginal effects are well defined:

$$\mathcal{A}_K(K, \mathbf{x}) \equiv \frac{\partial \mathcal{A}(K, \mathbf{x})}{\partial K}, \quad \mathcal{A}_{x_j}(K, \mathbf{x}) \equiv \frac{\partial \mathcal{A}(K, \mathbf{x})}{\partial x_j}, \quad j = 1, \dots, d. \quad (225)$$

These derivatives describe how the capital increment responds locally to the existing capital stock and to each feature of the accumulation environment. The gradient with respect to the feature vector is

$$\nabla_{\mathbf{x}} \mathcal{A}(K, \mathbf{x}) = (\mathcal{A}_{x_1}(K, \mathbf{x}), \dots, \mathcal{A}_{x_d}(K, \mathbf{x}))^\top. \quad (226)$$

The second-order derivatives are also well defined. Let

$$\mathbf{y} \equiv (K, x_1, \dots, x_d)^\top. \quad (227)$$

The Hessian matrix of the accumulation function is

$$\nabla^2 \mathcal{A}(\mathbf{y}) = \left[\frac{\partial^2 \mathcal{A}(\mathbf{y})}{\partial y_a \partial y_b} \right]_{a,b=0}^d. \quad (228)$$

This matrix governs curvature, complementarity, substitution, and local stability. In particular, the cross-partial derivative

$$\frac{\partial^2 \mathcal{A}(K, \mathbf{x})}{\partial x_j \partial x_\ell} \quad (229)$$

measures whether features x_j and x_ℓ reinforce or offset one another in the accumulation process.

Because $\mathcal{S}$ is compact and $\mathcal{A} \in C^2(\mathcal{S})$, all first- and second-order derivatives are uniformly bounded. Hence there exist finite constants $\bar{A}_1$ and $\bar{A}_2$ such that

$$\sup_{\mathbf{y} \in \mathcal{S}} \|\nabla \mathcal{A}(\mathbf{y})\| \leq \bar{A}_1 < \infty, \quad \sup_{\mathbf{y} \in \mathcal{S}} \|\nabla^2 \mathcal{A}(\mathbf{y})\| \leq \bar{A}_2 < \infty. \quad (230)$$

These bounds imply that the accumulation function cannot exhibit arbitrarily large local responses within the admissible domain.

A direct consequence is Lipschitz continuity. For any two states $\mathbf{y}, \mathbf{y}' \in \mathcal{S}$, the mean-value theorem implies

$$|\mathcal{A}(\mathbf{y}) - \mathcal{A}(\mathbf{y}')| \leq \bar{A}_1 \|\mathbf{y} - \mathbf{y}'\|. \quad (231)$$

Thus, small changes in capital or features induce only small changes in the capital increment. This regularity is important for both theoretical stability and empirical estimation.

Smoothness also justifies local Taylor approximations. Around a reference state $\mathbf{y}_0 = (K_0, \mathbf{x}_0^\top)^\top$, the accumulation function admits the second-order expansion

$$\mathcal{A}(\mathbf{y}) = \mathcal{A}(\mathbf{y}_0) + \nabla \mathcal{A}(\mathbf{y}_0)^\top (\mathbf{y} - \mathbf{y}_0) + \frac{1}{2} (\mathbf{y} - \mathbf{y}_0)^\top \nabla^2 \mathcal{A}(\tilde{\mathbf{y}}) (\mathbf{y} - \mathbf{y}_0), \quad (232)$$

for some $\tilde{\mathbf{y}}$ on the line segment between $\mathbf{y}_0$ and $\mathbf{y}$. The remainder term is bounded by

$$\left| \frac{1}{2} (\mathbf{y} - \mathbf{y}_0)^\top \nabla^2 \mathcal{A}(\tilde{\mathbf{y}}) (\mathbf{y} - \mathbf{y}_0) \right| \leq \frac{\bar{A}_2}{2} \|\mathbf{y} - \mathbf{y}_0\|^2. \quad (233)$$

This provides a rigorous basis for using linear and quadratic specifications as local approximations to the general accumulation function.

The boundedness condition in Equation (224) places an upper limit on the absolute one-period increment of capital:

$$-\bar{A} \leq \mathcal{A}(K, \mathbf{x}) \leq \bar{A}, \quad (K, \mathbf{x}) \in \mathcal{S}. \quad (234)$$

Consequently, the next-period capital stock satisfies

$$K_t - \bar{A} \leq K_{t+1} \leq K_t + \bar{A}. \quad (235)$$

This excludes explosive one-period movements and ensures that accumulation, devaluation, and crisis are modeled as finite transitions within the specified domain.

The state transition map is

$$T_K(K, \mathbf{x}) = K + \mathcal{A}(K, \mathbf{x}). \quad (236)$$

Since $\mathcal{A} \in C^2(\mathcal{S})$, the transition map also satisfies

$$T_K \in C^2(\mathcal{S}). \quad (237)$$

Its derivative with respect to capital is

$$\frac{\partial T_K(K, \mathbf{x})}{\partial K} = 1 + \mathcal{A}_K(K, \mathbf{x}). \quad (238)$$

This derivative is central to local dynamic stability. Around a steady state $(K^*, \mathbf{x}^*)$ satisfying

$$\mathcal{A}(K^*, \mathbf{x}^*) = 0, \quad (239)$$

the local capital dynamics are stable when

$$|1 + \mathcal{A}_K(K^*, \mathbf{x}^*)| < 1. \quad (240)$$

For empirical implementation, smoothness and boundedness also restrict the complexity of admissible specifications. In a parametric model $\mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta})$, the maintained condition is

$$\mathcal{A}(\cdot, \cdot; \boldsymbol{\theta}) \in C^2(\mathcal{S}), \quad \sup_{(K, \mathbf{x}) \in \mathcal{S}} |\mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta})| < \infty. \quad (241)$$

In a nonparametric model, the analogous restriction is

$$g \in \mathcal{G} \subset C^2(\mathcal{S}) \cap L_2(\mathcal{S}), \quad \|g\|_\infty \leq \bar{A}. \quad (242)$$

Thus, both parametric and nonparametric formulations are governed by the same regularity principle: the accumulation law should be sufficiently smooth to support marginal analysis and sufficiently bounded to preserve finite economic dynamics.

The smoothness and boundedness assumptions therefore provide the analytical foundation for the subsequent theory. They ensure that marginal effects, elasticities, Hessians, local approximations, stability conditions, and optimization problems are all well defined on the admissible economic state space:

$$\mathcal{A} \in C^2(\mathcal{S}), \quad \|\mathcal{A}\|_\infty \leq \bar{A} < \infty. \quad (243)$$

3.3 Econometric Orthogonality

The theoretical accumulation function becomes empirically meaningful only when its disturbance term satisfies a suitable orthogonality condition. In the time-series formulation, we impose

$$\mathbb{E}[\varepsilon_t \mid \mathbf{z}_t] = 0, \quad \mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] = 0. \quad (244)$$

The first condition states that the innovation ε_t is mean independent of the augmented feature vector $\mathbf{z}_t = (1, \mathbf{x}_t^\top)^\top$. The second condition is the corresponding panel-data restriction: after conditioning on current capital, observed features, unit fixed effects, and time effects, the residual component u_{it} has zero conditional mean.

For the linear multi-feature formula, the stochastic specification is

$$K_{t+1} - K_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t + \varepsilon_t. \quad (245)$$

Using the augmented notation, this becomes

$$\Delta K_t = \boldsymbol{\theta}^\top \mathbf{z}_t + \varepsilon_t, \quad \boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}^\top)^\top. \quad (246)$$

The orthogonality condition in Equation (244) implies the population moment condition

$$\mathbb{E} \left[\mathbf{z}_t \left(\Delta K_t - \boldsymbol{\theta}^\top \mathbf{z}_t \right) \right] = \mathbf{0}. \quad (247)$$

This condition identifies the coefficient vector $\boldsymbol{\theta}$ when the second-moment matrix is nonsingular:

$$\mathbb{E}[\mathbf{z}_t \mathbf{z}_t^\top] \text{ is nonsingular.} \quad (248)$$

Under this rank condition, the population coefficient is

$$\boldsymbol{\theta}_0 = \left(\mathbb{E}[\mathbf{z}_t \mathbf{z}_t^\top] \right)^{-1} \mathbb{E}[\mathbf{z}_t \Delta K_t]. \quad (249)$$

In the general nonlinear case, the stochastic accumulation equation is

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) + \varepsilon_t. \quad (250)$$

Equivalently,

$$\Delta K_t = \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) + \varepsilon_t. \quad (251)$$

The conditional mean restriction becomes

$$\mathbb{E}[\Delta K_t - \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}_0) \mid K_t, \mathbf{x}_t] = 0. \quad (252)$$

This states that, at the true parameter value $\boldsymbol{\theta}_0$, the accumulation function equals the conditional mean of the capital increment:

$$\mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}_0) = \mathbb{E}[\Delta K_t \mid K_t, \mathbf{x}_t]. \quad (253)$$

The orthogonality condition is not merely a technical assumption. It defines which component of the observed capital increment is attributed to the systematic accumulation mechanism and which component is treated as an unanticipated disturbance. Writing

$$\varepsilon_t = \Delta K_t - \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}_0), \quad (254)$$

orthogonality requires that the systematic component has exhausted all predictable variation in ΔK_t with respect to the conditioning variables:

$$\mathbb{E}[\varepsilon_t \mid K_t, \mathbf{x}_t] = 0. \quad (255)$$

In panel data, the accumulation equation may be written as

$$K_{i,t+1} - K_{it} = \mathcal{A}(K_{it}, \mathbf{x}_{it}; \boldsymbol{\theta}) + \alpha_i + \tau_t + u_{it}, \quad (256)$$

where α_i captures time-invariant unit heterogeneity and τ_t captures common time shocks. The maintained orthogonality restriction is

$$\mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] = 0. \quad (257)$$

This permits the observed features to be correlated with permanent unit characteristics and common historical shocks, provided those components are properly conditioned on or controlled for.

For a linear fixed-effects specification,

$$K_{i,t+1} - K_{it} = \alpha_i + \tau_t + \boldsymbol{\beta}^\top \mathbf{x}_{it} + u_{it}, \quad (258)$$

the identifying moment condition is

$$\mathbb{E}[\tilde{\mathbf{x}}_{it} (\widetilde{\Delta K_{it}} - \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_{it})] = \mathbf{0}, \quad (259)$$

where tildes denote variables residualized with respect to unit and time effects. The corresponding rank condition is

$$\mathbb{E}[\tilde{\mathbf{x}}_{it} \tilde{\mathbf{x}}_{it}^\top] \text{ is nonsingular.} \quad (260)$$

The orthogonality assumption also clarifies the main threats to identification. If a feature is correlated with an omitted determinant of capital accumulation, then

$$\mathbb{E}[\varepsilon_t \mid \mathbf{z}_t] \neq 0. \quad (261)$$

Similarly, in panel data, if time-varying unobserved shocks affect both features and capital accumulation, then

$$\mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] \neq 0. \quad (262)$$

Such violations may arise from reverse causality, omitted variables, measurement error, simultaneity, anticipation effects, or endogenous policy responses.

A common example is reverse causality from expected accumulation to current features. If agents choose $\mathbf{x}_{it}$ in anticipation of future capital growth, then

$$\mathbf{x}_{it} = \boldsymbol{\Pi} \Delta K_{i,t+1} + \mathbf{v}_{it}, \quad (263)$$

which generally implies

$$\text{Cov}(\mathbf{x}_{it}, u_{it}) \neq \mathbf{0}. \quad (264)$$

In that case, the estimated coefficients do not have a structural accumulation interpretation without additional identification strategies.

When strict conditional orthogonality is not credible, the weaker moment condition

$$\mathbb{E}[\mathbf{q}_{it} u_{it}] = \mathbf{0} \quad (265)$$

may be imposed for a vector of instruments $\mathbf{q}_{it}$. Valid instruments must satisfy relevance and exogeneity:

$$\text{rank}\left(\mathbb{E}[\mathbf{q}_{it}\mathbf{q}_{it}^\top]\right) = d, \quad \mathbb{E}[\mathbf{q}_{it}u_{it}] = \mathbf{0}. \quad (266)$$

This provides a route from the accumulation formula to instrumental-variable or generalized method of moments estimation.

The orthogonality condition therefore gives the multi-feature capital formula its econometric interpretation. It states that, after conditioning on the chosen state variables and appropriate fixed effects, the remaining disturbance contains no systematic information about capital accumulation. Under this condition, the accumulation function can be interpreted as a structural or conditional-mean law of motion rather than a purely descriptive curve:

$$\mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta}_0) = \mathbb{E}[\Delta K \mid K, \mathbf{x}]. \quad (267)$$

3.4 Infinite-Dimensional Operator Conditions

The finite-dimensional formulation can be extended to settings in which the state of accumulation is represented not by a finite vector, but by a function. This is useful when the determinants of capital accumulation are distributed over a continuum, such as space, sectors, technologies, tasks, skills, environmental states, or institutional configurations. Let H be a real Hilbert space equipped with inner product $\langle \cdot, \cdot \rangle_H$ and norm $\|\cdot\|_H$. The infinite-dimensional interaction structure is governed by a linear operator $\mathcal{K} : H \rightarrow H$ satisfying

$$\mathcal{K} : H \rightarrow H \quad \text{is bounded and self-adjoint.} \quad (268)$$

For the associated inverse problem, we impose

$$\mathcal{K}^{-1} \text{ exists} \quad \vee \quad (\mathcal{K}^\dagger \text{ exists and } b \in \text{Range}(\mathcal{K})). \quad (269)$$

Here, $\mathcal{K}^{-1}$ denotes the ordinary inverse, while $\mathcal{K}^\dagger$ denotes the Moore–Penrose generalized inverse. The vector $b \in H$ represents the first-order linear component of the infinite-dimensional accumulation functional.

The boundedness of $\mathcal{K}$ means that there exists a finite constant $\bar{\kappa} < \infty$ such that

$$\|\mathcal{K}f\|_H \leq \bar{\kappa}\|f\|_H, \quad \forall f \in H. \quad (270)$$

Equivalently,

$$\|\mathcal{K}\|_{\text{op}} \equiv \sup_{\|f\|_H \leq 1} \|\mathcal{K}f\|_H < \infty. \quad (271)$$

This condition ensures that the operator cannot magnify small perturbations in the feature function without bound. Economically, boundedness rules out infinite sensitivity of capital accumulation to infinitesimal changes in the underlying functional state.

Self-adjointness means that

$$\langle f, \mathcal{K}g \rangle_H = \langle \mathcal{K}f, g \rangle_H, \quad \forall f, g \in H. \quad (272)$$

This condition guarantees that the quadratic interaction term is symmetric:

$$\langle f, \mathcal{K}g \rangle_H = \langle g, \mathcal{K}f \rangle_H. \quad (273)$$

Thus, the operator represents a well-defined symmetric interaction structure among the continuum of accumulation-relevant features.

A canonical infinite-dimensional accumulation functional takes the form

$$\mathcal{A}(x) = a_0 + \langle b, x \rangle_H + \langle x, \mathcal{K}x \rangle_H, \quad x \in H. \quad (274)$$

The linear term $\langle b, x \rangle_H$ captures the direct contribution of the feature function x , while the quadratic term $\langle x, \mathcal{K}x \rangle_H$ captures pairwise interactions across the feature domain. By boundedness of $\mathcal{K}$, the quadratic term is finite:

$$|\langle x, \mathcal{K}x \rangle_H| \leq \|x\|_H \|\mathcal{K}x\|_H \leq \|\mathcal{K}\|_{\text{op}} \|x\|_H^2. \quad (275)$$

The first variation of Equation (274) is

$$D\mathcal{A}(x)[h] = \langle b, h \rangle_H + \langle h, \mathcal{K}x \rangle_H + \langle x, \mathcal{K}h \rangle_H, \quad h \in H. \quad (276)$$

Using self-adjointness of $\mathcal{K}$, this becomes

$$D\mathcal{A}(x)[h] = \langle b + 2\mathcal{K}x, h \rangle_H. \quad (277)$$

Hence the functional gradient is

$$\nabla_H \mathcal{A}(x) = b + 2\mathcal{K}x. \quad (278)$$

An interior critical point $x^* \in H$ satisfies

$$\nabla_H \mathcal{A}(x^*) = 0, \quad b + 2\mathcal{K}x^* = 0. \quad (279)$$

Equivalently,

$$\mathcal{K}x^* = -\frac{1}{2}b. \quad (280)$$

This is an operator equation. Its solvability depends on the inverse properties of $\mathcal{K}$.

If $\mathcal{K}^{-1}$ exists as a bounded inverse on H , then the critical point is uniquely given by

$$x^* = -\frac{1}{2}\mathcal{K}^{-1}b. \quad (281)$$

In this case, the infinite-dimensional accumulation problem behaves like a well-posed quadratic problem: every admissible linear component $b \in H$ corresponds to a unique stationary functional state.

If $\mathcal{K}^{-1}$ does not exist, the equation may still be solvable provided that b lies in the range of $\mathcal{K}$:

$$b \in \text{Range}(\mathcal{K}). \quad (282)$$

When this condition holds and the Moore–Penrose generalized inverse $\mathcal{K}^\dagger$ exists, the minimum-norm solution is

$$x^* = -\frac{1}{2}\mathcal{K}^\dagger b. \quad (283)$$

This solution selects, among all possible solutions, the one with the smallest Hilbert-space norm:

$$x^* \in \arg \min \left\{ \|x\|_H : \mathcal{K}x = -\frac{1}{2}b \right\}. \quad (284)$$

The role of the range condition is essential. If

$$b \notin \text{Range}(\mathcal{K}), \quad (285)$$

then the operator equation

$$\mathcal{K}x = -\frac{1}{2}b \quad (286)$$

has no exact solution in H . Economically, this means that the linear directional force represented by b cannot be balanced by the quadratic interaction structure encoded in $\mathcal{K}$. The accumulation functional then lacks an interior critical point in the unconstrained Hilbert-space domain.

The curvature of the infinite-dimensional functional is determined by $\mathcal{K}$. The second variation is

$$D^2\mathcal{A}(x)[h, h] = 2\langle h, \mathcal{K}h \rangle_H. \quad (287)$$

If

$$\langle h, \mathcal{K}h \rangle_H \leq 0, \quad \forall h \in H, \quad (288)$$

then $\mathcal{A}$ is concave on H . If

$$\langle h, \mathcal{K}h \rangle_H \geq 0, \quad \forall h \in H, \quad (289)$$

then $\mathcal{A}$ is convex on H . Strict definiteness gives strict local classification:

$$\langle h, \mathcal{K}h \rangle_H < 0 \quad \forall h \neq 0 \implies x^* \text{ is a strict local maximum,} \quad (290)$$

whereas

$$\langle h, \mathcal{K}h \rangle_H > 0 \quad \forall h \neq 0 \implies x^* \text{ is a strict local minimum.} \quad (291)$$

When $H = L^2(\Omega, \mu)$, a common representation of $\mathcal{K}$ is an integral operator:

$$(\mathcal{K}f)(\omega) = \int_{\Omega} \kappa(\omega, \omega') f(\omega') d\mu(\omega'). \quad (292)$$

Self-adjointness corresponds to kernel symmetry:

$$\kappa(\omega, \omega') = \kappa(\omega', \omega) \quad \text{for } \mu \otimes \mu\text{-a.e. } (\omega, \omega'). \quad (293)$$

A sufficient boundedness condition is Hilbert–Schmidt integrability:

$$\int_{\Omega} \int_{\Omega} |\kappa(\omega, \omega')|^2 d\mu(\omega) d\mu(\omega') < \infty. \quad (294)$$

Under this condition, $\mathcal{K}$ is compact and bounded, and the quadratic interaction term can be written as

$$\langle x, \mathcal{K}x \rangle_H = \int_{\Omega} \int_{\Omega} x(\omega) \kappa(\omega, \omega') x(\omega') d\mu(\omega) d\mu(\omega'). \quad (295)$$

The operator conditions therefore provide the mathematical foundation for the infinite-dimensional capital formula. Boundedness ensures finite and stable functional interactions. Self-adjointness ensures symmetric quadratic structure and well-defined curvature. The inverse or generalized-inverse condition ensures that critical points of the infinite-dimensional accumulation functional are well posed:

$$\mathcal{K}^{-1} \text{ exists} \quad \text{or} \quad \left(\mathcal{K}^{\dagger} \text{ exists and } b \in \text{Range}(\mathcal{K}) \right). \quad (296)$$

Together, these conditions make the transition from finite-dimensional features to functional capital accumulation analytically coherent.

4 Nonlinear Coupling Formula

4.1 Second-Order Expansion

The linear multi-feature formula provides a transparent benchmark, but it cannot capture nonlinear feedback, diminishing or increasing marginal effects, or interaction among accumulation-relevant features. To introduce these mechanisms, consider a smooth accumulation function $\mathcal{A}(\mathbf{x})$ defined over the feature space $\mathcal{X} \subset \mathbb{R}^d$. Around a reference point, the function can be locally approximated by a second-order expansion:

$$\mathcal{A}(\mathbf{x}) \approx a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}. \quad (297)$$

Here, a_0 is a scalar baseline term, $\mathbf{a} \in \mathbb{R}^d$ is the vector of first-order effects, and $\mathbf{H} \in \mathbb{R}^{d \times d}$ is the Hessian or interaction matrix governing second-order curvature.

More generally, if the expansion is taken around a point $\mathbf{x}_0 \in \mathcal{X}$, Taylor's theorem gives

$$\mathcal{A}(\mathbf{x}) = \mathcal{A}(\mathbf{x}_0) + \nabla \mathcal{A}(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^\top \nabla^2 \mathcal{A}(\mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0) + R_3(\mathbf{x}, \mathbf{x}_0). \quad (298)$$

The compact expression in Equation (297) is obtained by normalizing the reference point to $\mathbf{x}_0 = \mathbf{0}$ or by absorbing the reference-state terms into the coefficients:

$$a_0 = \mathcal{A}(\mathbf{x}_0), \quad \mathbf{a} = \nabla \mathcal{A}(\mathbf{x}_0), \quad \mathbf{H} = \nabla^2 \mathcal{A}(\mathbf{x}_0). \quad (299)$$

The first-order term captures the direct marginal contribution of each feature:

$$\mathbf{a}^\top \mathbf{x} = \sum_{j=1}^d a_j x_j. \quad (300)$$

If $a_j > 0$, feature x_j increases accumulation locally, holding all other features fixed. If $a_j < 0$, it depresses accumulation locally. This part of the expansion corresponds to the linear multi-feature formula.

The second-order term introduces curvature and cross-feature coupling:

$$\frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x} = \frac{1}{2} \sum_{j=1}^d \sum_{\ell=1}^d H_{j\ell} x_j x_\ell. \quad (301)$$

The diagonal entries H_{jj} determine the own curvature of each feature, while the off-diagonal entries $H_{j\ell}$ determine interactions between different features. Thus, the second-order formula allows the effect of one feature to depend on the level of another.

The local marginal effect of feature x_j is

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \approx a_j + \sum_{\ell=1}^d H_{j\ell} x_\ell. \quad (302)$$

In vector form,

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}) \approx \mathbf{a} + \mathbf{H} \mathbf{x}. \quad (303)$$

This expression shows why the second-order approximation is richer than the linear formula: marginal effects are no longer constant, but vary with the state of the feature vector.

The second derivative with respect to two features is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j \partial x_\ell} \approx H_{j\ell}. \quad (304)$$

When $j = \ell$, this term measures own-feature curvature. When $j \neq \ell$, it measures cross-feature coupling. A positive cross-partial indicates local complementarity:

$$H_{j\ell} > 0 \implies x_j \text{ and } x_\ell \text{ are locally complementary.} \quad (305)$$

A negative cross-partial indicates local substitution:

$$H_{j\ell} < 0 \implies x_j \text{ and } x_\ell \text{ are locally substitutive.} \quad (306)$$

The curvature of the accumulation function is governed by the definiteness of $\mathbf{H}$. If

$$\mathbf{H} \preceq 0, \quad (307)$$

then the quadratic approximation is concave, so marginal returns are globally non-increasing within the local approximation. If

$$\mathbf{H} \succeq 0, \quad (308)$$

then the quadratic approximation is convex, allowing increasing marginal effects and self-reinforcing accumulation.

The stationary point of the quadratic approximation satisfies

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}^*) \approx \mathbf{a} + \mathbf{H}\mathbf{x}^* = \mathbf{0}. \quad (309)$$

If $\mathbf{H}$ is nonsingular, the stationary point is

$$\mathbf{x}^* = -\mathbf{H}^{-1}\mathbf{a}. \quad (310)$$

The nature of this point depends on the curvature matrix:

$$\mathbf{H} \prec 0 \implies \mathbf{x}^* \text{ is a strict local maximum,} \quad (311)$$

whereas

$$\mathbf{H} \succ 0 \implies \mathbf{x}^* \text{ is a strict local minimum.} \quad (312)$$

The approximation error is represented by the remainder term. If $\mathcal{A} \in C^3(\mathcal{X})$, then for $\mathbf{x}$ near $\mathbf{x}_0$,

$$R_3(\mathbf{x}, \mathbf{x}_0) = O(\|\mathbf{x} - \mathbf{x}_0\|^3). \quad (313)$$

Thus, the second-order expansion is locally accurate when the feature vector remains close to the reference state. Its purpose is not to replace the general function globally, but to provide a tractable local representation of nonlinear accumulation.

Substituting the second-order approximation into the capital law of motion gives

$$K_{t+1} = K_t + a_0 + \mathbf{a}^\top \mathbf{x}_t + \frac{1}{2} \mathbf{x}_t^\top \mathbf{H} \mathbf{x}_t. \quad (314)$$

The implied capital increment is

$$\Delta K_t = a_0 + \mathbf{a}^\top \mathbf{x}_t + \frac{1}{2} \mathbf{x}_t^\top \mathbf{H} \mathbf{x}_t. \quad (315)$$

This specification is the minimal nonlinear extension of the finite-feature capital formula. It preserves interpretability while allowing accumulation to depend on curvature, interaction, and state-dependent marginal effects.

The second-order expansion therefore plays a central role in the general framework. It bridges the linear formula and the fully nonparametric accumulation function:

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x} + R_3(\mathbf{x}). \quad (316)$$

The matrix $\mathbf{H}$ is the key object: it encodes whether the determinants of capital accumulation operate independently, reinforce one another, or offset one another within the local economic state space.

4.2 Interaction Matrix

The second-order expansion can be written explicitly in terms of an interaction matrix. Let

$$\mathbf{H} = (h_{ij})_{i,j=1}^d, \quad \mathcal{A}(\mathbf{x}) = a_0 + \sum_i a_i x_i + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d h_{ij} x_i x_j. \quad (317)$$

The matrix $\mathbf{H}$ collects all second-order effects among the features in $\mathbf{x}$. Its diagonal entries describe own-feature curvature, while its off-diagonal entries describe pairwise interaction effects.

The vector form and the summation form are equivalent:

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} = \sum_{i=1}^d \sum_{j=1}^d h_{ij} x_i x_j. \quad (318)$$

Hence the accumulation function can be written compactly as

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}. \quad (319)$$

This representation is useful because it separates the baseline term, first-order feature effects, and second-order coupling structure.

The diagonal entries of $\mathbf{H}$ determine the curvature associated with each individual feature:

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i^2} = h_{ii}. \quad (320)$$

If $h_{ii} < 0$, feature x_i exhibits diminishing marginal contribution to capital accumulation. If $h_{ii} > 0$, feature x_i exhibits increasing marginal contribution. If $h_{ii} = 0$, the local effect of x_i is linear, conditional on the other features.

The off-diagonal entries determine cross-feature coupling:

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} = \frac{1}{2} (h_{ij} + h_{ji}), \quad i \neq j. \quad (321)$$

Thus, only the symmetric part of $\mathbf{H}$ affects the quadratic form. This is because

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} = \mathbf{x}^\top \left(\frac{\mathbf{H} + \mathbf{H}^\top}{2} \right) \mathbf{x}. \quad (322)$$

Without loss of generality, the interaction matrix may therefore be taken as symmetric:

$$\mathbf{H} = \mathbf{H}^\top. \quad (323)$$

Under this normalization, the cross-partial derivative becomes

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} = h_{ij}, \quad i \neq j. \quad (324)$$

The first-order marginal effect of feature x_i is

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_i} = a_i + \frac{1}{2} \sum_{j=1}^d h_{ij} x_j + \frac{1}{2} \sum_{j=1}^d h_{ji} x_j. \quad (325)$$

If $\mathbf{H}$ is symmetric, this simplifies to

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_i} = a_i + \sum_{j=1}^d h_{ij} x_j. \quad (326)$$

Hence the marginal contribution of each feature depends on the entire vector of economic conditions.

In vector notation, the gradient is

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}) = \mathbf{a} + \frac{1}{2}(\mathbf{H} + \mathbf{H}^\top) \mathbf{x}. \quad (327)$$

Under the symmetric normalization, this becomes

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}) = \mathbf{a} + \mathbf{H} \mathbf{x}. \quad (328)$$

The Hessian is

$$\nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}) = \frac{1}{2}(\mathbf{H} + \mathbf{H}^\top), \quad (329)$$

and, when $\mathbf{H}$ is symmetric,

$$\nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}) = \mathbf{H}. \quad (330)$$

The sign of h_{ij} has a direct economic interpretation. When

$$h_{ij} > 0, \quad i \neq j, \quad (331)$$

features x_i and x_j are locally complementary: an increase in one raises the marginal effect of the other on capital accumulation. When

$$h_{ij} < 0, \quad i \neq j, \quad (332)$$

the features are locally substitutive: an increase in one lowers the marginal effect of the other. When

$$h_{ij} = 0, \quad i \neq j, \quad (333)$$

there is no local second-order interaction between the two features.

The interaction matrix also determines the curvature of the accumulation surface. If

$$\mathbf{H} \preceq 0, \quad (334)$$

then the quadratic approximation is concave and the accumulation function exhibits globally diminishing joint returns over the local domain. If

$$\mathbf{H} \succeq 0, \quad (335)$$

then the quadratic approximation is convex and can generate increasing joint returns. If $\mathbf{H}$ is indefinite, the accumulation surface contains both expansionary and contractionary directions:

$$\exists \mathbf{u}, \mathbf{v} \in \mathbb{R}^d \quad \text{such that} \quad \mathbf{u}^\top \mathbf{H} \mathbf{u} > 0, \quad \mathbf{v}^\top \mathbf{H} \mathbf{v} < 0. \quad (336)$$

This case is especially relevant for capital accumulation, because some feature combinations may reinforce accumulation while others generate fragility, overaccumulation, or crisis tendencies.

The eigenvalue decomposition provides another interpretation of the interaction matrix. If $\mathbf{H}$ is symmetric, then

$$\mathbf{H} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_d), \quad (337)$$

where $\mathbf{Q}$ is orthonormal and $\lambda_1, \dots, \lambda_d$ are real eigenvalues. The quadratic term becomes

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} = \sum_{k=1}^d \lambda_k (\mathbf{q}_k^\top \mathbf{x})^2. \quad (338)$$

Each eigenvector $\mathbf{q}_k$ represents a principal direction of feature variation, and each eigenvalue λ_k measures the curvature of accumulation along that direction.

A positive eigenvalue indicates a direction of increasing returns:

$$\lambda_k > 0 \implies \text{convex accumulation response along } \mathbf{q}_k. \quad (339)$$

A negative eigenvalue indicates a direction of diminishing returns or instability:

$$\lambda_k < 0 \implies \text{concave accumulation response along } \mathbf{q}_k. \quad (340)$$

Thus, the eigenstructure of $\mathbf{H}$ summarizes the dominant nonlinear directions of the accumulation system.

The stationary condition for the quadratic approximation is

$$\mathbf{a} + \mathbf{H} \mathbf{x}^* = \mathbf{0}, \quad (341)$$

assuming $\mathbf{H}$ is symmetric. If $\mathbf{H}$ is nonsingular, the stationary feature vector is

$$\mathbf{x}^* = -\mathbf{H}^{-1} \mathbf{a}. \quad (342)$$

The classification of $\mathbf{x}^*$ depends entirely on the definiteness of $\mathbf{H}$:

$$\mathbf{H} \prec 0 \implies \mathbf{x}^* \text{ is a strict local maximum,} \quad (343)$$

$$\mathbf{H} \succ 0 \implies \mathbf{x}^* \text{ is a strict local minimum,} \quad (344)$$

and

$$\mathbf{H} \text{ indefinite} \implies \mathbf{x}^* \text{ is a saddle point.} \quad (345)$$

Substituting the interaction-matrix formula into the capital law of motion yields

$$K_{t+1} = K_t + a_0 + \sum_{i=1}^d a_i x_{it} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d h_{ij} x_{it} x_{jt}. \quad (346)$$

The corresponding capital increment is

$$\Delta K_t = a_0 + \sum_{i=1}^d a_i x_{it} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d h_{ij} x_{it} x_{jt}. \quad (347)$$

For empirical implementation, a stochastic version is

$$\Delta K_t = a_0 + \sum_{i=1}^d a_i x_{it} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d h_{ij} x_{it} x_{jt} + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t | \mathbf{x}_t] = 0. \quad (348)$$

This equation can be estimated as a linear regression in expanded regressors that include both original features and pairwise products:

$$\{x_{1t}, \dots, x_{dt}, x_{1t}x_{1t}, x_{1t}x_{2t}, \dots, x_{dt}x_{dt}\}. \quad (349)$$

The interaction matrix therefore provides a tractable way to move beyond additive accumulation while preserving interpretability.

In summary, $\mathbf{H}$ is the central object of nonlinear finite-dimensional capital accumulation. It determines curvature, complementarity, substitution, critical points, and the principal directions of expansion or contraction:

$$\mathbf{H} = \nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}) \quad \text{within the second-order approximation.} \quad (350)$$

Through this matrix, the model captures the idea that the determinants of capital accumulation do not operate in isolation, but through a structured system of nonlinear couplings.

4.3 Symmetric and Asymmetric Coupling

The interaction matrix in the quadratic accumulation formula need not be symmetric as written. However, only its symmetric component contributes to the scalar quadratic form. Any matrix $\mathbf{H} \in \mathbb{R}^{d \times d}$ can be decomposed uniquely into a symmetric part and an asymmetric part:

$$\mathbf{H}_S = \frac{1}{2}(\mathbf{H} + \mathbf{H}^\top), \quad \mathbf{H}_A = \frac{1}{2}(\mathbf{H} - \mathbf{H}^\top), \quad \mathbf{x}^\top \mathbf{H}_A \mathbf{x} = 0. \quad (351)$$

Here, $\mathbf{H}_S$ is symmetric and $\mathbf{H}_A$ is skew-symmetric:

$$\mathbf{H}_S^\top = \mathbf{H}_S, \quad \mathbf{H}_A^\top = -\mathbf{H}_A. \quad (352)$$

The original interaction matrix is recovered as

$$\mathbf{H} = \mathbf{H}_S + \mathbf{H}_A. \quad (353)$$

The key property is that the asymmetric component vanishes inside the quadratic form. Since $\mathbf{x}^\top \mathbf{H}_A \mathbf{x}$ is a scalar, it equals its transpose:

$$\mathbf{x}^\top \mathbf{H}_A \mathbf{x} = \left(\mathbf{x}^\top \mathbf{H}_A \mathbf{x} \right)^\top = \mathbf{x}^\top \mathbf{H}_A^\top \mathbf{x} = -\mathbf{x}^\top \mathbf{H}_A \mathbf{x}. \quad (354)$$

Therefore,

$$2\mathbf{x}^\top \mathbf{H}_A \mathbf{x} = 0, \quad \mathbf{x}^\top \mathbf{H}_A \mathbf{x} = 0. \quad (355)$$

It follows that

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} = \mathbf{x}^\top \mathbf{H}_S \mathbf{x} + \mathbf{x}^\top \mathbf{H}_A \mathbf{x} = \mathbf{x}^\top \mathbf{H}_S \mathbf{x}. \quad (356)$$

Hence, the second-order accumulation formula can be written without loss of generality as

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H}_S \mathbf{x}. \quad (357)$$

The asymmetric matrix $\mathbf{H}_A$ does not affect the level of the accumulation function. For this reason, when the objective is to model the scalar capital increment ΔK_t , the relevant second-order object is the symmetric interaction matrix $\mathbf{H}_S$.

The same conclusion appears in the gradient. For the general quadratic form,

$$\nabla_{\mathbf{x}} \left(\frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x} \right) = \frac{1}{2} (\mathbf{H} + \mathbf{H}^\top) \mathbf{x} = \mathbf{H}_S \mathbf{x}. \quad (358)$$

Therefore,

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}) = \mathbf{a} + \mathbf{H}_S \mathbf{x}. \quad (359)$$

The Hessian is

$$\nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}) = \mathbf{H}_S. \quad (360)$$

Thus, marginal effects, curvature, complementarity, substitution, and critical-point classification are all determined by $\mathbf{H}_S$, not by $\mathbf{H}_A$.

The cross-feature effect between x_i and x_j is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} = (\mathbf{H}_S)_{ij} = \frac{1}{2} (h_{ij} + h_{ji}), \quad i \neq j. \quad (361)$$

This equation shows that the economically relevant pairwise interaction is the average of the two directional entries h_{ij} and h_{ji} . If h_{ij} and h_{ji} differ, the scalar quadratic accumulation function cannot separately identify the two directional effects; it identifies only their symmetric component.

The diagonal entries are unchanged by symmetrization:

$$(\mathbf{H}_S)_{ii} = h_{ii}, \quad (\mathbf{H}_A)_{ii} = 0. \quad (362)$$

Thus, own-feature curvature is always symmetric, while asymmetry can arise only in off-diagonal directional terms.

The economic interpretation is straightforward. The symmetric component captures reciprocal interaction in accumulation. If

$$(\mathbf{H}_S)_{ij} > 0, \quad (363)$$

then features x_i and x_j are locally complementary: each raises the marginal contribution of the other to capital accumulation. If

$$(\mathbf{H}_S)_{ij} < 0, \quad (364)$$

then the two features are locally substitutive: an increase in one lowers the marginal contribution of the other. If

$$(\mathbf{H}_S)_{ij} = 0, \quad (365)$$

then there is no local second-order coupling between the two features in the scalar accumulation function.

The asymmetric component has a different status. Since it does not affect $\mathcal{A}(\mathbf{x})$, it cannot be interpreted as curvature of the scalar accumulation surface. Instead, if retained in an extended model, it may represent directional adjustment, ordering effects, or non-conservative dynamics outside the scalar potential formulation. For example, an asymmetric interaction may matter in a vector-valued dynamic system:

$$\mathbf{x}_{t+1} = \mathbf{G}(\mathbf{x}_t), \quad (366)$$

where directional effects need not be reducible to a scalar quadratic potential. In the present scalar accumulation formula, however, such directional effects are not identified by $\mathbf{x}^\top \mathbf{H} \mathbf{x}$.

For empirical estimation, this observation reduces the number of distinct second-order coefficients. A fully unrestricted matrix $\mathbf{H}$ contains d^2 entries, but the symmetric interaction matrix contains only

$$\frac{d(d+1)}{2} \quad (367)$$

distinct coefficients. The quadratic specification can therefore be written as

$$\mathcal{A}(\mathbf{x}) = a_0 + \sum_{i=1}^d a_i x_i + \frac{1}{2} \sum_{i=1}^d h_{ii} x_i^2 + \sum_{1 \leq i < j \leq d} h_{ij}^S x_i x_j, \quad (368)$$

where

$$h_{ij}^S = \frac{1}{2}(h_{ij} + h_{ji}). \quad (369)$$

Substituting into the capital law of motion gives

$$K_{t+1} = K_t + a_0 + \mathbf{a}^\top \mathbf{x}_t + \frac{1}{2} \mathbf{x}_t^\top \mathbf{H}_S \mathbf{x}_t. \quad (370)$$

Equivalently,

$$\Delta K_t = a_0 + \mathbf{a}^\top \mathbf{x}_t + \frac{1}{2} \mathbf{x}_t^\top \mathbf{H}_S \mathbf{x}_t. \quad (371)$$

The symmetric-asymmetric decomposition therefore clarifies the identifiable content of nonlinear coupling. In a scalar second-order accumulation function, all economically relevant curvature and pairwise interaction are encoded in $\mathbf{H}_S$:

$$\mathbf{H}_S = \nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}). \quad (372)$$

The asymmetric component $\mathbf{H}_A$ is orthogonal to the quadratic scalar form and therefore drops out:

$$\mathbf{x}^\top \mathbf{H}_A \mathbf{x} = 0. \quad (373)$$

Consequently, the second-order capital accumulation formula may impose $\mathbf{H} = \mathbf{H}^\top$ without loss of generality whenever the object of interest is the scalar increment of capital.

4.4 Higher-Order Coupling

Second-order coupling captures pairwise interactions among accumulation features. However, many economic mechanisms are not reducible to isolated pairs. The effect of one feature may depend jointly on two or more other features. To allow for such nonlinear structures, the accumulation function can be extended to include higher-order terms:

$$\mathcal{A}(\mathbf{x}) = a_0 + \sum_i a_i x_i + \sum_{i,j} a_{ij} x_i x_j + \sum_{i,j,\ell} a_{ij\ell} x_i x_j x_\ell. \quad (374)$$

The coefficients a_i capture first-order effects, a_{ij} capture pairwise interactions, and $a_{ij\ell}$ capture third-order coupling among features.

More explicitly, the third-order specification can be written as

$$\mathcal{A}(\mathbf{x}) = a_0 + \sum_{i=1}^d a_i x_i + \sum_{i=1}^d \sum_{j=1}^d a_{ij} x_i x_j + \sum_{i=1}^d \sum_{j=1}^d \sum_{\ell=1}^d a_{ij\ell} x_i x_j x_\ell. \quad (375)$$

This representation generalizes the quadratic approximation by permitting the marginal effect of one feature to vary with combinations of other features. In the context of capital accumulation, such terms are useful when production, finance, technology, labor, demand, and ecological constraints jointly determine the feasibility and intensity of accumulation.

The first-order marginal effect of feature x_m is

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_m} = a_m + \sum_{j=1}^d a_{mj} x_j + \sum_{i=1}^d a_{im} x_i + \sum_{j=1}^d \sum_{\ell=1}^d a_{mj\ell} x_j x_\ell + \sum_{i=1}^d \sum_{\ell=1}^d a_{im\ell} x_i x_\ell + \sum_{i=1}^d \sum_{j=1}^d a_{ijm} x_i x_j. \quad (376)$$

If the coefficient arrays are symmetrized, this expression simplifies to

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_m} = a_m + 2 \sum_{j=1}^d a_{mj} x_j + 3 \sum_{j=1}^d \sum_{\ell=1}^d a_{mj\ell} x_j x_\ell. \quad (377)$$

Thus, the marginal contribution of x_m is no longer linear in the feature vector; it may itself be a nonlinear function of the accumulation state.

The second-order cross-partial derivative between two features x_m and x_n is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_m \partial x_n} = 2a_{mn} + 6 \sum_{\ell=1}^d a_{mnl} x_\ell, \quad (378)$$

under symmetric coefficient normalization. Equation (378) shows that complementarity and substitution are state dependent. Unlike the quadratic model, where the cross-partial is constant, higher-order coupling allows two features to be complementary in one region of the state space and substitutive in another.

This state dependence can be expressed as

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_m \partial x_n} > 0 \implies x_m \text{ and } x_n \text{ are locally complementary at } \mathbf{x}, \quad (379)$$

and

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_m \partial x_n} < 0 \implies x_m \text{ and } x_n \text{ are locally substitutive at } \mathbf{x}. \quad (380)$$

The qualification “at $\mathbf{x}$ ” is essential: the sign of the interaction may change with the surrounding economic environment.

Third-order terms capture conditional interactions. The coefficient $a_{ij\ell}$ measures how the pairwise relation between x_i and x_j depends on the level of x_ℓ . Formally,

$$\frac{\partial}{\partial x_\ell} \left(\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} \right) = 6a_{ij\ell} \quad (381)$$

under symmetric normalization. If $a_{ij\ell} > 0$, an increase in x_ℓ strengthens the complementarity between x_i and x_j or weakens their substitutability. If $a_{ij\ell} < 0$, an increase in x_ℓ weakens their complementarity or strengthens their substitutability.

Economically, higher-order coupling allows the model to represent mechanisms such as

$$\text{technology} \times \text{labor organization} \times \text{market demand}, \quad (382)$$

or

$$\text{finance} \times \text{leverage} \times \text{macroeconomic instability}. \quad (383)$$

In such cases, the effect of technology on accumulation may depend jointly on labor organization and demand conditions, while the effect of finance may depend on leverage and aggregate instability.

The capital law of motion becomes

$$K_{t+1} = K_t + a_0 + \sum_{i=1}^d a_i x_{it} + \sum_{i=1}^d \sum_{j=1}^d a_{ij} x_{it} x_{jt} + \sum_{i=1}^d \sum_{j=1}^d \sum_{\ell=1}^d a_{ij\ell} x_{it} x_{jt} x_{\ell t}. \quad (384)$$

Equivalently,

$$\Delta K_t = a_0 + \sum_{i=1}^d a_i x_{it} + \sum_{i=1}^d \sum_{j=1}^d a_{ij} x_{it} x_{jt} + \sum_{i=1}^d \sum_{j=1}^d \sum_{\ell=1}^d a_{ij\ell} x_{it} x_{jt} x_{\ell t}. \quad (385)$$

A stochastic empirical version is

$$\Delta K_t = a_0 + \sum_{i=1}^d a_i x_{it} + \sum_{i=1}^d \sum_{j=1}^d a_{ij} x_{it} x_{jt} + \sum_{i=1}^d \sum_{j=1}^d \sum_{\ell=1}^d a_{ij\ell} x_{it} x_{jt} x_{\ell t} + \varepsilon_t, \quad (386)$$

with

$$\mathbb{E}[\varepsilon_t \mid \mathbf{x}_t] = 0. \quad (387)$$

This model can be estimated by expanding the regressor set to include linear, quadratic, and cubic terms. The number of raw terms, however, grows rapidly with dimension:

$$d + d^2 + d^3. \quad (388)$$

Under full symmetry of the quadratic and cubic coefficient arrays, the number of distinct polynomial terms up to degree three is

$$1 + d + \binom{d+1}{2} + \binom{d+2}{3}. \quad (389)$$

This growth motivates regularization, sparsity restrictions, or structured feature selection when d is large.

A regularized estimator may be written as

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \left\{ \frac{1}{n} \sum_{t=1}^n (\Delta K_t - \mathcal{A}(\mathbf{x}_t; \boldsymbol{\theta}))^2 + \lambda \mathcal{P}(\boldsymbol{\theta}) \right\}, \quad (390)$$

where $\mathcal{P}(\boldsymbol{\theta})$ penalizes model complexity. Common choices include an ℓ_1 penalty for sparse interactions,

$$\mathcal{P}(\boldsymbol{\theta}) = \sum_i |a_i| + \sum_{i,j} |a_{ij}| + \sum_{i,j,\ell} |a_{ij\ell}|, \quad (391)$$

or a grouped penalty that selects entire interaction blocks.

Higher-order coupling therefore extends the nonlinear capital formula from pairwise curvature to conditional interaction. It allows the accumulation effect of a feature to depend on broader configurations of the economic state:

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_i} = \text{function of } (x_1, \dots, x_d). \quad (392)$$

This provides a flexible bridge between the interpretable quadratic model and the fully general nonparametric accumulation function. Its main advantage is economic richness; its main cost is dimensional complexity.

4.5 General Polynomial Approximation

The higher-order coupling formula can be written more systematically using multi-index notation. Let

$$\nu = (\nu_1, \dots, \nu_d) \in \mathbb{N}_0^d \quad (393)$$

be a multi-index, and define

$$|\nu| = \sum_{j=1}^d \nu_j, \quad \mathbf{x}^\nu = \prod_{j=1}^d x_j^{\nu_j}. \quad (394)$$

A general polynomial approximation of degree p is then

$$\mathcal{A}(\mathbf{x}) \approx \sum_{|\nu| \leq p} a_\nu \mathbf{x}^\nu, \quad \mathbf{x}^\nu = \prod_{j=1}^d x_j^{\nu_j}, \quad |\nu| = \sum_{j=1}^d \nu_j. \quad (395)$$

Here, each coefficient a_ν measures the contribution of the monomial $\mathbf{x}^\nu$ to the capital accumulation function.

The polynomial approximation nests the previous linear, quadratic, and cubic formulations. When $p = 1$, Equation (395) reduces to the linear multi-feature formula:

$$\mathcal{A}(\mathbf{x}) \approx a_0 + \sum_{j=1}^d a_{\mathbf{e}_j} x_j, \quad (396)$$

where $\mathbf{0} = (0, \dots, 0)$ and $\mathbf{e}_j$ is the j th coordinate vector. When $p = 2$, it becomes a quadratic coupling model:

$$\mathcal{A}(\mathbf{x}) \approx a_0 + \sum_{j=1}^d a_{\mathbf{e}_j} x_j + \sum_{i=1}^d \sum_{j=1}^d a_{\mathbf{e}_i + \mathbf{e}_j} x_i x_j. \quad (397)$$

When $p = 3$, it includes third-order conditional interactions:

$$\mathcal{A}(\mathbf{x}) \approx a_0 + \sum_j a_{\mathbf{e}_j} x_j + \sum_{i,j} a_{\mathbf{e}_i + \mathbf{e}_j} x_i x_j + \sum_{i,j,\ell} a_{\mathbf{e}_i + \mathbf{e}_j + \mathbf{e}_\ell} x_i x_j x_\ell. \quad (398)$$

The number of distinct monomials of degree at most p in d variables is

$$N(d, p) = \binom{d+p}{p}. \quad (399)$$

Thus, the polynomial approximation becomes increasingly flexible as p increases, but the number of coefficients grows rapidly with both dimension and degree. This creates a trade-off between expressive power and statistical discipline.

The first-order marginal effect of feature x_j is

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \approx \sum_{|\nu| \leq p} a_\nu \nu_j x_j^{\nu_j-1} \prod_{\ell \neq j} x_\ell^{\nu_\ell}. \quad (400)$$

Equivalently,

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \approx \sum_{\substack{|\nu| \leq p \\ \nu_j \geq 1}} a_\nu \nu_j \mathbf{x}^{\nu - \mathbf{e}_j}. \quad (401)$$

This expression shows that marginal effects are generally state dependent. The effect of x_j depends on the levels of all features appearing in the same monomials.

The second-order cross-partial derivative is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} \approx \sum_{\substack{|\nu| \leq p \\ \nu_i \geq 1, \nu_j \geq 1}} a_\nu \nu_i \nu_j \mathbf{x}^{\nu - \mathbf{e}_i - \mathbf{e}_j}, \quad i \neq j. \quad (402)$$

For the own second derivative,

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} \approx \sum_{\substack{|\nu| \leq p \\ \nu_j \geq 2}} a_\nu \nu_j (\nu_j - 1) \mathbf{x}^{\nu - 2\mathbf{e}_j}. \quad (403)$$

These formulas make clear that curvature, complementarity, and substitution can vary across the state space.

A local complementarity condition between features x_i and x_j is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} > 0, \quad (404)$$

whereas a local substitution condition is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} < 0. \quad (405)$$

Unlike the quadratic model, the sign of the cross-partial may change with $\mathbf{x}$. Thus, polynomial approximation permits regime-dependent interaction patterns without requiring an explicit threshold model.

The polynomial approximation can be embedded directly in the capital law of motion:

$$K_{t+1} = K_t + \sum_{|\nu| \leq p} a_\nu \mathbf{x}_t^\nu. \quad (406)$$

The corresponding capital increment is

$$\Delta K_t = \sum_{|\nu| \leq p} a_\nu \mathbf{x}_t^\nu. \quad (407)$$

With an error term, the empirical specification becomes

$$\Delta K_t = \sum_{|\nu| \leq p} a_\nu \mathbf{x}_t^\nu + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathbf{x}_t] = 0. \quad (408)$$

For estimation, define the polynomial feature map

$$\boldsymbol{\phi}_p(\mathbf{x}) = (\mathbf{x}^\nu)_{|\nu| \leq p} \in \mathbb{R}^{N(d,p)}. \quad (409)$$

Then the approximation can be written as a linear model in transformed features:

$$\mathcal{A}(\mathbf{x}) \approx \mathbf{a}^\top \boldsymbol{\phi}_p(\mathbf{x}), \quad (410)$$

where $\mathbf{a}$ stacks the coefficients a_ν . This representation is linear in parameters but nonlinear in the original economic features.

A least-squares estimator solves

$$\hat{\mathbf{a}} = \arg \min_{\mathbf{a}} \sum_{t=1}^T \left[\Delta K_t - \mathbf{a}^\top \boldsymbol{\phi}_p(\mathbf{x}_t) \right]^2. \quad (411)$$

When the number of polynomial terms is large relative to the sample size, a regularized estimator is preferable:

$$\hat{\mathbf{a}}_\lambda = \arg \min_{\mathbf{a}} \left\{ \frac{1}{T} \sum_{t=1}^T \left[\Delta K_t - \mathbf{a}^\top \boldsymbol{\phi}_p(\mathbf{x}_t) \right]^2 + \lambda \mathcal{P}(\mathbf{a}) \right\}. \quad (412)$$

For example, an ℓ_2 penalty yields ridge-type shrinkage:

$$\mathcal{P}(\mathbf{a}) = \|\mathbf{a}\|_2^2, \quad (413)$$

while an ℓ_1 penalty encourages sparsity:

$$\mathcal{P}(\mathbf{a}) = \|\mathbf{a}\|_1. \quad (414)$$

The approximation error is

$$R_p(\mathbf{x}) = \mathcal{A}(\mathbf{x}) - \sum_{|\nu| \leq p} a_\nu \mathbf{x}^\nu. \quad (415)$$

If $\mathcal{A}$ is sufficiently smooth and the feature space is compact, higher-degree polynomial approximations can reduce this error over the admissible domain:

$$\sup_{\mathbf{x} \in \mathcal{X}} |R_p(\mathbf{x})| \rightarrow 0 \quad \text{as} \quad p \rightarrow \infty, \quad (416)$$

under standard approximation conditions. In finite samples, however, the degree p must be chosen to balance approximation accuracy against variance and interpretability.

The general polynomial approximation therefore offers a structured path from simple linear accumulation to flexible nonlinear accumulation. It preserves a clear algebraic interpretation of interactions while allowing higher-order state dependence:

$$\mathcal{A}(\mathbf{x}) \approx \sum_{|\nu| \leq p} a_\nu \mathbf{x}^\nu. \quad (417)$$

In this form, the accumulation law can represent direct effects, pairwise coupling, conditional interactions, and more complex nonlinear responses within a single unified specification.

4.6 General Nonlinear Mapping

The polynomial approximation provides a structured algebraic representation of nonlinear accumulation. A still more general approach is to represent the accumulation function through a nonlinear composition of transformed features. Let $\sigma(\cdot)$ denote a nonlinear activation function, let $\mathbf{b}_m \in \mathbb{R}^d$ be a feature-loading vector, let $c_m \in \mathbb{R}$ be a shift parameter, let $w_m \in \mathbb{R}$ be an aggregation weight, and let $\Phi(\cdot)$ be an outer transformation. The general nonlinear mapping is

$$\mathcal{A}(\mathbf{x}) = \Phi\left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x} + c_m)\right). \quad (418)$$

This formulation treats capital accumulation as the result of multiple latent nonlinear channels. Each channel transforms the feature vector through a scalar index $\mathbf{b}_m^\top \mathbf{x} + c_m$, applies the nonlinear response σ , and then contributes to the aggregate accumulation signal with weight w_m .

Define the latent index and hidden response as

$$q_m(\mathbf{x}) \equiv \mathbf{b}_m^\top \mathbf{x} + c_m, \quad h_m(\mathbf{x}) \equiv \sigma(q_m(\mathbf{x})). \quad (419)$$

Then the inner aggregation is

$$S(\mathbf{x}) \equiv \sum_{m=1}^M w_m h_m(\mathbf{x}) = \sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x} + c_m). \quad (420)$$

The accumulation function can therefore be written compactly as

$$\mathcal{A}(\mathbf{x}) = \Phi(S(\mathbf{x})). \quad (421)$$

This specification nests several familiar cases. If the outer transformation is the identity map,

$$\Phi(u) = u, \quad (422)$$

then

$$\mathcal{A}(\mathbf{x}) = \sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x} + c_m). \quad (423)$$

This is a single-hidden-layer nonlinear representation. If, in addition, $\sigma(u) = u$, then the model collapses to an affine function:

$$\mathcal{A}(\mathbf{x}) = \sum_{m=1}^M w_m (\mathbf{b}_m^\top \mathbf{x} + c_m) = \boldsymbol{\beta}^\top \mathbf{x} + \alpha, \quad (424)$$

where

$$\boldsymbol{\beta} = \sum_{m=1}^M w_m \mathbf{b}_m, \quad \alpha = \sum_{m=1}^M w_m c_m. \quad (425)$$

Thus, the linear multi-feature formula appears as a special case.

The first-order marginal effect of feature x_j is obtained by the chain rule:

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} = \Phi'(S(\mathbf{x})) \sum_{m=1}^M w_m \sigma'(q_m(\mathbf{x})) b_{mj}, \quad (426)$$

where b_{mj} is the j th component of $\mathbf{b}_m$. In vector form,

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}) = \Phi'(S(\mathbf{x})) \sum_{m=1}^M w_m \sigma'(q_m(\mathbf{x})) \mathbf{b}_m. \quad (427)$$

The marginal effect is therefore state dependent. Unlike the linear model, the effect of x_j varies with the full feature configuration through the latent indices $q_m(\mathbf{x})$ and the aggregate signal $S(\mathbf{x})$.

The second-order derivative between features x_i and x_j is

$$\begin{aligned} \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} &= \Phi''(S(\mathbf{x})) \left[\sum_{m=1}^M w_m \sigma'(q_m(\mathbf{x})) b_{mi} \right] \left[\sum_{m=1}^M w_m \sigma'(q_m(\mathbf{x})) b_{mj} \right] \\ &+ \Phi'(S(\mathbf{x})) \sum_{m=1}^M w_m \sigma''(q_m(\mathbf{x})) b_{mi} b_{mj}. \end{aligned} \quad (428)$$

This expression shows that interaction arises from two sources. The first term comes from the curvature of the outer transformation Φ . The second term comes from the curvature of the inner activation σ . Therefore, even when the original feature vector enters only through linear indices, the resulting accumulation function may display rich nonlinear coupling.

The Hessian can be written more compactly as

$$\nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}) = \Phi''(S(\mathbf{x})) \nabla_{\mathbf{x}} S(\mathbf{x}) \nabla_{\mathbf{x}} S(\mathbf{x})^\top + \Phi'(S(\mathbf{x})) \nabla_{\mathbf{x}}^2 S(\mathbf{x}), \quad (429)$$

where

$$\nabla_{\mathbf{x}} S(\mathbf{x}) = \sum_{m=1}^M w_m \sigma'(q_m(\mathbf{x})) \mathbf{b}_m, \quad (430)$$

and

$$\nabla_{\mathbf{x}}^2 S(\mathbf{x}) = \sum_{m=1}^M w_m \sigma''(q_m(\mathbf{x})) \mathbf{b}_m \mathbf{b}_m^\top. \quad (431)$$

The Hessian is generally state dependent, which allows the model to represent changing complementarity, substitution, and curvature across different accumulation regimes.

The capital law of motion associated with the nonlinear mapping is

$$K_{t+1} = K_t + \Phi \left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x}_t + c_m) \right). \quad (432)$$

Equivalently, the capital increment is

$$\Delta K_t = \Phi \left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x}_t + c_m) \right). \quad (433)$$

This specification allows the accumulation increment to respond flexibly to the full configuration of economic features.

The sign of the nonlinear mapping determines the accumulation regime:

$$\Phi(S(\mathbf{x}_t)) \begin{cases} > 0, & \text{expanded accumulation,} \\ = 0, & \text{stationary reproduction,} \\ < 0, & \text{devaluation or contraction.} \end{cases} \quad (434)$$

The boundary between expansion and contraction is

$$\Phi(S(\mathbf{x})) = 0. \quad (435)$$

When Φ is monotone and has a unique zero at u_0 , this boundary can be written as

$$S(\mathbf{x}) = u_0. \quad (436)$$

Unlike the linear model, this boundary may be curved, disconnected, or highly state dependent.

The outer transformation Φ can be chosen to impose economically meaningful restrictions. If accumulation is required to be nonnegative, one may choose

$$\Phi(u) = \exp(u) \quad (437)$$

or

$$\Phi(u) = \log(1 + \exp(u)). \quad (438)$$

If both positive and negative accumulation are admissible, the identity map is appropriate:

$$\Phi(u) = u. \quad (439)$$

If bounded capital increments are required, one may use a bounded outer transformation:

$$\Phi(u) = \bar{A} \tanh(u), \quad |\mathcal{A}(\mathbf{x})| \leq \bar{A}. \quad (440)$$

The activation function σ determines the local nonlinear response of each latent channel. For example, a smooth sigmoidal response can represent saturation:

$$\sigma(u) = \frac{1}{1 + \exp(-u)}. \quad (441)$$

A rectified response can represent threshold-like activation:

$$\sigma(u) = \max\{0, u\}. \quad (442)$$

A hyperbolic tangent response can represent bounded positive and negative effects:

$$\sigma(u) = \tanh(u). \quad (443)$$

The choice of σ therefore determines whether the nonlinear mechanism is smooth, threshold-like, saturating, or sign-changing.

For empirical implementation, let the parameter vector be

$$\boldsymbol{\theta} = \left(w_1, \dots, w_M, \mathbf{b}_1^\top, \dots, \mathbf{b}_M^\top, c_1, \dots, c_M \right)^\top. \quad (444)$$

A stochastic specification is

$$\Delta K_t = \Phi \left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x}_t + c_m) \right) + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathbf{x}_t] = 0. \quad (445)$$

The corresponding least-squares estimator solves

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{t=1}^T \left[\Delta K_t - \Phi \left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x}_t + c_m) \right) \right]^2. \quad (446)$$

Regularization may be added to control the complexity of the nonlinear mapping:

$$\hat{\boldsymbol{\theta}}_\lambda = \arg \min_{\boldsymbol{\theta}} \left\{ \frac{1}{T} \sum_{t=1}^T [\Delta K_t - \mathcal{A}(\mathbf{x}_t; \boldsymbol{\theta})]^2 + \lambda \mathcal{P}(\boldsymbol{\theta}) \right\}. \quad (447)$$

The number of hidden channels M governs the expressive capacity of the model. When M is small, the specification is a low-dimensional nonlinear index model. When M is large, the mapping can approximate a broad class of smooth accumulation functions on compact domains:

$$\mathcal{A}(\mathbf{x}) \approx \Phi \left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x} + c_m) \right), \quad M \text{ large}. \quad (448)$$

Thus, the nonlinear mapping provides a flexible bridge between structured economic theory and general function approximation.

The main advantage of this formulation is that it can represent nonlinear thresholds, saturation, heterogeneous marginal effects, and high-dimensional feature interactions without explicitly enumerating all polynomial terms. Its main cost is interpretability: the parameters $(w_m, \mathbf{b}_m, c_m)$ are less directly interpretable than the coefficients of a linear or quadratic model. For this reason, the nonlinear mapping is best viewed as the most flexible finite-dimensional specification:

$$\mathcal{A}(\mathbf{x}) = \Phi \left(\sum_{m=1}^M w_m \sigma(\mathbf{b}_m^\top \mathbf{x} + c_m) \right), \quad (449)$$

while the linear, polynomial, and interaction-matrix formulations serve as more interpretable special cases.

5 Monotonicity and Marginal Effects

5.1 First-Order Marginal Effects

The accumulation function $\mathcal{A}(\mathbf{x})$ maps the vector of economic features into the net increment of capital. To interpret how each feature contributes to accumulation, we define the first-order marginal effect of feature x_j as

$$ME_j(\mathbf{x}) \equiv \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j}, \quad j = 1, \dots, d. \quad (450)$$

The marginal effect $ME_j(\mathbf{x})$ measures the local change in the capital accumulation increment induced by a small change in x_j , holding the other features fixed.

Formally, for a small perturbation Δx_j , the local change in the accumulation function is approximated by

$$\mathcal{A}(x_1, \dots, x_j + \Delta x_j, \dots, x_d) - \mathcal{A}(x_1, \dots, x_j, \dots, x_d) \approx ME_j(\mathbf{x}) \Delta x_j. \quad (451)$$

Thus, $ME_j(\mathbf{x})$ gives the local slope of the accumulation surface along the j th feature dimension.

The sign of the marginal effect determines the local direction of influence:

$$ME_j(\mathbf{x}) \begin{cases} > 0, & \text{feature } x_j \text{ locally increases accumulation,} \\ = 0, & \text{feature } x_j \text{ is locally neutral,} \\ < 0, & \text{feature } x_j \text{ locally decreases accumulation.} \end{cases} \quad (452)$$

This sign interpretation is local because the marginal effect may vary across the state space. A feature may be accumulation-enhancing in one region and accumulation-reducing in another.

For the linear multi-feature formula,

$$\mathcal{A}(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}, \quad (453)$$

the marginal effect is constant:

$$ME_j(\mathbf{x}) = \beta_j. \quad (454)$$

The linear model therefore imposes that the effect of each feature is independent of the level of all other features.

For the quadratic coupling formula,

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad (455)$$

with symmetric $\mathbf{H}$, the marginal effect is

$$ME_j(\mathbf{x}) = a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell. \quad (456)$$

Hence, the effect of x_j depends on the entire feature vector. This is the basic mechanism through which nonlinear coupling generates state-dependent marginal effects.

For a Cobb-Douglas-type accumulation function,

$$\mathcal{A}(\mathbf{x}) = A_0 \prod_{j=1}^d x_j^{\beta_j}, \quad x_j > 0, \quad (457)$$

the marginal effect is

$$ME_j(\mathbf{x}) = \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} = \beta_j \frac{\mathcal{A}(\mathbf{x})}{x_j}. \quad (458)$$

Although β_j is constant as an elasticity, the level marginal effect varies with both the accumulation level and the scale of x_j .

In dynamic notation, if the capital law of motion is

$$K_{t+1} = K_t + \mathcal{A}(\mathbf{x}_t), \quad (459)$$

then the effect of feature x_{jt} on next-period capital is

$$\frac{\partial K_{t+1}}{\partial x_{jt}} = \frac{\partial \mathcal{A}(\mathbf{x}_t)}{\partial x_{jt}} = ME_j(\mathbf{x}_t). \quad (460)$$

Thus, marginal effects can be interpreted either as effects on the capital increment or as effects on next-period capital, holding current capital fixed.

The marginal effect is especially useful for distinguishing direct effects from total accumulation outcomes. A feature may have a positive marginal effect but still be associated with low accumulation if other features are unfavorable. Conversely, a negative marginal effect may occur in a state where total accumulation remains positive. Therefore, the sign of $\mathcal{A}(\mathbf{x})$ and the sign of $ME_j(\mathbf{x})$ describe different objects:

$$\mathcal{A}(\mathbf{x}) \quad \text{determines the accumulation regime,} \quad ME_j(\mathbf{x}) \quad \text{determines the local contribution of feature } j. \quad (461)$$

For empirical analysis, one may evaluate marginal effects at representative states. The marginal effect at the mean feature vector is

$$ME_j(\bar{\mathbf{x}}) = \left. \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \right|_{\mathbf{x}=\bar{\mathbf{x}}}, \quad (462)$$

where

$$\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t. \quad (463)$$

The average marginal effect over the observed sample is

$$AME_j = \frac{1}{T} \sum_{t=1}^T ME_j(\mathbf{x}_t). \quad (464)$$

In panel data, the corresponding average marginal effect is

$$AME_j = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T ME_j(\mathbf{x}_{it}). \quad (465)$$

The first-order marginal effect therefore provides the basic local interpretation of the accumulation function. It tells whether a feature raises, lowers, or leaves unchanged the capital increment at a given economic state:

$$ME_j(\mathbf{x}) = \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j}. \quad (466)$$

This object is the foundation for monotonicity restrictions, elasticity analysis, nonlinear interaction tests, and policy counterfactuals.

5.2 Gradient Representation

The collection of first-order marginal effects can be written compactly as the gradient of the accumulation function. The gradient vector is

$$\nabla \mathcal{A}(\mathbf{x}) = \begin{bmatrix} \partial \mathcal{A} / \partial x_1 \\ \vdots \\ \partial \mathcal{A} / \partial x_d \end{bmatrix}. \quad (467)$$

Equivalently,

$$\nabla \mathcal{A}(\mathbf{x}) = (ME_1(\mathbf{x}), \dots, ME_d(\mathbf{x}))^\top. \quad (468)$$

The gradient gives the direction in feature space along which the accumulation function increases most rapidly.

For a small vector perturbation $\Delta \mathbf{x}$, the first-order Taylor approximation gives

$$\mathcal{A}(\mathbf{x} + \Delta \mathbf{x}) - \mathcal{A}(\mathbf{x}) \approx \nabla \mathcal{A}(\mathbf{x})^\top \Delta \mathbf{x}. \quad (469)$$

Thus, the total local change in accumulation is the inner product between the gradient and the feature perturbation.

If the perturbation is in direction $\mathbf{u}$ with $\|\mathbf{u}\| = 1$, the directional derivative is

$$D_{\mathbf{u}}\mathcal{A}(\mathbf{x}) = \nabla\mathcal{A}(\mathbf{x})^\top \mathbf{u}. \quad (470)$$

The direction of steepest local increase is

$$\mathbf{u}^* = \frac{\nabla\mathcal{A}(\mathbf{x})}{\|\nabla\mathcal{A}(\mathbf{x})\|} \quad \text{when} \quad \nabla\mathcal{A}(\mathbf{x}) \neq \mathbf{0}. \quad (471)$$

The corresponding maximal directional derivative is

$$\max_{\|\mathbf{u}\|=1} D_{\mathbf{u}}\mathcal{A}(\mathbf{x}) = \|\nabla\mathcal{A}(\mathbf{x})\|. \quad (472)$$

In the linear model,

$$\mathcal{A}(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}, \quad (473)$$

the gradient is constant:

$$\nabla\mathcal{A}(\mathbf{x}) = \boldsymbol{\beta}. \quad (474)$$

This means that the direction of strongest accumulation is the same at every point in the feature space.

In the quadratic model,

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad \mathbf{H} = \mathbf{H}^\top, \quad (475)$$

the gradient is

$$\nabla\mathcal{A}(\mathbf{x}) = \mathbf{a} + \mathbf{H}\mathbf{x}. \quad (476)$$

The direction and magnitude of marginal accumulation therefore vary with the state vector. This state dependence is central to nonlinear accumulation analysis.

A stationary point of the accumulation surface satisfies

$$\nabla\mathcal{A}(\mathbf{x}^*) = \mathbf{0}. \quad (477)$$

At such a point, no infinitesimal first-order feature perturbation changes the accumulation function. Whether the point is a local maximum, local minimum, or saddle point depends on the Hessian matrix, which is analyzed in the next section.

The gradient representation is also useful for constrained accumulation problems. Suppose the feasible feature set is defined by

$$\mathcal{F} = \{\mathbf{x} \in \mathbb{R}^d : g_m(\mathbf{x}) \leq 0, \ m = 1, \dots, M\}. \quad (478)$$

At an interior optimum, the necessary first-order condition is

$$\nabla\mathcal{A}(\mathbf{x}^*) = \mathbf{0}. \quad (479)$$

At a constrained optimum, the gradient of accumulation is balanced against the gradients of the binding constraints:

$$\nabla\mathcal{A}(\mathbf{x}^*) = \sum_{m=1}^M \lambda_m \nabla g_m(\mathbf{x}^*), \quad \lambda_m \geq 0. \quad (480)$$

Thus, the gradient links marginal accumulation to resource, ecological, financial, or institutional constraints.

The gradient also provides a basis for empirical comparison across units or periods. For unit i at time t , define

$$\nabla \mathcal{A}(\mathbf{x}_{it}) = (ME_1(\mathbf{x}_{it}), \dots, ME_d(\mathbf{x}_{it}))^\top. \quad (481)$$

Variation in this gradient across i and t captures heterogeneity in the local structure of accumulation:

$$\nabla \mathcal{A}(\mathbf{x}_{it}) \neq \nabla \mathcal{A}(\mathbf{x}_{i't'}) \implies \text{heterogeneous marginal accumulation effects.} \quad (482)$$

Finally, the norm of the gradient measures local sensitivity:

$$S(\mathbf{x}) \equiv \|\nabla \mathcal{A}(\mathbf{x})\|. \quad (483)$$

A large value of $S(\mathbf{x})$ indicates that accumulation is highly sensitive to small changes in the feature vector. A small value indicates local insensitivity or saturation:

$$S(\mathbf{x}) \approx 0 \implies \mathcal{A}(\mathbf{x}) \text{ is locally flat.} \quad (484)$$

The gradient representation therefore consolidates all first-order marginal effects into a single geometric object:

$$\nabla \mathcal{A}(\mathbf{x}) = \left(\frac{\partial \mathcal{A}}{\partial x_1}, \dots, \frac{\partial \mathcal{A}}{\partial x_d} \right)^\top. \quad (485)$$

It is the central tool for analyzing monotonicity, local sensitivity, directional change, and constrained optimal allocation in the multi-feature capital accumulation framework.

5.3 Elasticities

Marginal effects measure changes in the accumulation increment in level units. When features are measured in different units, however, it is often more informative to use elasticities. For any feature $x_j > 0$ and any state with $\mathcal{A}(\mathbf{x}) \neq 0$, define

$$\varepsilon_j(\mathbf{x}) \equiv \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \frac{x_j}{\mathcal{A}(\mathbf{x})}, \quad \boldsymbol{\varepsilon}(\mathbf{x}) = \frac{\text{diag}(\mathbf{x}) \nabla \mathcal{A}(\mathbf{x})}{\mathcal{A}(\mathbf{x})}. \quad (486)$$

The elasticity $\varepsilon_j(\mathbf{x})$ measures the percentage change in the capital accumulation increment associated with a one-percent change in feature x_j , holding all other features fixed.

Equivalently, the elasticity can be written as a log-derivative:

$$\varepsilon_j(\mathbf{x}) = \frac{\partial \log |\mathcal{A}(\mathbf{x})|}{\partial \log x_j}, \quad \mathcal{A}(\mathbf{x}) \neq 0, \quad x_j > 0. \quad (487)$$

Thus, elasticities provide a scale-free measure of sensitivity. This is particularly useful when the feature vector contains variables with different economic units, such as wages, leverage, output, energy use, institutional indices, or technological intensity.

For the linear accumulation function,

$$\mathcal{A}(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}, \quad (488)$$

the elasticity is

$$\varepsilon_j(\mathbf{x}) = \beta_j \frac{x_j}{\alpha + \boldsymbol{\beta}^\top \mathbf{x}}. \quad (489)$$

Although the marginal effect β_j is constant, the elasticity varies with the level of x_j and with the total accumulation increment.

For the Cobb–Douglas-type formula,

$$\mathcal{A}(\mathbf{x}) = A_0 \prod_{j=1}^d x_j^{\beta_j}, \quad (490)$$

the elasticity is constant:

$$\varepsilon_j(\mathbf{x}) = \beta_j. \quad (491)$$

This constant-elasticity property is one of the main reasons why Cobb–Douglas specifications are analytically convenient.

For the quadratic coupling formula,

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad \mathbf{H} = \mathbf{H}^\top, \quad (492)$$

the elasticity becomes

$$\varepsilon_j(\mathbf{x}) = \left(a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell \right) \frac{x_j}{a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}}. \quad (493)$$

Hence, both the numerator and denominator vary with the state vector. The elasticity is therefore state dependent and reflects nonlinear interaction among features.

The vector of elasticities can also be used to decompose local proportional changes. For a small proportional feature perturbation $d \log \mathbf{x}$, the proportional change in the accumulation increment is approximated by

$$d \log |\mathcal{A}(\mathbf{x})| \approx \boldsymbol{\varepsilon}(\mathbf{x})^\top d \log \mathbf{x}. \quad (494)$$

This expression shows that $\boldsymbol{\varepsilon}(\mathbf{x})$ is the proportional counterpart of the gradient $\nabla \mathcal{A}(\mathbf{x})$.

The aggregate elasticity of accumulation with respect to a common proportional scaling of all features is

$$E(\mathbf{x}) \equiv \sum_{j=1}^d \varepsilon_j(\mathbf{x}). \quad (495)$$

If all features are scaled by a common factor $\lambda > 0$, then $E(\mathbf{x})$ measures the local percentage response of accumulation to this joint scaling. In homogeneous models, this object corresponds to the degree of returns to scale.

Elasticities are undefined when $\mathcal{A}(\mathbf{x}) = 0$. This is not merely a technical issue. At the stationary reproduction boundary, proportional changes relative to the accumulation increment are not well defined:

$$\mathcal{A}(\mathbf{x}) = 0 \implies \varepsilon_j(\mathbf{x}) \text{ is undefined.} \quad (496)$$

Near this boundary, level marginal effects may be more stable and economically interpretable than elasticities.

5.4 Expected Sign Restrictions

Economic theory often implies that some features should have a predictable directional effect on capital accumulation. Let $\mathcal{J}_+$ denote the set of features expected to raise accumulation, and let $\mathcal{J}_-$ denote the set of features expected to reduce accumulation. The sign restrictions are

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \geq 0 \quad (j \in \mathcal{J}_+), \quad \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \leq 0 \quad (j \in \mathcal{J}_-). \quad (497)$$

These inequalities impose monotonicity over the relevant economic domain.

The positive-restriction set may include features such as productive capacity, technological efficiency, effective demand, infrastructure, or market access:

$$j \in \mathcal{J}_+ \implies ME_j(\mathbf{x}) \geq 0 \quad \text{for all } \mathbf{x} \in \mathcal{X}. \quad (498)$$

The negative-restriction set may include features such as ecological pressure, financial fragility, debt burden, resource scarcity, or crisis intensity:

$$j \in \mathcal{J}_- \implies ME_j(\mathbf{x}) \leq 0 \quad \text{for all } \mathbf{x} \in \mathcal{X}. \quad (499)$$

In the linear model,

$$\mathcal{A}(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}, \quad (500)$$

the sign restrictions reduce to coefficient restrictions:

$$\beta_j \geq 0 \quad (j \in \mathcal{J}_+), \quad \beta_j \leq 0 \quad (j \in \mathcal{J}_-). \quad (501)$$

Thus, monotonicity is global and easy to verify.

In the quadratic model,

$$ME_j(\mathbf{x}) = a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell. \quad (502)$$

The sign restrictions become

$$a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell \geq 0 \quad \forall \mathbf{x} \in \mathcal{X}, \quad j \in \mathcal{J}_+, \quad (503)$$

and

$$a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell \leq 0 \quad \forall \mathbf{x} \in \mathcal{X}, \quad j \in \mathcal{J}_-. \quad (504)$$

These are stronger restrictions than simple coefficient signs because the marginal effect depends on the full state vector.

For a general smooth accumulation function, the monotonicity region for feature j is

$$\mathcal{M}_j^+ = \left\{ \mathbf{x} \in \mathcal{X} : \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \geq 0 \right\}, \quad (505)$$

and the negative monotonicity region is

$$\mathcal{M}_j^- = \left\{ \mathbf{x} \in \mathcal{X} : \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \leq 0 \right\}. \quad (506)$$

Global monotonicity requires

$$\mathcal{M}_j^+ = \mathcal{X} \quad \text{for } j \in \mathcal{J}_+, \quad \mathcal{M}_j^- = \mathcal{X} \quad \text{for } j \in \mathcal{J}_-. \quad (507)$$

These restrictions are useful for both theoretical discipline and empirical regularization. A constrained estimator can impose them directly:

$$\hat{\mathcal{A}} = \arg \min_{\mathcal{A} \in \mathcal{G}} \left\{ \frac{1}{n} \sum_{t=1}^n [\Delta K_t - \mathcal{A}(\mathbf{x}_t)]^2 \right\} \quad (508)$$

subject to

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \geq 0 \quad (j \in \mathcal{J}_+), \quad \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \leq 0 \quad (j \in \mathcal{J}_-), \quad \forall \mathbf{x} \in \mathcal{X}. \quad (509)$$

Such restrictions can improve interpretability and reduce implausible extrapolation outside dense regions of the data.

The sign restrictions should not be imposed mechanically. Some features may have theoretically ambiguous effects. For example, financial development may support accumulation at moderate levels but generate fragility at high levels. Technological change may raise productivity while displacing labor or intensifying overaccumulation. In such cases, global monotonicity is too restrictive, and a non-monotone specification is more appropriate.

5.5 Non-Monotone Effects

Many determinants of capital accumulation need not have a constant direction of influence. A feature may raise accumulation up to a point and reduce it afterward, or it may be harmful at low levels but beneficial after crossing a threshold. Such cases are captured by non-monotone marginal effects:

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} = 0 \iff x_j = x_j^*(\mathbf{x}_{-j}), \quad \left. \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} \right|_{x_j=x_j^*} \neq 0. \quad (510)$$

Here, $\mathbf{x}_{-j}$ denotes the vector of all features except x_j , and $x_j^*(\mathbf{x}_{-j})$ is the critical value of x_j conditional on the other features.

The critical value satisfies the first-order condition

$$ME_j(x_j^*, \mathbf{x}_{-j}) = \frac{\partial \mathcal{A}(x_j^*, \mathbf{x}_{-j})}{\partial x_j} = 0. \quad (511)$$

The second-order condition determines whether this critical value is a local maximum or a local minimum in the x_j direction. If

$$\left. \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} \right|_{x_j=x_j^*} < 0, \quad (512)$$

then $x_j^*(\mathbf{x}_{-j})$ is a local accumulation-maximizing level of feature j . If

$$\left. \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} \right|_{x_j=x_j^*} > 0, \quad (513)$$

then $x_j^*(\mathbf{x}_{-j})$ is a local minimum or turning point from negative to positive marginal effects.

A simple quadratic example is

$$\mathcal{A}(x_j, \mathbf{x}_{-j}) = \gamma_0 + \gamma_1 x_j + \gamma_2 x_j^2 + \boldsymbol{\delta}^\top \mathbf{x}_{-j}, \quad (514)$$

where

$$\frac{\partial \mathcal{A}}{\partial x_j} = \gamma_1 + 2\gamma_2 x_j. \quad (515)$$

The critical point is

$$x_j^* = -\frac{\gamma_1}{2\gamma_2}, \quad \gamma_2 \neq 0. \quad (516)$$

If $\gamma_2 < 0$, the relationship is inverted-U shaped; if $\gamma_2 > 0$, it is U shaped:

$$\gamma_2 < 0 \implies \text{inverted-U effect}, \quad \gamma_2 > 0 \implies \text{U-shaped effect}. \quad (517)$$

With interactions, the turning point depends on other features. Consider

$$\mathcal{A}(\mathbf{x}) = a_0 + a_j x_j + \frac{1}{2} h_{jj} x_j^2 + \sum_{\ell \neq j} h_{j\ell} x_j x_\ell + R(\mathbf{x}_{-j}). \quad (518)$$

Then

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} = a_j + h_{jj} x_j + \sum_{\ell \neq j} h_{j\ell} x_\ell. \quad (519)$$

If $h_{jj} \neq 0$, the conditional turning point is

$$x_j^*(\mathbf{x}_{-j}) = -\frac{a_j + \sum_{\ell \neq j} h_{j\ell} x_\ell}{h_{jj}}. \quad (520)$$

This expression shows that the critical level of one feature shifts with the levels of other features.

The non-monotone case is important for capital accumulation because many economic variables have bounded or regime-dependent effects. For example, leverage may initially increase accumulation by relaxing financing constraints, but excessive leverage may raise fragility and reduce future accumulation. Similarly, technological intensity may increase productivity, but under some conditions may generate displacement, overcapacity, or falling profitability. These mechanisms can be summarized as

$$ME_j(\mathbf{x}) > 0 \quad \text{for } x_j < x_j^*(\mathbf{x}_{-j}), \quad ME_j(\mathbf{x}) < 0 \quad \text{for } x_j > x_j^*(\mathbf{x}_{-j}), \quad (521)$$

in the inverted-U case.

More generally, the sign of the marginal effect partitions the feature space into regions:

$$\mathcal{X}_j^+ = \{\mathbf{x} \in \mathcal{X} : ME_j(\mathbf{x}) > 0\}, \quad \mathcal{X}_j^- = \{\mathbf{x} \in \mathcal{X} : ME_j(\mathbf{x}) < 0\}, \quad (522)$$

with the turning boundary

$$\mathcal{B}_j = \{\mathbf{x} \in \mathcal{X} : ME_j(\mathbf{x}) = 0\}. \quad (523)$$

A globally monotone model has either $\mathcal{X}_j^+ = \mathcal{X}$ or $\mathcal{X}_j^- = \mathcal{X}$. A non-monotone model allows both regions to be nonempty:

$$\mathcal{X}_j^+ \neq \emptyset, \quad \mathcal{X}_j^- \neq \emptyset. \quad (524)$$

For empirical analysis, non-monotonicity can be tested by examining whether the estimated marginal effect changes sign over the observed state space:

$$\exists \mathbf{x}, \mathbf{x}' \in \mathcal{X} \quad \text{such that} \quad \widehat{ME}_j(\mathbf{x}) > 0 \quad \text{and} \quad \widehat{ME}_j(\mathbf{x}') < 0. \quad (525)$$

The estimated turning point can then be obtained from

$$\widehat{x}_j^*(\mathbf{x}_{-j}) : \quad \widehat{ME}_j(\widehat{x}_j^*, \mathbf{x}_{-j}) = 0. \quad (526)$$

Non-monotone effects therefore generalize simple sign restrictions by allowing the direction of accumulation effects to depend on the economic state. This is essential for a theory of capital accumulation in which features may be productive within one range, neutral at a critical point, and destabilizing beyond that point:

$$ME_j(\mathbf{x}) = \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \quad \text{may change sign over } \mathcal{X}. \quad (527)$$

6 Second-Order Properties

6.1 Hessian Matrix

First-order marginal effects describe the local contribution of each feature to capital accumulation. Second-order properties describe how these marginal effects themselves change across the feature space. The central object is the Hessian matrix of the accumulation function:

$$\nabla^2 \mathcal{A}(\mathbf{x}) = \left[\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} \right]_{i,j=1}^d. \quad (528)$$

The Hessian collects all own-curvature and cross-feature interaction effects. Its diagonal elements measure how the marginal effect of a feature changes with its own level, while its off-diagonal elements measure how the marginal effect of one feature changes with the level of another.

The diagonal entries are

$$[\nabla^2 \mathcal{A}(\mathbf{x})]_{jj} = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} = \frac{\partial ME_j(\mathbf{x})}{\partial x_j}. \quad (529)$$

These terms determine whether feature x_j exhibits diminishing, constant, or increasing marginal returns in the accumulation process.

The off-diagonal entries are

$$[\nabla^2 \mathcal{A}(\mathbf{x})]_{ij} = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial ME_i(\mathbf{x})}{\partial x_j}, \quad i \neq j. \quad (530)$$

They measure cross-feature effects. If the off-diagonal entry is positive, an increase in x_j raises the marginal contribution of x_i to accumulation. If it is negative, an increase in x_j lowers the marginal contribution of x_i .

When $\mathcal{A} \in C^2(\mathcal{X})$, the Hessian is symmetric by equality of mixed partial derivatives:

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j \partial x_i}. \quad (531)$$

Therefore,

$$\nabla^2 \mathcal{A}(\mathbf{x}) = [\nabla^2 \mathcal{A}(\mathbf{x})]^\top. \quad (532)$$

The Hessian governs the second-order local approximation of the accumulation function. Around a reference point $\mathbf{x}_0$, Taylor's theorem gives

$$\mathcal{A}(\mathbf{x}) = \mathcal{A}(\mathbf{x}_0) + \nabla \mathcal{A}(\mathbf{x}_0)^\top (\mathbf{x} - \mathbf{x}_0) + \frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^\top \nabla^2 \mathcal{A}(\mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0) + R_3(\mathbf{x}, \mathbf{x}_0). \quad (533)$$

Thus, the Hessian determines the local curvature of the accumulation surface around $\mathbf{x}_0$.

For the quadratic coupling model,

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad \mathbf{H} = \mathbf{H}^\top, \quad (534)$$

the Hessian is constant:

$$\nabla^2 \mathcal{A}(\mathbf{x}) = \mathbf{H}. \quad (535)$$

For a general nonlinear model, by contrast, the Hessian may vary with $\mathbf{x}$:

$$\nabla^2 \mathcal{A}(\mathbf{x}) \neq \nabla^2 \mathcal{A}(\mathbf{x}') \quad \text{for } \mathbf{x} \neq \mathbf{x}'. \quad (536)$$

This state dependence is important because curvature, complementarity, and substitution may change across different accumulation regimes.

6.2 Concavity and Convexity

The Hessian matrix provides a direct criterion for concavity and convexity. If the Hessian is negative semidefinite over the relevant domain, the accumulation function is concave. If it is positive semidefinite, the accumulation function is convex:

$$\nabla^2 \mathcal{A}(\mathbf{x}) \preceq 0 \Rightarrow \mathcal{A} \text{ concave}, \quad \nabla^2 \mathcal{A}(\mathbf{x}) \succeq 0 \Rightarrow \mathcal{A} \text{ convex}. \quad (537)$$

These conditions are understood to hold for all $\mathbf{x} \in \mathcal{X}$ when global concavity or convexity is claimed. Negative semidefiniteness means that

$$\mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}) \mathbf{u} \leq 0 \quad \forall \mathbf{u} \in \mathbb{R}^d, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (538)$$

In this case, the accumulation function exhibits diminishing joint returns: moving in any direction of the feature space weakly reduces the marginal increase in accumulation.

Positive semidefiniteness means that

$$\mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}) \mathbf{u} \geq 0 \quad \forall \mathbf{u} \in \mathbb{R}^d, \quad \forall \mathbf{x} \in \mathcal{X}. \quad (539)$$

In this case, the accumulation function exhibits increasing joint returns: moving in any direction weakly raises the marginal increase in accumulation.

Strict concavity is obtained when

$$\mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}) \mathbf{u} < 0 \quad \forall \mathbf{u} \neq \mathbf{0}. \quad (540)$$

Strict convexity is obtained when

$$\mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}) \mathbf{u} > 0 \quad \forall \mathbf{u} \neq \mathbf{0}. \quad (541)$$

Concavity has an important economic interpretation. If $\mathcal{A}$ is concave, then the accumulation increment generated by a convex combination of two states is at least as large as the corresponding convex combination of their accumulation increments:

$$\mathcal{A}(\lambda \mathbf{x} + (1 - \lambda) \mathbf{x}') \geq \lambda \mathcal{A}(\mathbf{x}) + (1 - \lambda) \mathcal{A}(\mathbf{x}'), \quad \lambda \in [0, 1]. \quad (542)$$

Convexity reverses the inequality:

$$\mathcal{A}(\lambda \mathbf{x} + (1 - \lambda) \mathbf{x}') \leq \lambda \mathcal{A}(\mathbf{x}) + (1 - \lambda) \mathcal{A}(\mathbf{x}'). \quad (543)$$

If the Hessian is indefinite, the accumulation surface contains both concave and convex directions:

$$\exists \mathbf{u}, \mathbf{v} \in \mathbb{R}^d \text{ such that } \mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}) \mathbf{u} > 0, \quad \mathbf{v}^\top \nabla^2 \mathcal{A}(\mathbf{x}) \mathbf{v} < 0. \quad (544)$$

This case is economically important because accumulation may be reinforced along some feature directions while being exhausted or destabilized along others.

6.3 Diminishing Marginal Returns

A feature exhibits diminishing marginal returns when its marginal contribution to accumulation decreases as the feature itself increases. For feature x_j , this condition is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} < 0, \quad \frac{\partial ME_j(\mathbf{x})}{\partial x_j} < 0. \quad (545)$$

The two expressions are equivalent because $ME_j(\mathbf{x}) = \partial \mathcal{A}(\mathbf{x}) / \partial x_j$.

Under diminishing marginal returns, a positive increase in x_j still may raise accumulation, but the size of the marginal gain becomes smaller as x_j grows:

$$x'_j > x_j \implies ME_j(x'_j, \mathbf{x}_{-j}) < ME_j(x_j, \mathbf{x}_{-j}). \quad (546)$$

This property is local to the feature dimension j and conditional on the remaining features $\mathbf{x}_{-j}$.

For a quadratic specification,

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad (547)$$

diminishing marginal returns in feature x_j require

$$h_{jj} < 0. \quad (548)$$

For a Cobb–Douglas-type function,

$$\mathcal{A}(\mathbf{x}) = A_0 \prod_{j=1}^d x_j^{\beta_j}, \quad x_j > 0, \quad (549)$$

the own second derivative is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} = \beta_j(\beta_j - 1) \frac{\mathcal{A}(\mathbf{x})}{x_j^2}. \quad (550)$$

When $\mathcal{A}(\mathbf{x}) > 0$, diminishing marginal returns require

$$0 < \beta_j < 1. \quad (551)$$

Economically, diminishing marginal returns may arise from saturation, capacity constraints, congestion, limited demand, resource scarcity, or institutional limits. It captures the idea that increasing one feature without proportionally adjusting the rest of the accumulation structure may eventually yield smaller gains.

6.4 Increasing Marginal Returns

A feature exhibits increasing marginal returns when its marginal contribution to accumulation rises as the feature itself increases. The formal condition is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} > 0, \quad \frac{\partial ME_j(\mathbf{x})}{\partial x_j} > 0. \quad (552)$$

In this case, the accumulation effect of an additional unit of x_j becomes larger at higher levels of x_j .

Equivalently,

$$x'_j > x_j \implies ME_j(x'_j, \mathbf{x}_{-j}) > ME_j(x_j, \mathbf{x}_{-j}). \quad (553)$$

Increasing marginal returns can generate self-reinforcing accumulation, especially when the feature represents technology, network scale, market reach, learning capacity, or financial leverage under favorable conditions.

For the quadratic model, increasing marginal returns in feature x_j require

$$h_{jj} > 0. \quad (554)$$

For the Cobb–Douglas-type model with positive accumulation,

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2} > 0 \iff \beta_j > 1 \text{ or } \beta_j < 0. \quad (555)$$

In most productive-feature interpretations where $\beta_j > 0$, increasing marginal returns correspond to $\beta_j > 1$.

Increasing marginal returns should be interpreted carefully. They may describe genuine scale economies, network externalities, learning effects, or technological complementarities. They may also indicate unstable accumulation, speculative amplification, or unsustainable leverage. Therefore, the same mathematical property can have either productive or destabilizing economic meaning depending on the feature and historical context.

6.5 Complementarity and Substitution

Second-order cross-partial derivatives describe how the marginal effect of one feature changes with the level of another feature. For two distinct features x_i and x_j , the basic classification is

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} > 0 \Rightarrow \text{complementarity}, \quad \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} < 0 \Rightarrow \text{substitution}. \quad (556)$$

Complementarity means that increasing feature x_j raises the marginal effect of feature x_i :

$$\frac{\partial ME_i(\mathbf{x})}{\partial x_j} = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} > 0. \quad (557)$$

In this case, the two features reinforce each other in the accumulation process. For example, technological intensity may raise the accumulation effect of skilled labor; market expansion may raise the effect of productive capacity; infrastructure may raise the return to industrial investment.

Substitution means that increasing feature x_j lowers the marginal effect of feature x_i :

$$\frac{\partial ME_i(\mathbf{x})}{\partial x_j} = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} < 0. \quad (558)$$

In this case, the two features partially offset each other. For example, automation may reduce the accumulation contribution of labor intensity, or financial substitution may reduce the role of internal surplus generation.

For the quadratic coupling model with symmetric $\mathbf{H}$,

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} = h_{ij}. \quad (559)$$

Thus,

$$h_{ij} > 0 \Rightarrow \text{complementarity}, \quad h_{ij} < 0 \Rightarrow \text{substitution}. \quad (560)$$

For a general nonlinear model, complementarity and substitution may be state-dependent:

$$\frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} \gtrless 0 \quad \text{depending on } \mathbf{x}. \quad (561)$$

Therefore, two features may be complementary in one region of the state space and substitutive in another:

$$\exists \mathbf{x}, \mathbf{x}' \in \mathcal{X} \quad \text{such that} \quad CPE_{ij}(\mathbf{x}) > 0, \quad CPE_{ij}(\mathbf{x}') < 0. \quad (562)$$

This possibility is central for analyzing regime-dependent accumulation mechanisms.

6.6 Cross-Partial Effects

To summarize cross-feature interactions, define the cross-partial effect between features x_i and x_j as

$$CPE_{ij}(\mathbf{x}) \equiv \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j}, \quad CPE_{ij} = CPE_{ji} \text{ if } \mathcal{A} \in C^2. \quad (563)$$

The equality $CPE_{ij} = CPE_{ji}$ follows from the symmetry of mixed partial derivatives under twice continuous differentiability.

The cross-partial effect can be interpreted in two equivalent ways:

$$CPE_{ij}(\mathbf{x}) = \frac{\partial ME_i(\mathbf{x})}{\partial x_j} = \frac{\partial ME_j(\mathbf{x})}{\partial x_i}. \quad (564)$$

Thus, it measures both how x_j changes the marginal effect of x_i and how x_i changes the marginal effect of x_j .

The sign classification is

$$CPE_{ij}(\mathbf{x}) \begin{cases} > 0, & \text{local complementarity,} \\ = 0, & \text{local separability,} \\ < 0, & \text{local substitution.} \end{cases} \quad (565)$$

Local separability means that, at the state $\mathbf{x}$, the marginal effect of one feature does not depend on the other:

$$CPE_{ij}(\mathbf{x}) = 0 \implies \frac{\partial ME_i(\mathbf{x})}{\partial x_j} = 0. \quad (566)$$

The matrix of cross-partial effects is precisely the Hessian:

$$\mathbf{CPE}(\mathbf{x}) = [\mathbf{CPE}_{ij}(\mathbf{x})]_{i,j=1}^d = \nabla^2 \mathcal{A}(\mathbf{x}). \quad (567)$$

The diagonal elements of this matrix are own-second-order effects, while the off-diagonal elements are cross-feature interaction effects:

$$\mathbf{CPE}_{jj}(\mathbf{x}) = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j^2}, \quad \mathbf{CPE}_{ij}(\mathbf{x}) = \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_i \partial x_j} \quad (i \neq j). \quad (568)$$

For empirical analysis, one may evaluate cross-partial effects at observed states and then compute average interaction effects:

$$\mathbf{ACPE}_{ij} = \frac{1}{T} \sum_{t=1}^T \mathbf{CPE}_{ij}(\mathbf{x}_t). \quad (569)$$

In panel data, the corresponding average is

$$\mathbf{ACPE}_{ij} = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T \mathbf{CPE}_{ij}(\mathbf{x}_{it}). \quad (570)$$

The variation of cross-partial effects over the state space can be used to detect nonlinear regime dependence:

$$\text{Var}_{\mathbf{x} \in \mathcal{X}} [\mathbf{CPE}_{ij}(\mathbf{x})] > 0 \implies \text{state-dependent interaction}. \quad (571)$$

If the sign of $\mathbf{CPE}_{ij}$ changes over $\mathcal{X}$, the relationship between the two features is not globally complementary or globally substitutive:

$$\min_{\mathbf{x} \in \mathcal{X}} \mathbf{CPE}_{ij}(\mathbf{x}) < 0 < \max_{\mathbf{x} \in \mathcal{X}} \mathbf{CPE}_{ij}(\mathbf{x}). \quad (572)$$

The second-order properties of $\mathcal{A}$ therefore determine the local geometry of capital accumulation. The Hessian identifies curvature, diminishing or increasing marginal returns, complementarity, substitution, and regime-dependent interaction:

$$\nabla^2 \mathcal{A}(\mathbf{x}) = [\mathbf{CPE}_{ij}(\mathbf{x})]_{i,j=1}^d. \quad (573)$$

This matrix is the natural extension of first-order marginal analysis to a multi-feature theory in which capital accumulation depends not only on individual determinants, but also on their nonlinear relations.

7 Bounds and Constraints

The accumulation function describes the potential increment of capital, but capital accumulation never occurs in an unconstrained space. It is bounded by the existing scale of capital, the availability of inputs, the financial capacity of the unit, ecological limits, market demand, and profitability conditions. This section formalizes these restrictions and defines the feasible set over which accumulation can take place.

7.1 Lower Bounds

We first impose lower bounds on capital and on each accumulation-relevant feature:

$$K_t \geq \underline{K} > 0, \quad x_{jt} \geq \underline{x}_j, \quad j = 1, \dots, d. \quad (574)$$

The lower bound on capital ensures that the accumulation process begins from a strictly positive value magnitude. This excludes the degenerate case in which the capital circuit has no value to advance.

The condition

$$K_t \geq \underline{K} > 0 \quad (575)$$

also guarantees that proportional growth rates are well defined:

$$g_t = \frac{K_{t+1} - K_t}{K_t} = \frac{\mathcal{A}(K_t, \mathbf{x}_t)}{K_t}. \quad (576)$$

Without a positive lower bound, the growth-rate representation may become undefined or economically meaningless.

The feature lower bounds

$$x_{jt} \geq \underline{x}_j, \quad j = 1, \dots, d, \quad (577)$$

define the minimum admissible level of each determinant of accumulation. Depending on the interpretation of x_{jt} , these bounds may represent minimum labor input, minimum technological capacity, minimum demand, minimum resource availability, or non-negativity of economic quantities.

When logarithmic or Cobb–Douglas-type specifications are used, the lower bounds must be strictly positive:

$$\underline{x}_j > 0, \quad j = 1, \dots, d. \quad (578)$$

This condition ensures that

$$\log x_{jt} \quad \text{and} \quad x_{jt}^{\beta_j} \quad (579)$$

are well defined over the admissible feature space.

The lower-bound restrictions therefore serve both an economic and a mathematical purpose. Economically, they define the minimum viable conditions for capital accumulation. Mathematically, they ensure that the accumulation function and its transformations are well defined.

7.2 Positive Accumulation Condition

The accumulation function may take positive, zero, or negative values:

$$\begin{aligned} \mathcal{A}(K_t, \mathbf{x}_t) &\in \mathbb{R}, \\ \mathcal{A}(K_t, \mathbf{x}_t) > 0 &\Rightarrow \text{expanded accumulation,} \\ \mathcal{A}(K_t, \mathbf{x}_t) = 0 &\Rightarrow \text{stationary reproduction,} \\ \mathcal{A}(K_t, \mathbf{x}_t) < 0 &\Rightarrow \text{devaluation / crisis.} \end{aligned} \quad (580)$$

This sign classification is central to the interpretation of the model.

If

$$\mathcal{A}(K_t, \mathbf{x}_t) > 0, \quad (581)$$

then

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t) > K_t. \quad (582)$$

The capital stock expands, and the circuit corresponds to enlarged reproduction. This is the formal analogue of the classical movement from M to M' with $M' > M$.

If

$$\mathcal{A}(K_t, \mathbf{x}_t) = 0, \quad (583)$$

then

$$K_{t+1} = K_t. \quad (584)$$

The process reproduces the existing capital stock without expansion. This case corresponds to stationary reproduction or value preservation.

If

$$\mathcal{A}(K_t, \mathbf{x}_t) < 0, \quad (585)$$

then

$$K_{t+1} < K_t. \quad (586)$$

The capital stock contracts. This regime represents devaluation, crisis, or failed accumulation. In the classical language, the circuit fails to realize the advanced value as an expanded value.

The three regimes can be summarized by the sign partition

$$\mathcal{S}^+ = \{(K, \mathbf{x}) : \mathcal{A}(K, \mathbf{x}) > 0\}, \quad \mathcal{S}^0 = \{(K, \mathbf{x}) : \mathcal{A}(K, \mathbf{x}) = 0\}, \quad \mathcal{S}^- = \{(K, \mathbf{x}) : \mathcal{A}(K, \mathbf{x}) < 0\}. \quad (587)$$

Here, $\mathcal{S}^0$ is the boundary between accumulation and devaluation. This boundary is important because small changes in features near it may determine whether the system expands or enters crisis.

7.3 Upper Bounds

The accumulation process is also subject to upper bounds:

$$K_t \leq \bar{K}, \quad x_{jt} \leq \bar{x}_j, \quad j = 1, \dots, d, \quad \mathcal{A}(K_t, \mathbf{x}_t) \leq \bar{A}. \quad (588)$$

The upper bound on capital restricts the analysis to a finite economic domain. It may represent the observed support of the data, the scale of an industry, the capacity of a region, or the historically relevant range of a national economy.

The feature upper bounds

$$x_{jt} \leq \bar{x}_j, \quad j = 1, \dots, d, \quad (589)$$

represent finite limits on the determinants of accumulation. These may arise from limited labor supply, finite productive capacity, technological frontiers, resource scarcity, credit limits, ecological ceilings, or market saturation.

The bound on the accumulation increment,

$$\mathcal{A}(K_t, \mathbf{x}_t) \leq \bar{A}, \quad (590)$$

rules out infinite one-period expansion. If the accumulation function is continuous on a compact state space, such a bound follows naturally:

$$\sup_{(K, \mathbf{x}) \in \mathcal{S}} |\mathcal{A}(K, \mathbf{x})| < \infty. \quad (591)$$

To preserve the admissible capital interval, the transition must satisfy

$$\underline{K} \leq K_t + \mathcal{A}(K_t, \mathbf{x}_t) \leq \overline{K}. \quad (592)$$

Equivalently,

$$\underline{K} - K_t \leq \mathcal{A}(K_t, \mathbf{x}_t) \leq \overline{K} - K_t. \quad (593)$$

Thus, the allowable increment depends on the current position of the capital stock within its bounded interval.

7.4 Resource Constraints

Capital accumulation requires the allocation of scarce resources. Let $\mathbf{p}_t$ denote the vector of feature prices or resource costs, and let B_t denote the available budget. The budget constraint is

$$\mathbf{p}_t^\top \mathbf{x}_t \leq B_t. \quad (594)$$

This condition states that the total cost of the chosen feature vector cannot exceed the available resources.

More generally, accumulation may be subject to multiple resource constraints:

$$\sum_{j=1}^d a_{rj} x_{jt} \leq R_{rt}, \quad r = 1, \dots, R. \quad (595)$$

Here, a_{rj} measures the amount of resource r required per unit of feature x_j , and R_{rt} is the available quantity of resource r at time t .

The resource constraints can be written in matrix form. Let

$$\mathbf{A}_R = (a_{rj})_{r=1, \dots, R; j=1, \dots, d}, \quad \mathbf{R}_t = (R_{1t}, \dots, R_{Rt})^\top. \quad (596)$$

Then

$$\mathbf{A}_R \mathbf{x}_t \leq \mathbf{R}_t. \quad (597)$$

These constraints define the physically or institutionally feasible region of production and accumulation.

The shadow value of a resource constraint will be important in optimization. If $\lambda_{rt} \geq 0$ is the multiplier associated with resource r , then a binding constraint satisfies

$$\lambda_{rt} \left(R_{rt} - \sum_{j=1}^d a_{rj} x_{jt} \right) = 0. \quad (598)$$

A positive multiplier indicates that relaxing the constraint would raise the maximum attainable accumulation increment.

7.5 Ecological Constraints

Capital accumulation may also be constrained by ecological limits. Let $E_t(\mathbf{x}_t)$ denote the ecological burden associated with the feature vector, and let $\overline{E}_t$ denote the admissible ecological ceiling:

$$E_t(\mathbf{x}_t) \leq \overline{E}_t, \quad E_t(\mathbf{x}_t) = \sum_{j=1}^d \eta_j x_{jt} + \sum_{i,j} \eta_{ij} x_{it} x_{jt}. \quad (599)$$

The linear coefficients η_j capture the direct ecological intensity of each feature. The quadratic coefficients η_{ij} capture interaction effects in ecological burden.

The ecological function may be interpreted broadly. It may represent carbon emissions, energy use, water extraction, pollution, land degradation, material throughput, or a composite environmental-pressure index. The constraint

$$E_t(\mathbf{x}_t) \leq \bar{E}_t \quad (600)$$

requires accumulation choices to remain within the ecological limit.

The marginal ecological burden of feature x_j is

$$\frac{\partial E_t(\mathbf{x}_t)}{\partial x_{jt}} = \eta_j + \sum_i \eta_{ij} x_{it} + \sum_i \eta_{ji} x_{it}. \quad (601)$$

If the interaction matrix (η_{ij}) is symmetric, this simplifies to

$$\frac{\partial E_t(\mathbf{x}_t)}{\partial x_{jt}} = \eta_j + 2 \sum_i \eta_{ij} x_{it}. \quad (602)$$

This derivative measures the additional ecological pressure created by a marginal increase in feature x_j .

The ecological constraint introduces a direct trade-off between accumulation and environmental feasibility. A feature may raise capital accumulation while also increasing ecological burden:

$$\frac{\partial \mathcal{A}}{\partial x_j} > 0, \quad \frac{\partial E_t}{\partial x_j} > 0. \quad (603)$$

In that case, unconstrained accumulation may be environmentally infeasible, and the optimal feature allocation must account for the ecological shadow cost.

7.6 Market Constraints

Capital accumulation also requires realization in the market. Let $D_t(P_t)$ denote market demand at price P_t , and let $Q_t(\mathbf{x}_t)$ denote output generated by the feature vector. The market-clearing feasibility condition is

$$D_t(P_t) \geq Q_t(\mathbf{x}_t), \quad \Pi_t = P_t Q_t - C_t(\mathbf{x}_t) \geq 0. \quad (604)$$

The first inequality requires output not to exceed effective demand at the prevailing price. The second inequality requires nonnegative profit.

The demand constraint

$$D_t(P_t) \geq Q_t(\mathbf{x}_t) \quad (605)$$

captures the realization condition. Even if production generates commodity output, capital accumulation depends on whether that output can be sold. If

$$Q_t(\mathbf{x}_t) > D_t(P_t), \quad (606)$$

then the system produces more output than can be realized at price P_t . This may generate unsold inventories, price reductions, profit compression, or crisis.

The profit condition is

$$\Pi_t = P_t Q_t - C_t(\mathbf{x}_t) \geq 0. \quad (607)$$

Here, $C_t(\mathbf{x}_t)$ denotes the cost of producing output Q_t using the feature vector $\mathbf{x}_t$. Nonnegative profit is a minimal viability condition for continued accumulation.

The marginal profit effect of feature x_j is

$$\frac{\partial \Pi_t}{\partial x_{jt}} = P_t \frac{\partial Q_t(\mathbf{x}_t)}{\partial x_{jt}} - \frac{\partial C_t(\mathbf{x}_t)}{\partial x_{jt}}, \quad (608)$$

when price is taken as given. If price also responds to output, then the marginal profit effect becomes

$$\frac{\partial \Pi_t}{\partial x_{jt}} = \left(P_t + Q_t \frac{\partial P_t}{\partial Q_t} \right) \frac{\partial Q_t(\mathbf{x}_t)}{\partial x_{jt}} - \frac{\partial C_t(\mathbf{x}_t)}{\partial x_{jt}}. \quad (609)$$

This expression incorporates the possibility that expanding output depresses price when demand is downward sloping.

Market constraints therefore connect production to realization. They ensure that the accumulation function does not treat output expansion as sufficient for capital expansion unless the resulting commodities can be sold profitably.

7.7 Feasible Set

Combining the preceding restrictions, the feasible set of admissible feature choices at time t is

$$\mathcal{F}_t = \{\mathbf{x} \in \mathbb{R}_+^d : \mathbf{p}_t^\top \mathbf{x} \leq B_t, E_t(\mathbf{x}) \leq \bar{E}_t, D_t(P_t) \geq Q_t(\mathbf{x})\}. \quad (610)$$

This set contains all nonnegative feature vectors satisfying the budget, ecological, and market-realization constraints.

A more complete feasible set can include the coordinate bounds and resource constraints:

$$\mathcal{F}_t = \left\{ \mathbf{x} \in \mathbb{R}_+^d : \underline{x}_j \leq x_j \leq \bar{x}_j, \mathbf{p}_t^\top \mathbf{x} \leq B_t, \mathbf{A}_R \mathbf{x} \leq \mathbf{R}_t, E_t(\mathbf{x}) \leq \bar{E}_t, D_t(P_t) \geq Q_t(\mathbf{x}), \Pi_t(\mathbf{x}) \geq 0 \right\}. \quad (611)$$

This expression makes explicit that accumulation is constrained by multiple simultaneous conditions.

The constrained capital law of motion is therefore

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t), \quad \mathbf{x}_t \in \mathcal{F}_t. \quad (612)$$

If $\mathbf{x}_t \notin \mathcal{F}_t$, then the corresponding accumulation path is not economically admissible, even if the unconstrained value of $\mathcal{A}(K_t, \mathbf{x}_t)$ is positive.

The feasible set also defines the constrained expansion region:

$$\mathcal{F}_t^+ = \{\mathbf{x} \in \mathcal{F}_t : \mathcal{A}(K_t, \mathbf{x}) > 0\}. \quad (613)$$

Similarly, the feasible stationary and devaluation regions are

$$\mathcal{F}_t^0 = \{\mathbf{x} \in \mathcal{F}_t : \mathcal{A}(K_t, \mathbf{x}) = 0\}, \quad \mathcal{F}_t^- = \{\mathbf{x} \in \mathcal{F}_t : \mathcal{A}(K_t, \mathbf{x}) < 0\}. \quad (614)$$

The distinction between unconstrained and feasible accumulation is crucial. An unconstrained state may generate positive accumulation,

$$\mathcal{A}(K_t, \mathbf{x}) > 0, \quad (615)$$

but if

$$\mathbf{x} \notin \mathcal{F}_t, \quad (616)$$

then it is not attainable under the resource, ecological, and market constraints. The feasible accumulation problem must therefore be formulated over $\mathcal{F}_t$, not over the entire feature space.

In summary, the bounds and constraints define the economically admissible domain of the multi-feature capital accumulation formula:

$$\mathbf{x}_t \in \mathcal{F}_t, \quad K_t \in [\underline{K}, \overline{K}], \quad K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t). \quad (617)$$

They ensure that the model does not describe accumulation as a purely technical mapping, but as a constrained process shaped by material, financial, ecological, and market limits.

8 Optimization and Critical Points

The accumulation function $\mathcal{A}(\mathbf{x})$ describes how a given configuration of economic features generates a capital increment. Once this mapping is specified, a natural question is which feature configuration maximizes accumulation. This section formulates the corresponding optimization problem, first without constraints and then under budget, resource, ecological, and market restrictions.

8.1 Unconstrained Maximization

In the unconstrained case, the accumulation-maximizing feature vector solves

$$\max_{\mathbf{x} \in \mathbb{R}^d} \mathcal{A}(\mathbf{x}), \quad \nabla \mathcal{A}(\mathbf{x}^*) = \mathbf{0}. \quad (618)$$

The first expression states the objective: choose the feature vector $\mathbf{x}$ that maximizes the capital accumulation increment. The second expression is the first-order necessary condition for an interior optimum.

The gradient condition means that every first-order marginal effect vanishes at the critical point:

$$\frac{\partial \mathcal{A}(\mathbf{x}^*)}{\partial x_j} = 0, \quad j = 1, \dots, d. \quad (619)$$

Equivalently,

$$ME_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, d. \quad (620)$$

At such a point, no infinitesimal change in any individual feature increases accumulation to first order.

For the quadratic specification

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad \mathbf{H} = \mathbf{H}^\top, \quad (621)$$

the first-order condition is

$$\mathbf{a} + \mathbf{H} \mathbf{x}^* = \mathbf{0}. \quad (622)$$

If $\mathbf{H}$ is nonsingular, the unique critical point is

$$\mathbf{x}^* = -\mathbf{H}^{-1} \mathbf{a}. \quad (623)$$

The unconstrained problem is analytically useful, but economically limited. Capital accumulation generally occurs under resource, budget, ecological, and market constraints. Therefore, the unconstrained solution should be interpreted as a benchmark. It identifies the feature configuration that would maximize accumulation if no feasibility restrictions were imposed.

8.2 Constrained Maximization

In the constrained case, the accumulation-maximizing feature vector must lie inside a feasible set. Let $\mathcal{F}$ denote the admissible set of feature vectors. The constrained maximization problem is

$$\max_{\mathbf{x} \in \mathcal{F}} \mathcal{A}(\mathbf{x}), \quad g_m(\mathbf{x}) \leq 0, \quad m = 1, \dots, M. \quad (624)$$

The functions $g_m(\mathbf{x})$ represent constraints. These may include budget constraints, resource constraints, ecological ceilings, market-realization conditions, non-negativity restrictions, or institutional limits.

The feasible set can be written as

$$\mathcal{F} = \left\{ \mathbf{x} \in \mathbb{R}^d : g_m(\mathbf{x}) \leq 0, \quad m = 1, \dots, M \right\}. \quad (625)$$

A solution $\mathbf{x}^*$ is feasible only if

$$g_m(\mathbf{x}^*) \leq 0, \quad m = 1, \dots, M. \quad (626)$$

Unlike the unconstrained case, the optimum need not satisfy $\nabla \mathcal{A}(\mathbf{x}^*) = \mathbf{0}$. If the optimum lies on the boundary of the feasible set, the gradient of accumulation may be nonzero. In that case, further increases in accumulation are blocked by one or more binding constraints:

$$g_m(\mathbf{x}^*) = 0 \quad \text{for some } m. \quad (627)$$

The constrained problem therefore captures a central economic idea: accumulation may be limited not by the absence of profitable directions, but by the impossibility of moving further in those directions without violating material, financial, ecological, or market constraints.

8.3 Lagrangian Formulation

The constrained optimization problem can be analyzed using the Lagrangian:

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \mathcal{A}(\mathbf{x}) - \sum_{m=1}^M \lambda_m g_m(\mathbf{x}), \quad \boldsymbol{\lambda} \geq \mathbf{0}. \quad (628)$$

Here, λ_m is the Lagrange multiplier associated with constraint $g_m(\mathbf{x}) \leq 0$. The multiplier measures the shadow value of relaxing the corresponding constraint.

The gradient of the Lagrangian with respect to $\mathbf{x}$ is

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \nabla \mathcal{A}(\mathbf{x}) - \sum_{m=1}^M \lambda_m \nabla g_m(\mathbf{x}). \quad (629)$$

At an interior constrained optimum, this gradient must vanish:

$$\nabla \mathcal{A}(\mathbf{x}^*) = \sum_{m=1}^M \lambda_m^* \nabla g_m(\mathbf{x}^*). \quad (630)$$

This equation states that the marginal accumulation vector is balanced by a weighted combination of the gradients of the binding constraints.

The multipliers have a direct economic interpretation. If

$$\lambda_m^* > 0, \quad (631)$$

then constraint m is binding and scarce. Relaxing it would increase the maximum attainable accumulation. If

$$\lambda_m^* = 0, \quad (632)$$

then constraint m is locally nonbinding and does not affect the optimum.

8.4 Karush-Kuhn-Tucker Conditions

Under standard regularity conditions, a constrained optimum satisfies the Karush–Kuhn–Tucker conditions:

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) = \mathbf{0}, \quad g_m(\mathbf{x}^*) \leq 0, \quad \lambda_m^* \geq 0, \quad \lambda_m^* g_m(\mathbf{x}^*) = 0. \quad (633)$$

These conditions consist of stationarity, primal feasibility, dual feasibility, and complementary slackness.

Stationarity requires

$$\nabla \mathcal{A}(\mathbf{x}^*) - \sum_{m=1}^M \lambda_m^* \nabla g_m(\mathbf{x}^*) = \mathbf{0}. \quad (634)$$

This condition balances marginal accumulation against the marginal cost of constraints.

Primal feasibility requires

$$g_m(\mathbf{x}^*) \leq 0, \quad m = 1, \dots, M. \quad (635)$$

This ensures that the chosen feature vector is economically admissible.

Dual feasibility requires

$$\lambda_m^* \geq 0, \quad m = 1, \dots, M. \quad (636)$$

Since the constraints are written as upper-bound inequalities, the shadow values of relaxing them must be nonnegative.

Complementary slackness requires

$$\lambda_m^* g_m(\mathbf{x}^*) = 0, \quad m = 1, \dots, M. \quad (637)$$

This condition implies that a constraint has a positive shadow value only if it binds:

$$\lambda_m^* > 0 \implies g_m(\mathbf{x}^*) = 0. \quad (638)$$

If a constraint is slack,

$$g_m(\mathbf{x}^*) < 0, \quad (639)$$

then its multiplier must be zero:

$$\lambda_m^* = 0. \quad (640)$$

The KKT system is especially useful because it connects the geometry of the accumulation function with the economic scarcity of constraints. It gives a formal way to identify whether accumulation is limited by budget, resources, ecology, market demand, or profitability.

8.5 Budget-Constrained Capital Allocation

A central constrained problem is the allocation of features under a budget. Let $\mathbf{p}$ denote the vector of feature costs and let B denote the available budget. The problem is

$$\max_{\mathbf{x} \geq \mathbf{0}} \mathcal{A}(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{p}^\top \mathbf{x} \leq B, \quad \frac{\partial \mathcal{A} / \partial x_j}{p_j} = \lambda \text{ if } x_j > 0. \quad (641)$$

The budget constraint requires the total cost of the feature vector not to exceed the available budget.

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \lambda, \boldsymbol{\nu}) = \mathcal{A}(\mathbf{x}) - \lambda(\mathbf{p}^\top \mathbf{x} - B) + \boldsymbol{\nu}^\top \mathbf{x}, \quad (642)$$

where $\lambda \geq 0$ is the multiplier on the budget constraint and $\boldsymbol{\nu} \geq \mathbf{0}$ are multipliers on the non-negativity constraints $x_j \geq 0$.

For an interior feature with $x_j^* > 0$, the non-negativity constraint does not bind, so $\nu_j^* = 0$. The first-order condition is

$$\frac{\partial \mathcal{A}(\mathbf{x}^*)}{\partial x_j} = \lambda^* p_j. \quad (643)$$

Equivalently,

$$\frac{\partial \mathcal{A}(\mathbf{x}^*) / \partial x_j}{p_j} = \lambda^*, \quad x_j^* > 0. \quad (644)$$

Thus, at the optimum, every actively used feature has the same marginal accumulation contribution per unit cost.

For a feature not used at the optimum, $x_j^* = 0$, the Kuhn–Tucker condition implies

$$\frac{\partial \mathcal{A}(\mathbf{x}^*) / \partial x_j}{p_j} \leq \lambda^*. \quad (645)$$

A non-used feature has marginal accumulation per unit cost no greater than the shadow value of the budget.

If the budget constraint binds, then

$$\mathbf{p}^\top \mathbf{x}^* = B, \quad \lambda^* > 0. \quad (646)$$

If it does not bind, then

$$\mathbf{p}^\top \mathbf{x}^* < B, \quad \lambda^* = 0. \quad (647)$$

The multiplier λ^* measures the marginal increase in maximum accumulation produced by an additional unit of budget.

8.6 Ecological-Constrained Capital Allocation

Accumulation may also be constrained by ecological limits. Consider the problem

$$\max_{\mathbf{x} \geq \mathbf{0}} \mathcal{A}(\mathbf{x}) \quad \text{s.t.} \quad E(\mathbf{x}) \leq \bar{E}, \quad \nabla \mathcal{A}(\mathbf{x}^*) = \mu \nabla E(\mathbf{x}^*) + \nu \mathbf{p}. \quad (648)$$

Here, $E(\mathbf{x})$ denotes ecological burden and $\bar{E}$ denotes the ecological ceiling. The vector $\mathbf{p}$ may represent an additional budget or resource-cost constraint, with multiplier ν .

With both ecological and budget constraints, the problem can be written as

$$\max_{\mathbf{x} \geq \mathbf{0}} \mathcal{A}(\mathbf{x}) \quad \text{s.t.} \quad E(\mathbf{x}) \leq \bar{E}, \quad \mathbf{p}^\top \mathbf{x} \leq B. \quad (649)$$

The Lagrangian is

$$\mathcal{L}(\mathbf{x}, \mu, \nu) = \mathcal{A}(\mathbf{x}) - \mu (E(\mathbf{x}) - \bar{E}) - \nu (\mathbf{p}^\top \mathbf{x} - B), \quad \mu, \nu \geq 0. \quad (650)$$

The stationarity condition is

$$\nabla \mathcal{A}(\mathbf{x}^*) = \mu^* \nabla E(\mathbf{x}^*) + \nu^* \mathbf{p}. \quad (651)$$

This condition says that the marginal accumulation vector is balanced by two shadow-cost vectors: the marginal ecological burden and the marginal financial or resource cost. If the ecological constraint binds, then

$$E(\mathbf{x}^*) = \bar{E}, \quad \mu^* > 0. \quad (652)$$

If it is slack, then

$$E(\mathbf{x}^*) < \bar{E}, \quad \mu^* = 0. \quad (653)$$

The multiplier μ^* measures the shadow value of relaxing the ecological ceiling. A high value of μ^* indicates that the ecological constraint is a major limit on further accumulation:

$$\frac{\partial \mathcal{A}^*}{\partial \bar{E}} = \mu^*, \quad (654)$$

where $\mathcal{A}^*$ denotes the optimized value of the accumulation function under the constraint.

If

$$\frac{\partial \mathcal{A}}{\partial x_j} > 0 \quad \text{and} \quad \frac{\partial E}{\partial x_j} > 0, \quad (655)$$

then feature x_j raises accumulation but also increases ecological burden. The optimal allocation must then compare the marginal accumulation benefit with the ecological shadow cost:

$$\frac{\partial \mathcal{A}}{\partial x_j} = \mu^* \frac{\partial E}{\partial x_j} + \nu^* p_j. \quad (656)$$

8.7 Classification of Critical Points

A critical point of the unconstrained accumulation function satisfies

$$\nabla \mathcal{A}(\mathbf{x}^*) = \mathbf{0}, \quad \nabla^2 \mathcal{A}(\mathbf{x}^*) \prec 0 \Rightarrow \text{strict local max}, \quad \nabla^2 \mathcal{A}(\mathbf{x}^*) \succ 0 \Rightarrow \text{strict local min}. \quad (657)$$

The classification depends on the definiteness of the Hessian matrix at the critical point.

If the Hessian is negative definite,

$$\mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}^*) \mathbf{u} < 0 \quad \forall \mathbf{u} \neq \mathbf{0}, \quad (658)$$

then $\mathbf{x}^*$ is a strict local maximum. This means that for all sufficiently small nonzero perturbations $\mathbf{h}$,

$$\mathcal{A}(\mathbf{x}^* + \mathbf{h}) < \mathcal{A}(\mathbf{x}^*). \quad (659)$$

If the Hessian is positive definite,

$$\mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}^*) \mathbf{u} > 0 \quad \forall \mathbf{u} \neq \mathbf{0}, \quad (660)$$

then $\mathbf{x}^*$ is a strict local minimum:

$$\mathcal{A}(\mathbf{x}^* + \mathbf{h}) > \mathcal{A}(\mathbf{x}^*) \quad (661)$$

for all sufficiently small nonzero perturbations $\mathbf{h}$.

If the Hessian is indefinite, then $\mathbf{x}^*$ is a saddle point:

$$\exists \mathbf{u}, \mathbf{v} \quad \text{such that} \quad \mathbf{u}^\top \nabla^2 \mathcal{A}(\mathbf{x}^*) \mathbf{u} > 0, \quad \mathbf{v}^\top \nabla^2 \mathcal{A}(\mathbf{x}^*) \mathbf{v} < 0. \quad (662)$$

At a saddle point, accumulation increases in some directions and decreases in others. Economically, this corresponds to a configuration that is stable with respect to some feature perturbations but unstable with respect to others.

If the Hessian is only semidefinite, the second-order test is inconclusive:

$$\nabla^2 \mathcal{A}(\mathbf{x}^*) \preceq 0 \quad \text{or} \quad \nabla^2 \mathcal{A}(\mathbf{x}^*) \succeq 0 \quad \text{with zero eigenvalues.} \quad (663)$$

In that case, higher-order terms or global properties of $\mathcal{A}$ are needed to classify the critical point.

For constrained problems, the relevant second-order condition applies to feasible directions. Let $\mathcal{T}(\mathbf{x}^*)$ denote the tangent space of feasible directions at the optimum. A sufficient condition for a strict constrained local maximum is

$$\mathbf{h}^\top \nabla_{\mathbf{x}}^2 \mathcal{L}(\mathbf{x}^*, \boldsymbol{\lambda}^*) \mathbf{h} < 0 \quad \forall \mathbf{h} \in \mathcal{T}(\mathbf{x}^*), \quad \mathbf{h} \neq \mathbf{0}. \quad (664)$$

This condition states that the Lagrangian is locally concave along all feasible directions.

The classification of critical points therefore links marginal analysis, curvature, and feasibility. A feature vector is not accumulation-maximizing merely because marginal effects vanish; the curvature of the accumulation surface and the binding constraints determine whether the point is a maximum, minimum, saddle, or boundary optimum:

$$\text{critical point} + \text{Hessian structure} + \text{feasibility constraints} \implies \text{economic classification.} \quad (665)$$

9 Thresholds and Regime Shifts

The preceding sections model capital accumulation as a smooth function of economic features. In many historical and empirical settings, however, accumulation does not respond smoothly over the entire state space. Financial fragility, network effects, ecological pressure, policy changes, and crisis conditions may generate discontinuities, kinks, or regime-dependent slopes. This section introduces threshold and regime-shift specifications that allow the accumulation law to change when an observable or latent state variable crosses a critical value.

9.1 Single-Threshold Formula

A simple way to model regime dependence is to allow the accumulation equation to change when a threshold variable q_t exceeds a critical value τ :

$$K_{t+1} = K_t + \alpha_1 + \beta_1^\top \mathbf{x}_t + \left(\alpha_2 + \beta_2^\top \mathbf{x}_t \right) \mathbf{1}\{q_t > \tau\}. \quad (666)$$

Here, $\mathbf{1}\{q_t > \tau\}$ is an indicator function equal to one when the threshold condition is satisfied and zero otherwise. The variable q_t may represent leverage, network size, ecological pressure, demand stress, capacity utilization, or any state variable capable of changing the accumulation regime.

When $q_t \leq \tau$, the accumulation law is

$$K_{t+1} = K_t + \alpha_1 + \beta_1^\top \mathbf{x}_t. \quad (667)$$

When $q_t > \tau$, it becomes

$$K_{t+1} = K_t + (\alpha_1 + \alpha_2) + (\beta_1 + \beta_2)^\top \mathbf{x}_t. \quad (668)$$

Thus, α_2 measures the shift in the intercept after the threshold is crossed, while β_2 measures the change in marginal effects.

The corresponding accumulation function is

$$\mathcal{A}(K_t, \mathbf{x}_t, q_t) = \alpha_1 + \beta_1^\top \mathbf{x}_t + \left(\alpha_2 + \beta_2^\top \mathbf{x}_t \right) \mathbf{1}\{q_t > \tau\}. \quad (669)$$

This function is piecewise linear in $\mathbf{x}_t$ and discontinuous in q_t if $\alpha_2 + \beta_2^\top \mathbf{x}_t \neq 0$ at $q_t = \tau$.

If continuity at the threshold is desired, the threshold specification may instead be written in kink form:

$$\mathcal{A}_t = \alpha + \beta^\top \mathbf{x}_t + \gamma(q_t - \tau)_+, \quad (q_t - \tau)_+ \equiv \max\{q_t - \tau, 0\}. \quad (670)$$

This formulation allows the slope to change after the threshold without creating a discrete jump in the level of accumulation.

The threshold τ can be treated as known from theory or estimated from data. In empirical applications, an estimator may solve

$$\hat{\tau} = \arg \min_{\tau \in \mathcal{T}} \sum_{t=1}^T \left[K_{t+1} - K_t - \alpha_1 - \beta_1^\top \mathbf{x}_t - \left(\alpha_2 + \beta_2^\top \mathbf{x}_t \right) \mathbf{1}\{q_t > \tau\} \right]^2. \quad (671)$$

The admissible set $\mathcal{T}$ is usually restricted to an interior range of observed threshold values to avoid regimes with too few observations.

The single-threshold model therefore formalizes the idea that the same feature vector may generate different accumulation outcomes depending on the surrounding regime:

$$\mathcal{A}_L(\mathbf{x}_t) \neq \mathcal{A}_H(\mathbf{x}_t), \quad q_t \leq \tau \quad \text{versus} \quad q_t > \tau. \quad (672)$$

9.2 Financial-Leverage Threshold

A particularly important threshold variable is financial leverage. Let D_t denote debt and let A_t denote total assets. The leverage ratio is

$$q_t \equiv \ell_t = \frac{D_t}{A_t}, \quad \mathcal{A}_t = \mathcal{A}_L(\mathbf{x}_t) \mathbf{1}\{\ell_t \leq \tau_\ell\} + \mathcal{A}_H(\mathbf{x}_t) \mathbf{1}\{\ell_t > \tau_\ell\}. \quad (673)$$

Here, τ_ℓ is the leverage threshold separating a low-leverage regime from a high-leverage regime.

In the low-leverage regime,

$$\ell_t \leq \tau_\ell \implies \mathcal{A}_t = \mathcal{A}_L(\mathbf{x}_t). \quad (674)$$

In this region, debt may support accumulation by relaxing financing constraints, enabling investment, or smoothing liquidity needs.

In the high-leverage regime,

$$\ell_t > \tau_\ell \implies \mathcal{A}_t = \mathcal{A}_H(\mathbf{x}_t). \quad (675)$$

Here, additional debt may reduce accumulation by increasing interest burdens, rollover risk, default probability, or financial fragility.

A simple parametric version is

$$\mathcal{A}_L(\mathbf{x}_t) = \alpha_L + \beta_L^\top \mathbf{x}_t, \quad \mathcal{A}_H(\mathbf{x}_t) = \alpha_H + \beta_H^\top \mathbf{x}_t. \quad (676)$$

The difference between the regimes is

$$\mathcal{A}_H(\mathbf{x}_t) - \mathcal{A}_L(\mathbf{x}_t) = (\alpha_H - \alpha_L) + (\beta_H - \beta_L)^\top \mathbf{x}_t. \quad (677)$$

If this difference is negative, high leverage depresses accumulation relative to the low-leverage regime.

The marginal effect of feature x_j is regime dependent:

$$\frac{\partial \mathcal{A}_t}{\partial x_{jt}} = \frac{\partial \mathcal{A}_L(\mathbf{x}_t)}{\partial x_{jt}} \mathbf{1}\{\ell_t \leq \tau_\ell\} + \frac{\partial \mathcal{A}_H(\mathbf{x}_t)}{\partial x_{jt}} \mathbf{1}\{\ell_t > \tau_\ell\}. \quad (678)$$

Thus, the same feature may support accumulation in a financially stable regime but weaken accumulation once leverage exceeds the critical threshold.

The leverage threshold captures a central nonlinear mechanism of capitalist accumulation: credit can expand accumulation up to a point, but excessive financial exposure can transform the same mechanism into a source of crisis:

$$\ell_t \leq \tau_\ell \implies \text{finance as expansion}, \quad \ell_t > \tau_\ell \implies \text{finance as fragility}. \quad (679)$$

9.3 Platform-Network Threshold

Network-based accumulation often displays threshold behavior. Let n_t denote network size, user scale, platform participation, or connectivity. The platform-network threshold model is

$$q_t \equiv n_t, \quad \mathcal{A}_t = \gamma_0 + \gamma_1 n_t + \gamma_2 n_t^2 \mathbf{1}\{n_t > \tau_n\}. \quad (680)$$

Here, τ_n is the network threshold after which nonlinear network effects become active.

When $n_t \leq \tau_n$, accumulation responds linearly to network size:

$$\mathcal{A}_t = \gamma_0 + \gamma_1 n_t. \quad (681)$$

When $n_t > \tau_n$, the quadratic network term enters:

$$\mathcal{A}_t = \gamma_0 + \gamma_1 n_t + \gamma_2 n_t^2. \quad (682)$$

If $\gamma_2 > 0$, network scale generates increasing returns after the threshold is crossed. If $\gamma_2 < 0$, the post-threshold region exhibits congestion, saturation, or diseconomies of scale.

The marginal effect of network size is

$$\frac{\partial \mathcal{A}_t}{\partial n_t} = \gamma_1 + 2\gamma_2 n_t \mathbf{1}\{n_t > \tau_n\}, \quad (683)$$

away from the threshold point. Therefore, the marginal effect jumps in slope once the network passes τ_n .

A continuous version of the platform-network model can be written as

$$\mathcal{A}_t = \gamma_0 + \gamma_1 n_t + \gamma_2 (n_t - \tau_n)_+^2. \quad (684)$$

The corresponding marginal effect is

$$\frac{\partial \mathcal{A}_t}{\partial n_t} = \gamma_1 + 2\gamma_2 (n_t - \tau_n)_+. \quad (685)$$

This specification avoids a discontinuous jump in the level of accumulation while retaining a non-linear post-threshold effect.

The platform-network threshold captures the idea that accumulation in platform economies may be limited at small scale but self-reinforcing after a critical mass:

$$n_t > \tau_n \implies \text{network-mediated nonlinear accumulation.} \quad (686)$$

9.4 Ecological-Constraint Threshold

Ecological pressure may also generate threshold effects. Let E_t denote ecological burden, such as emissions, energy use, resource depletion, or a composite environmental pressure index. The ecological-threshold model is

$$q_t \equiv E_t, \quad \mathcal{A}_t = \delta_0 + \delta_1 E_t + \delta_2 (E_t - \tau_E)_+. \quad (687)$$

Here, τ_E is the ecological threshold and $(E_t - \tau_E)_+ = \max\{E_t - \tau_E, 0\}$.

When ecological pressure is below the threshold,

$$E_t \leq \tau_E \implies \mathcal{A}_t = \delta_0 + \delta_1 E_t. \quad (688)$$

When ecological pressure exceeds the threshold,

$$E_t > \tau_E \implies \mathcal{A}_t = \delta_0 + \delta_1 E_t + \delta_2 (E_t - \tau_E). \quad (689)$$

The post-threshold slope is therefore

$$\frac{\partial \mathcal{A}_t}{\partial E_t} = \delta_1 + \delta_2 \quad \text{for } E_t > \tau_E. \quad (690)$$

The marginal effect of ecological pressure is piecewise:

$$\frac{\partial \mathcal{A}_t}{\partial E_t} = \delta_1 \mathbf{1}\{E_t \leq \tau_E\} + (\delta_1 + \delta_2) \mathbf{1}\{E_t > \tau_E\}. \quad (691)$$

If $\delta_2 < 0$, crossing the ecological threshold reduces the marginal contribution of further ecological pressure to accumulation. This may represent regulatory constraints, physical resource limits, climate damages, or ecological degradation.

A strict ecological constraint can also be represented by imposing

$$E_t \leq \bar{E}_t. \quad (692)$$

In that case, the threshold is not merely a change in slope; it defines a feasibility boundary. Once ecological burden reaches the ceiling, additional accumulation must either reduce ecological intensity or violate the constraint:

$$E_t = \bar{E}_t \implies \text{ecological constraint binds.} \quad (693)$$

The ecological-threshold formulation captures the possibility that environmental pressure may appear weakly constraining over one range but become a dominant limit once a critical level is reached:

$$E_t > \tau_E \implies \text{ecological regime shift in accumulation.} \quad (694)$$

9.5 Structural Break

Thresholds may also occur in historical time rather than in an economic state variable. A structural break model allows the relationship between features and an outcome Y_t to change at a break date T_b :

$$Y_t = \mathbf{x}_t^\top \beta_1 \mathbf{1}\{t \leq T_b\} + \mathbf{x}_t^\top \beta_2 \mathbf{1}\{t > T_b\} + u_t. \quad (695)$$

Here, β_1 governs the pre-break regime and β_2 governs the post-break regime.

In the context of capital accumulation, the outcome may be the capital increment:

$$Y_t = \Delta K_t. \quad (696)$$

Then the model becomes

$$\Delta K_t = \mathbf{x}_t^\top \beta_1 \mathbf{1}\{t \leq T_b\} + \mathbf{x}_t^\top \beta_2 \mathbf{1}\{t > T_b\} + u_t. \quad (697)$$

The break changes the marginal effects from β_1 to β_2 :

$$\frac{\partial \Delta K_t}{\partial \mathbf{x}_t} = \beta_1 \quad (t \leq T_b), \quad \frac{\partial \Delta K_t}{\partial \mathbf{x}_t} = \beta_2 \quad (t > T_b). \quad (698)$$

The magnitude of the structural change is

$$\Delta \beta = \beta_2 - \beta_1. \quad (699)$$

A null hypothesis of no structural break is

$$H_0 : \beta_1 = \beta_2. \quad (700)$$

Under the alternative,

$$H_1 : \beta_1 \neq \beta_2. \quad (701)$$

This tests whether the accumulation relationship is stable across the two historical periods.

The break date T_b may be known from historical context, such as a crisis, policy reform, war, technological transition, or institutional shift. If it is unknown, it can be estimated by minimizing the residual sum of squares:

$$\hat{T}_b = \arg \min_{T_b \in \mathcal{T}_b} \sum_t \hat{u}_t(T_b)^2. \quad (702)$$

The admissible set $\mathcal{T}_b$ excludes break dates too close to the beginning or end of the sample.

The structural-break model therefore formalizes the idea that the laws of capital accumulation may be historically contingent. The same features may have different effects before and after a major transformation:

$$\mathcal{A}_{\text{pre}}(\mathbf{x}) \neq \mathcal{A}_{\text{post}}(\mathbf{x}). \quad (703)$$

9.6 Regime-Switching Formula

In some settings, regimes are not directly observed or determined by a single observable threshold. Instead, the economy may switch among latent states. Let S_t denote the regime at time t . The regime-switching capital formula is

$$K_{t+1} = K_t + \alpha_{S_t} + \boldsymbol{\beta}_{S_t}^\top \mathbf{x}_t + \varepsilon_t, \quad \Pr(S_t = j \mid S_{t-1} = i) = p_{ij}. \quad (704)$$

Here, α_{S_t} and $\boldsymbol{\beta}_{S_t}$ are regime-specific coefficients, and p_{ij} is the probability of moving from regime i to regime j .

If there are J regimes, then

$$S_t \in \{1, \dots, J\}, \quad \sum_{j=1}^J p_{ij} = 1, \quad p_{ij} \geq 0. \quad (705)$$

The transition matrix is

$$\mathbf{P} = [p_{ij}]_{i,j=1}^J. \quad (706)$$

This matrix governs the persistence and mobility of accumulation regimes.

Conditional on regime $S_t = s$, the accumulation function is

$$\mathcal{A}_s(\mathbf{x}_t) = \alpha_s + \boldsymbol{\beta}_s^\top \mathbf{x}_t. \quad (707)$$

The capital law of motion becomes

$$K_{t+1} = K_t + \mathcal{A}_{S_t}(\mathbf{x}_t) + \varepsilon_t. \quad (708)$$

Thus, both the baseline accumulation rate and the marginal effects of features can vary across regimes.

A two-regime version may distinguish expansion and crisis:

$$S_t = 1 \implies \text{expansion regime}, \quad S_t = 2 \implies \text{crisis regime}. \quad (709)$$

In this case, one may have

$$\mathcal{A}_1(\mathbf{x}_t) > 0 \quad \text{for typical states}, \quad \mathcal{A}_2(\mathbf{x}_t) < 0 \quad \text{for typical states}. \quad (710)$$

The expected capital increment conditional on the information available at time $t-1$ is

$$\mathbb{E}[\Delta K_t \mid \mathcal{I}_{t-1}] = \sum_{s=1}^J \Pr(S_t = s \mid \mathcal{I}_{t-1}) \left(\alpha_s + \boldsymbol{\beta}_s^\top \mathbf{x}_t \right), \quad (711)$$

assuming $\mathbb{E}[\varepsilon_t \mid \mathcal{I}_{t-1}] = 0$. This shows that expected accumulation is a probability-weighted average of regime-specific accumulation laws.

Regime persistence is measured by the diagonal entries of the transition matrix:

$$p_{ss} = \Pr(S_t = s \mid S_{t-1} = s). \quad (712)$$

A large value of p_{ss} indicates that regime s is persistent. The expected duration of regime s is

$$\mathbb{E}[D_s] = \frac{1}{1 - p_{ss}}, \quad (713)$$

provided $p_{ss} < 1$.

The regime-switching formula is more flexible than a deterministic threshold model. It allows the economy to move between accumulation regimes in a probabilistic manner, including transitions that are not perfectly explained by observed variables. This is useful for modeling crises, recoveries, financial instability, policy regimes, or long-wave structural dynamics.

Together, threshold, structural-break, and regime-switching models extend the multi-feature capital formula beyond smooth accumulation. They allow the law of motion to change across historical, financial, ecological, network, and latent regimes:

$$K_{t+1} = K_t + \mathcal{A}_{S_t}(\mathbf{x}_t) + \varepsilon_t. \quad (714)$$

This formulation captures the possibility that capital accumulation is not governed by a single invariant function, but by a family of regime-dependent laws.

10 Dynamic Capital Accumulation

The preceding sections define the capital increment as a function of the current capital stock and the feature vector. This section studies the dynamic implications of that law of motion. The central question is how capital evolves over time when each period's capital stock becomes the base for the next period's accumulation process.

10.1 Discrete-Time Dynamics

The discrete-time capital accumulation process is written as

$$K_{t+1} = F(K_t, \mathbf{x}_t), \quad F(K_t, \mathbf{x}_t) = K_t + \mathcal{A}(K_t, \mathbf{x}_t). \quad (715)$$

Here, F is the transition function mapping the current state $(K_t, \mathbf{x}_t)$ into next-period capital. The accumulation function $\mathcal{A}$ determines the net increment, while K_t represents the capital stock carried into the period.

The transition equation can be equivalently written as

$$\Delta K_t \equiv K_{t+1} - K_t = \mathcal{A}(K_t, \mathbf{x}_t). \quad (716)$$

Thus, the capital stock evolves according to

$$K_{t+1} = K_t + \Delta K_t. \quad (717)$$

This expression is the dynamic counterpart of the classical circuit $M - C - M'$, where M' becomes the initial money capital for the next circuit.

The sign of $\mathcal{A}$ determines the direction of the transition:

$$F(K_t, \mathbf{x}_t) \begin{cases} > K_t, & \mathcal{A}(K_t, \mathbf{x}_t) > 0, \\ = K_t, & \mathcal{A}(K_t, \mathbf{x}_t) = 0, \\ < K_t, & \mathcal{A}(K_t, \mathbf{x}_t) < 0. \end{cases} \quad (718)$$

Positive accumulation expands the capital stock, zero accumulation reproduces it, and negative accumulation contracts it.

The transition function may be autonomous or non-autonomous. If the feature vector is fixed at $\mathbf{x}$, then the transition is

$$K_{t+1} = F(K_t, \mathbf{x}). \quad (719)$$

If the feature vector evolves over time, then

$$K_{t+1} = F_t(K_t) \equiv F(K_t, \mathbf{x}_t). \quad (720)$$

In the latter case, capital accumulation depends on the entire historical path of feature vectors.

The derivative of the transition function with respect to capital is

$$\frac{\partial F(K_t, \mathbf{x}_t)}{\partial K_t} = 1 + \frac{\partial \mathcal{A}(K_t, \mathbf{x}_t)}{\partial K_t}. \quad (721)$$

This derivative governs local amplification or damping. If it is large in absolute value, small differences in current capital generate large differences in future capital.

10.2 Recursive Growth

The dynamic system can be iterated forward. Starting from initial capital K_0 , the capital stock at time t is

$$K_t = F^{(t)}(K_0; \{\mathbf{x}_s\}_{s=0}^{t-1}), \quad K_{t+h} = F^{(h)}(K_t; \{\mathbf{x}_{t+s}\}_{s=0}^{h-1}). \quad (722)$$

The notation $F^{(t)}$ denotes the t -fold composition of the transition function along the realized sequence of feature vectors.

Explicitly, the first few iterations are

$$K_1 = F(K_0, \mathbf{x}_0), \quad (723)$$

$$K_2 = F(K_1, \mathbf{x}_1) = F(F(K_0, \mathbf{x}_0), \mathbf{x}_1), \quad (724)$$

and

$$K_3 = F(F(F(K_0, \mathbf{x}_0), \mathbf{x}_1), \mathbf{x}_2). \quad (725)$$

Thus, current capital is a cumulative outcome of past capital and past feature states.

Using the increment representation, recursive accumulation can be written as

$$K_t = K_0 + \sum_{s=0}^{t-1} \mathcal{A}(K_s, \mathbf{x}_s). \quad (726)$$

This expression emphasizes that long-run capital accumulation is the sum of period-by-period increments, each of which depends on the capital stock created by previous increments.

For an h -period forecast from time t , the recursive expression is

$$K_{t+h} = K_t + \sum_{s=0}^{h-1} \mathcal{A}(K_{t+s}, \mathbf{x}_{t+s}). \quad (727)$$

This representation is useful for studying medium-run accumulation paths, counterfactual feature sequences, and crisis trajectories.

If the feature vector is constant, $\mathbf{x}_t = \mathbf{x}$, then

$$K_t = F^{(t)}(K_0; \mathbf{x}). \quad (728)$$

If, in addition, the accumulation function is independent of K_t , then

$$K_t = K_0 + t\mathcal{A}(\mathbf{x}). \quad (729)$$

In contrast, if accumulation is proportional to capital,

$$\mathcal{A}(K_t, \mathbf{x}) = g(\mathbf{x})K_t, \quad (730)$$

then

$$K_t = K_0[1 + g(\mathbf{x})]^t. \quad (731)$$

The dynamic implications therefore depend critically on whether accumulation is additive or proportional in the existing capital stock.

10.3 Growth Rate Representation

The capital growth rate is defined as the capital increment relative to the current capital stock:

$$g_t \equiv \frac{K_{t+1} - K_t}{K_t} = \frac{\mathcal{A}(K_t, \mathbf{x}_t)}{K_t}, \quad \log \frac{K_{t+1}}{K_t} = \log(1 + g_t). \quad (732)$$

This representation translates the level accumulation function into a proportional growth concept. Since

$$K_{t+1} = K_t(1 + g_t), \quad (733)$$

iterating forward gives

$$K_T = K_0 \prod_{t=0}^{T-1} (1 + g_t). \quad (734)$$

Taking logarithms yields

$$\log K_T - \log K_0 = \sum_{t=0}^{T-1} \log(1 + g_t). \quad (735)$$

Thus, long-run proportional accumulation is governed by the cumulative sum of log growth rates. For small g_t , the log growth rate satisfies

$$\log(1 + g_t) \approx g_t. \quad (736)$$

Therefore,

$$\log K_T - \log K_0 \approx \sum_{t=0}^{T-1} g_t. \quad (737)$$

This approximation is useful when period-by-period accumulation rates are small.

The growth-rate representation also clarifies the difference between positive accumulation in levels and positive proportional growth:

$$g_t > 0 \iff \mathcal{A}(K_t, \mathbf{x}_t) > 0 \quad \text{given} \quad K_t > 0. \quad (738)$$

Similarly,

$$g_t = 0 \iff \mathcal{A}(K_t, \mathbf{x}_t) = 0, \quad (739)$$

and

$$g_t < 0 \iff \mathcal{A}(K_t, \mathbf{x}_t) < 0. \quad (740)$$

A necessary condition for positive next-period capital is

$$1 + g_t > 0. \quad (741)$$

Equivalently,

$$\mathcal{A}(K_t, \mathbf{x}_t) > -K_t. \quad (742)$$

If $g_t \leq -1$, the law of motion implies nonpositive capital, which falls outside the admissible domain of capitalist reproduction.

10.4 Steady State

A steady state is a capital-feature configuration that reproduces itself without changing the capital stock. Formally,

$$K^* = F(K^*, \mathbf{x}^*), \quad \mathcal{A}(K^*, \mathbf{x}^*) = 0. \quad (743)$$

The first condition says that next-period capital equals current capital. The second condition shows that the accumulation increment must be zero.

A steady state therefore lies on the stationary reproduction boundary:

$$\mathcal{S}^0 = \{(K, \mathbf{x}) : \mathcal{A}(K, \mathbf{x}) = 0\}. \quad (744)$$

If the system is at $(K^*, \mathbf{x}^*)$, then

$$K_{t+1} = K_t = K^*. \quad (745)$$

For a fixed feature vector $\mathbf{x}^*$, a steady-state capital stock solves

$$\mathcal{A}(K, \mathbf{x}^*) = 0. \quad (746)$$

Depending on the functional form of $\mathcal{A}$, this equation may have no solution, one solution, or multiple solutions:

$$\#\{K : \mathcal{A}(K, \mathbf{x}^*) = 0\} \in \{0, 1, 2, \dots\}. \quad (747)$$

If accumulation is positive below the steady state and negative above it,

$$\mathcal{A}(K, \mathbf{x}^*) > 0 \quad \text{for } K < K^*, \quad \mathcal{A}(K, \mathbf{x}^*) < 0 \quad \text{for } K > K^*, \quad (748)$$

then the steady state is locally attracting. If the inequalities are reversed, the steady state is locally repelling.

In economic terms, a steady state is not necessarily desirable. It may represent balanced reproduction, stagnation, or the limit of accumulation under constraints. Its interpretation depends on the surrounding dynamics and on whether the steady state is stable, unstable, or crisis-prone.

10.5 Stability Condition

Local stability determines whether small deviations from a steady state diminish or amplify over time. In the one-dimensional capital transition, the local stability condition is

$$\left| \frac{\partial F}{\partial K}(K^*, \mathbf{x}^*) \right| < 1, \quad \rho(\nabla_{\mathbf{s}} \mathbf{F}(\mathbf{s}^*)) < 1. \quad (749)$$

The first inequality applies when capital is the only dynamic state variable. The second applies to a multidimensional state vector $\mathbf{s}_t$ with transition map $\mathbf{F}$, where $\rho(\cdot)$ denotes the spectral radius.

Since

$$F(K, \mathbf{x}) = K + \mathcal{A}(K, \mathbf{x}), \quad (750)$$

the derivative with respect to capital is

$$\frac{\partial F}{\partial K} = 1 + \frac{\partial \mathcal{A}}{\partial K}. \quad (751)$$

Therefore, the one-dimensional stability condition becomes

$$\left| 1 + \frac{\partial \mathcal{A}}{\partial K}(K^*, \mathbf{x}^*) \right| < 1. \quad (752)$$

This is equivalent to

$$-2 < \frac{\partial \mathcal{A}}{\partial K}(K^*, \mathbf{x}^*) < 0. \quad (753)$$

The condition has a direct economic meaning. The derivative $\partial \mathcal{A} / \partial K$ must be negative enough to pull capital back toward the steady state, but not so negative that the system overshoots explosively. If

$$\frac{\partial \mathcal{A}}{\partial K}(K^*, \mathbf{x}^*) \geq 0, \quad (754)$$

then higher capital weakly raises future accumulation, producing local positive feedback. If

$$\frac{\partial \mathcal{A}}{\partial K}(K^*, \mathbf{x}^*) \leq -2, \quad (755)$$

then the corrective force is too strong and the system may oscillate or diverge.

For a multidimensional state vector

$$\mathbf{s}_t = (K_t, \mathbf{x}_t^\top)^\top, \quad (756)$$

suppose the system evolves according to

$$\mathbf{s}_{t+1} = \mathbf{F}(\mathbf{s}_t). \quad (757)$$

A steady state satisfies

$$\mathbf{s}^* = \mathbf{F}(\mathbf{s}^*). \quad (758)$$

Local stability requires all eigenvalues of the Jacobian $\nabla_{\mathbf{s}} \mathbf{F}(\mathbf{s}^*)$ to lie inside the unit circle:

$$\rho(\nabla_{\mathbf{s}} \mathbf{F}(\mathbf{s}^*)) < 1. \quad (759)$$

This condition allows capital dynamics to interact with feature dynamics. For example, if features evolve according to

$$\mathbf{x}_{t+1} = \mathbf{G}(K_t, \mathbf{x}_t), \quad (760)$$

then the joint system is

$$\mathbf{F}(\mathbf{s}_t) = \begin{pmatrix} K_t + \mathcal{A}(K_t, \mathbf{x}_t) \\ \mathbf{G}(K_t, \mathbf{x}_t) \end{pmatrix}. \quad (761)$$

The Jacobian of this system determines whether accumulation, technological change, financial conditions, and ecological constraints jointly converge or diverge.

10.6 Continuous-Time Dynamics

The discrete-time accumulation equation has a continuous-time analogue. Let $K(t)$ denote the capital stock at time t and let $\mathbf{x}(t)$ denote the feature vector. The continuous-time law of motion is

$$\dot{K}(t) = \mathcal{A}(K(t), \mathbf{x}(t)), \quad K(t) = K(0) + \int_0^t \mathcal{A}(K(u), \mathbf{x}(u)) du. \quad (762)$$

Here, $\dot{K}(t)$ is the instantaneous rate of change of capital.

The integral expression follows by integrating the differential equation:

$$K(t) - K(0) = \int_0^t \dot{K}(u) du = \int_0^t \mathcal{A}(K(u), \mathbf{x}(u)) du. \quad (763)$$

Thus, capital at time t is the initial capital stock plus the cumulative flow of accumulation over the interval $[0, t]$.

The instantaneous growth rate is

$$g(t) \equiv \frac{\dot{K}(t)}{K(t)} = \frac{\mathcal{A}(K(t), \mathbf{x}(t))}{K(t)}. \quad (764)$$

When $K(t) > 0$, the log capital stock evolves according to

$$\frac{d}{dt} \log K(t) = g(t). \quad (765)$$

Integrating gives

$$\log K(t) - \log K(0) = \int_0^t g(u) du. \quad (766)$$

A continuous-time steady state satisfies

$$\dot{K}(t) = 0 \iff \mathcal{A}(K^*, \mathbf{x}^*) = 0. \quad (767)$$

Local stability in one dimension requires

$$\frac{\partial \mathcal{A}}{\partial K}(K^*, \mathbf{x}^*) < 0. \quad (768)$$

This differs from the discrete-time condition because continuous-time dynamics do not require the derivative to lie within a unit interval; the linearized flow is stable whenever deviations decay exponentially.

If $\mathcal{A}(K, \mathbf{x}) = g(\mathbf{x})K$, then

$$\dot{K}(t) = g(\mathbf{x}(t))K(t). \quad (769)$$

The solution is

$$K(t) = K(0) \exp \left(\int_0^t g(\mathbf{x}(u)) du \right). \quad (770)$$

This is the continuous-time analogue of compound capital accumulation.

10.7 Crisis and Collapse Conditions

The dynamic framework also allows formal definitions of crisis and collapse. A crisis episode occurs when the accumulation function remains negative over a sustained interval:

$$\mathcal{A}(K_t, \mathbf{x}_t) < 0 \quad \forall t \in [t_0, t_1], \quad K_{t_1} < \underline{K}, \quad \exists t : \frac{\partial F}{\partial K} > 1 \text{ and } \mathcal{A} < 0. \quad (771)$$

The first condition states that capital contracts throughout the interval. The second states that the capital stock falls below its lower viability bound. The third identifies a locally unstable contractionary regime.

If

$$\mathcal{A}(K_t, \mathbf{x}_t) < 0 \quad t = t_0, \dots, t_1, \quad (772)$$

then

$$K_{t_1+1} = K_{t_0} + \sum_{t=t_0}^{t_1} \mathcal{A}(K_t, \mathbf{x}_t) < K_{t_0}. \quad (773)$$

The capital stock declines cumulatively over the crisis interval.

A collapse occurs when capital falls below the minimum viable threshold:

$$K_t < \underline{K}. \quad (774)$$

At this point, the modeled capital circuit is no longer self-sustaining within the admissible domain. The process may require external injection, restructuring, devaluation, or institutional intervention to re-enter the viable state space.

Local crisis amplification occurs when contraction is combined with dynamic instability:

$$\mathcal{A}(K_t, \mathbf{x}_t) < 0 \quad \text{and} \quad \frac{\partial F}{\partial K}(K_t, \mathbf{x}_t) > 1. \quad (775)$$

Since

$$\frac{\partial F}{\partial K} = 1 + \frac{\partial \mathcal{A}}{\partial K}, \quad (776)$$

this condition is equivalent to

$$\frac{\partial \mathcal{A}}{\partial K}(K_t, \mathbf{x}_t) > 0. \quad (777)$$

In such a regime, a higher capital stock reinforces the magnitude of the transition, and deviations may be amplified rather than dampened.

A debt-driven collapse may be represented by a leverage threshold:

$$\ell_t > \tau_\ell \implies \mathcal{A}(K_t, \mathbf{x}_t) < 0. \quad (778)$$

An ecological collapse may be represented by an ecological threshold:

$$E_t > \tau_E \implies \mathcal{A}(K_t, \mathbf{x}_t) < 0. \quad (779)$$

A demand-realization crisis may be represented by

$$Q_t(\mathbf{x}_t) > D_t(P_t) \implies \mathcal{A}(K_t, \mathbf{x}_t) < 0. \quad (780)$$

These examples show how financial, ecological, and market constraints can enter the dynamic crisis condition through the accumulation function.

The crisis region of the state space can be defined as

$$\mathcal{C} = \{(K, \mathbf{x}) : \mathcal{A}(K, \mathbf{x}) < 0\}. \quad (781)$$

The collapse region is

$$\mathcal{C}_{\text{collapse}} = \{(K, \mathbf{x}) : K + \mathcal{A}(K, \mathbf{x}) < \underline{K}\}. \quad (782)$$

The unstable crisis region is

$$\mathcal{C}_{\text{unstable}} = \left\{ (K, \mathbf{x}) : \mathcal{A}(K, \mathbf{x}) < 0, \quad \left| \frac{\partial F}{\partial K}(K, \mathbf{x}) \right| > 1 \right\}. \quad (783)$$

Dynamic capital accumulation can therefore be summarized as a transition system with expansionary, stationary, contractionary, and collapse regions:

$$K_{t+1} = F(K_t, \mathbf{x}_t) = K_t + \mathcal{A}(K_t, \mathbf{x}_t). \quad (784)$$

The long-run path of capital is determined not only by the sign of the current increment, but also by the feedback structure of the transition function and the evolution of the feature vector over time.

11 Econometric Specification

The theoretical accumulation function becomes empirically operational once it is embedded in an econometric specification. The purpose of this section is to translate the multi-feature capital accumulation formula into models that can be estimated using panel, time-series, or cross-sectional data. The index i denotes an observational unit, such as a firm, sector, region, or national economy, while t denotes time.

11.1 Panel Data Model

The most general panel specification writes next-period capital as current capital plus an accumulation function and an error term:

$$K_{i,t+1} = K_{it} + \mathcal{A}(K_{it}, \mathbf{x}_{it}) + u_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T. \quad (785)$$

Here, K_{it} is the capital stock of unit i at time t , $\mathbf{x}_{it}$ is the vector of accumulation-relevant features, and u_{it} collects unobserved shocks, measurement error, omitted variables, and other idiosyncratic disturbances.

Equivalently, the model may be written in first-difference form as

$$\Delta K_{it} \equiv K_{i,t+1} - K_{it} = \mathcal{A}(K_{it}, \mathbf{x}_{it}) + u_{it}. \quad (786)$$

This form makes clear that the econometric object of interest is the conditional law governing the increment of capital.

The maintained conditional mean restriction is

$$\mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}] = 0. \quad (787)$$

Under this restriction, the accumulation function has the interpretation

$$\mathcal{A}(K_{it}, \mathbf{x}_{it}) = \mathbb{E}[K_{i,t+1} - K_{it} \mid K_{it}, \mathbf{x}_{it}]. \quad (788)$$

Thus, $\mathcal{A}$ is the conditional mean of the capital increment.

The panel structure is important because capital accumulation is rarely homogeneous across units. Firms, sectors, regions, and national economies may differ in production technology, institutional environment, market power, financial access, labor relations, and historical position. These differences can be introduced either through observed features $\mathbf{x}_{it}$ or through unobserved unit-specific components in the error term.

11.2 Linear Fixed-Effects Model

A first estimable benchmark is the linear fixed-effects model:

$$K_{i,t+1} = \alpha_i + \tau_t + \rho K_{it} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + u_{it}. \quad (789)$$

The parameter α_i captures time-invariant unit heterogeneity, while τ_t captures common time shocks. The coefficient ρ measures capital persistence, and $\boldsymbol{\beta}$ measures the marginal contribution of the feature vector.

Subtracting K_{it} from both sides gives the implied accumulation equation:

$$\Delta K_{it} = \alpha_i + \tau_t + (\rho - 1)K_{it} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + u_{it}. \quad (790)$$

Thus, the implied accumulation function is

$$\mathcal{A}(K_{it}, \mathbf{x}_{it}) = \alpha_i + \tau_t + (\rho - 1)K_{it} + \boldsymbol{\beta}^\top \mathbf{x}_{it}. \quad (791)$$

The fixed effects allow the model to control for persistent differences across units:

$$\alpha_i = \text{time-invariant accumulation capacity of unit } i. \quad (792)$$

The time effects capture common shocks:

$$\tau_t = \text{historical, macroeconomic, or institutional shock common to all units.} \quad (793)$$

The identifying condition is

$$\mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] = 0. \quad (794)$$

This condition allows K_{it} and $\mathbf{x}_{it}$ to be correlated with unobserved unit and time effects, but not with the remaining idiosyncratic disturbance.

After removing unit and time effects, the within-transformed model becomes

$$\tilde{K}_{i,t+1} = \rho \tilde{K}_{it} + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_{it} + \tilde{u}_{it}, \quad (795)$$

where tildes denote residuals after projecting out unit and time effects. The corresponding moment condition is

$$\mathbb{E}[\tilde{\mathbf{w}}_{it} \tilde{u}_{it}] = \mathbf{0}, \quad \tilde{\mathbf{w}}_{it} = (\tilde{K}_{it}, \tilde{\mathbf{x}}_{it}^\top)^\top. \quad (796)$$

11.3 Interaction Fixed-Effects Model

The linear fixed-effects model assumes additive feature effects. To allow pairwise nonlinear coupling, we introduce interaction terms:

$$K_{i,t+1} = \alpha_i + \tau_t + \rho K_{it} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + \sum_{j<\ell} \gamma_{j\ell} x_{jit} x_{lit} + u_{it}. \quad (797)$$

The coefficients $\gamma_{j\ell}$ measure whether features x_j and x_ℓ are complementary or substitutive in the accumulation process.

The implied accumulation equation is

$$\Delta K_{it} = \alpha_i + \tau_t + (\rho - 1)K_{it} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + \sum_{j<\ell} \gamma_{j\ell} x_{jit} x_{lit} + u_{it}. \quad (798)$$

The marginal effect of feature x_{jit} is

$$\frac{\partial \Delta K_{it}}{\partial x_{jit}} = \beta_j + \sum_{\ell \neq j} \gamma_{j\ell} x_{lit}, \quad (799)$$

where $\gamma_{j\ell} = \gamma_{\ell j}$ under symmetric indexing.

The cross-partial effect is

$$\frac{\partial^2 \Delta K_{it}}{\partial x_{jit} \partial x_{lit}} = \gamma_{j\ell}. \quad (800)$$

Thus,

$$\gamma_{j\ell} > 0 \implies \text{complementarity}, \quad \gamma_{j\ell} < 0 \implies \text{substitution}. \quad (801)$$

This specification is a direct econometric counterpart of the second-order coupling formula. It remains linear in parameters but nonlinear in features. Therefore, it can be estimated using standard fixed-effects methods after constructing the interaction regressors.

11.4 Dynamic Panel Model

Capital accumulation is typically persistent. To capture dynamic dependence beyond the current capital stock, we introduce a lagged capital term:

$$K_{i,t+1} = \alpha_i + \tau_t + \phi K_{it} + \psi K_{i,t-1} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + u_{it}. \quad (802)$$

The coefficient ϕ measures the effect of current capital on future capital, while ψ captures additional persistence from lagged capital.

The implied accumulation equation is

$$\Delta K_{it} = \alpha_i + \tau_t + (\phi - 1)K_{it} + \psi K_{i,t-1} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + u_{it}. \quad (803)$$

This model allows the current increment to depend on both the current capital stock and the previous capital path.

The dynamic panel specification can capture delayed adjustment, investment gestation, depreciation dynamics, and persistent accumulation regimes. It is particularly useful when capital responds gradually to changes in technology, finance, labor organization, or demand.

The presence of lagged dependent variables and fixed effects creates a well-known endogeneity concern. Even if

$$\mathbb{E}[u_{it} \mid K_{it}, K_{i,t-1}, \mathbf{x}_{it}, \alpha_i, \tau_t] = 0, \quad (804)$$

within transformation may induce correlation between transformed lagged capital and transformed errors in short panels. A common solution is to use internal instruments based on deeper lags:

$$\mathbb{E}[K_{i,t-s}u_{it}] = 0, \quad s \geq 2. \quad (805)$$

This provides a route to instrumental-variable or GMM estimation.

The dynamic stability of the linear homogeneous part depends on the roots of the characteristic equation:

$$\lambda^2 - \phi\lambda - \psi = 0. \quad (806)$$

Local stability requires the roots to lie inside the unit circle:

$$|\lambda_1| < 1, \quad |\lambda_2| < 1. \quad (807)$$

11.5 Threshold Regression Model

To capture regime-dependent accumulation, we specify a panel threshold model:

$$K_{i,t+1} = \alpha_i + \tau_t + \mathbf{x}_{it}^\top \boldsymbol{\beta}_1 \mathbf{1}\{q_{it} \leq \tau\} + \mathbf{x}_{it}^\top \boldsymbol{\beta}_2 \mathbf{1}\{q_{it} > \tau\} + u_{it}. \quad (808)$$

Here, q_{it} is the threshold variable and τ is the threshold value. The slope vector is $\boldsymbol{\beta}_1$ in the low-threshold regime and $\boldsymbol{\beta}_2$ in the high-threshold regime.

The implied marginal effect of feature x_j is

$$\frac{\partial K_{i,t+1}}{\partial x_{jit}} = \beta_{1j} \mathbf{1}\{q_{it} \leq \tau\} + \beta_{2j} \mathbf{1}\{q_{it} > \tau\}. \quad (809)$$

Thus, the same feature may have different accumulation effects in different regimes.

The null hypothesis of no threshold effect is

$$H_0 : \boldsymbol{\beta}_1 = \boldsymbol{\beta}_2. \quad (810)$$

Under the alternative,

$$H_1 : \boldsymbol{\beta}_1 \neq \boldsymbol{\beta}_2. \quad (811)$$

This tests whether the accumulation law changes when q_{it} crosses τ .

If the threshold is unknown, it can be estimated by minimizing the residual sum of squares:

$$\hat{\tau} = \arg \min_{\tau \in \mathcal{T}} \sum_{i=1}^N \sum_{t=1}^T \hat{u}_{it}(\tau)^2. \quad (812)$$

The search set $\mathcal{T}$ is typically trimmed to ensure that both regimes contain enough observations:

$$\Pr(q_{it} \leq \tau) \geq \pi_0, \quad \Pr(q_{it} > \tau) \geq \pi_0, \quad (813)$$

for some small trimming constant $\pi_0 > 0$.

In applications, q_{it} may be leverage, ecological burden, market concentration, network size, capacity utilization, debt-service ratio, or policy regime intensity. The threshold model is therefore an empirical version of the regime-shift framework.

11.6 Nonparametric Regression Model

The most flexible empirical specification leaves the conditional mean function unspecified:

$$K_{i,t+1} = m(K_{it}, \mathbf{x}_{it}) + u_{it}, \quad \mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}] = 0. \quad (814)$$

Here, $m(\cdot)$ is the conditional mean of next-period capital:

$$m(K_{it}, \mathbf{x}_{it}) = \mathbb{E}[K_{i,t+1} \mid K_{it}, \mathbf{x}_{it}]. \quad (815)$$

The corresponding accumulation function is

$$\mathcal{A}(K_{it}, \mathbf{x}_{it}) = m(K_{it}, \mathbf{x}_{it}) - K_{it}. \quad (816)$$

Thus, estimating m is equivalent to estimating the accumulation law.

A sieve approximation writes

$$m(K, \mathbf{x}) \approx \sum_{r=1}^R \theta_r \psi_r(K, \mathbf{x}), \quad (817)$$

where $\{\psi_r\}_{r=1}^R$ are basis functions. The estimator solves

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{i=1}^N \sum_{t=1}^T \left[K_{i,t+1} - \sum_{r=1}^R \theta_r \psi_r(K_{it}, \mathbf{x}_{it}) \right]^2. \quad (818)$$

The estimated accumulation function is then

$$\hat{\mathcal{A}}(K, \mathbf{x}) = \hat{m}(K, \mathbf{x}) - K. \quad (819)$$

A regularized nonparametric estimator may be written as

$$\hat{m} = \arg \min_{m \in \mathcal{M}} \left\{ \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T [K_{i,t+1} - m(K_{it}, \mathbf{x}_{it})]^2 + \lambda \mathcal{P}(m) \right\}. \quad (820)$$

The penalty $\mathcal{P}(m)$ controls smoothness or complexity.

The nonparametric model is useful when theory suggests nonlinearities but does not specify their exact functional form. Its main advantage is flexibility; its main cost is the need for stronger data support and careful regularization, especially when the dimension of $\mathbf{x}_{it}$ is large.

11.7 Error Decomposition

The error term may contain several distinct components. A standard decomposition separates unit effects, time effects, and idiosyncratic shocks:

$$u_{it} = \mu_i + \lambda_t + \varepsilon_{it}, \quad u_{it} = \nu_{it} + \xi_{it}, \quad \mathbb{E}[\varepsilon_{it} \varepsilon_{js}] = \sigma_{ij,ts}. \quad (821)$$

Here, μ_i is a unit-specific component, λ_t is a common time-specific component, and ε_{it} is an idiosyncratic disturbance.

The decomposition

$$u_{it} = \mu_i + \lambda_t + \varepsilon_{it} \quad (822)$$

captures persistent heterogeneity and common shocks. The unit component μ_i may reflect persistent differences in technology, institutions, sectoral position, or organizational capacity. The time component λ_t may reflect macroeconomic shocks, policy shifts, crises, or global technological changes.

The alternative decomposition

$$u_{it} = \nu_{it} + \xi_{it} \quad (823)$$

can be used to distinguish structural shocks from measurement error. For example, ν_{it} may represent an economically meaningful accumulation shock, while ξ_{it} may represent data noise or accounting error.

The covariance structure is allowed to be general:

$$\mathbb{E}[\varepsilon_{it}\varepsilon_{js}] = \sigma_{ij,ts}. \quad (824)$$

This notation permits serial correlation, cross-sectional dependence, and heteroskedasticity. Serial correlation occurs when

$$\sigma_{ii,ts} \neq 0 \quad \text{for } t \neq s. \quad (825)$$

Cross-sectional dependence occurs when

$$\sigma_{ij,tt} \neq 0 \quad \text{for } i \neq j. \quad (826)$$

Heteroskedasticity occurs when

$$\sigma_{ii,tt} \neq \sigma_{jj,ss} \quad \text{for some } (i, t) \neq (j, s). \quad (827)$$

These error structures matter for inference. If errors are heteroskedastic or correlated within units, robust or clustered standard errors are required:

$$\widehat{\text{Var}}(\hat{\theta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{i=1}^N \mathbf{X}_i^\top \hat{\mathbf{u}}_i \hat{\mathbf{u}}_i^\top \mathbf{X}_i \right) (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (828)$$

If cross-sectional dependence is important, two-way clustered or spatially robust covariance estimators may be needed.

The econometric specification therefore connects the theoretical accumulation function to estimable models. Linear fixed effects provide a transparent benchmark; interaction models capture nonlinear coupling; dynamic panels capture persistence; threshold regressions capture regime dependence; nonparametric regressions capture flexible accumulation laws; and error decompositions clarify the assumptions required for valid inference:

$$K_{i,t+1} = K_{it} + \mathcal{A}(K_{it}, \mathbf{x}_{it}) + u_{it}. \quad (829)$$

12 Identification and Endogeneity

The econometric specifications in the previous section describe how the capital accumulation function can be estimated from data. Identification, however, requires more than a functional form. It requires assumptions under which variation in the observed features can be interpreted as variation in the determinants of capital accumulation rather than as the consequence of unobserved shocks, reverse causality, omitted variables, or measurement error. This section states the main identification conditions and the corresponding remedies when exogeneity fails.

12.1 Exogeneity Condition

The baseline identification condition is conditional exogeneity:

$$\mathbb{E}[u_{it} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] = 0. \quad (830)$$

This condition states that, after conditioning on current capital, observed features, unit fixed effects, and time effects, the remaining disturbance has zero conditional mean.

Under Equation (830), the accumulation function can be interpreted as the conditional mean component of capital dynamics:

$$\mathbb{E}[K_{i,t+1} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] = K_{it} + \mathcal{A}(K_{it}, \mathbf{x}_{it}) + \alpha_i + \tau_t. \quad (831)$$

Equivalently,

$$\mathbb{E}[\Delta K_{it} \mid K_{it}, \mathbf{x}_{it}, \alpha_i, \tau_t] = \mathcal{A}(K_{it}, \mathbf{x}_{it}) + \alpha_i + \tau_t. \quad (832)$$

For a linear fixed-effects specification,

$$K_{i,t+1} = \alpha_i + \tau_t + \rho K_{it} + \boldsymbol{\beta}^\top \mathbf{x}_{it} + u_{it}, \quad (833)$$

the exogeneity condition implies

$$\mathbb{E}[\mathbf{x}_{it} u_{it} \mid \alpha_i, \tau_t] = \mathbf{0}, \quad \mathbb{E}[K_{it} u_{it} \mid \alpha_i, \tau_t] = 0. \quad (834)$$

After removing fixed effects, the identifying moment condition becomes

$$\mathbb{E}[\tilde{\mathbf{w}}_{it} \tilde{u}_{it}] = \mathbf{0}, \quad \tilde{\mathbf{w}}_{it} = (\tilde{K}_{it}, \tilde{\mathbf{x}}_{it}^\top)^\top. \quad (835)$$

The exogeneity condition is strong. It requires that no unobserved time-varying determinant of capital accumulation is correlated with the observed features. It also rules out simultaneity between feature choice and future capital outcomes. When this condition holds, the estimated coefficients can be interpreted as structural or conditional marginal effects. When it fails, the estimates are generally biased and cannot be given a causal accumulation interpretation.

12.2 Endogeneity

Endogeneity arises when one or more observed features are correlated with the unobserved disturbance:

$$\text{Cov}(x_{jit}, u_{it}) \neq 0, \quad \mathbb{E}[u_{it} \mid \mathbf{x}_{it}] \neq 0. \quad (836)$$

In this case, variation in x_{jit} reflects not only the feature's systematic contribution to accumulation, but also unobserved shocks contained in u_{it} .

For the linear model

$$Y_{it} = \mathbf{x}_{it}^\top \boldsymbol{\beta} + u_{it}, \quad (837)$$

ordinary least squares identifies

$$\tilde{\boldsymbol{\beta}} = \left(\mathbb{E}[\mathbf{x}_{it} \mathbf{x}_{it}^\top] \right)^{-1} \mathbb{E}[\mathbf{x}_{it} Y_{it}]. \quad (838)$$

Substituting Equation (837) gives

$$\tilde{\boldsymbol{\beta}} = \boldsymbol{\beta} + \left(\mathbb{E}[\mathbf{x}_{it} \mathbf{x}_{it}^\top] \right)^{-1} \mathbb{E}[\mathbf{x}_{it} u_{it}]. \quad (839)$$

Thus, the bias term vanishes only when

$$\mathbb{E}[\mathbf{x}_{it}u_{it}] = \mathbf{0}. \quad (840)$$

Endogeneity may arise from several sources:

endogeneity $\in$ {reverse causality, omitted variables, measurement error, simultaneity, selection, anticipation}. (841)

Each source threatens the interpretation of β or $\mathcal{A}$ as a structural accumulation relationship.

In the capital accumulation context, endogeneity is especially likely because many features are chosen by firms, states, or institutions in response to expected future accumulation. Investment, leverage, labor intensity, technology adoption, and ecological intensity are not randomly assigned. They are strategic or institutional choices shaped by expectations, constraints, and shocks.

12.3 Reverse Causality

Reverse causality occurs when the feature vector is affected by future or expected capital accumulation. A simple representation is

$$x_{jit} = \pi_0 + \pi_1 K_{i,t+1} + v_{jit}, \quad \text{Cov}(v_{jit}, u_{it}) \neq 0. \quad (842)$$

Here, feature x_{jit} is not predetermined with respect to the capital outcome. Instead, it responds to the same future accumulation outcome that the model seeks to explain.

If the capital equation is

$$K_{i,t+1} = \beta_j x_{jit} + u_{it}, \quad (843)$$

and x_{jit} depends on $K_{i,t+1}$, then x_{jit} inherits part of the variation in u_{it} . This creates correlation between the regressor and the error:

$$\text{Cov}(x_{jit}, u_{it}) \neq 0. \quad (844)$$

Reverse causality is natural in accumulation settings. For example, firms expecting high future accumulation may increase investment, hire more labor, borrow more, or adopt new technologies before the capital expansion is observed. Similarly, regions expecting industrial growth may expand infrastructure or loosen credit constraints. In these cases, observed features partly reflect anticipated accumulation rather than exogenous determinants of it.

One way to express anticipatory behavior is

$$x_{jit} = \pi_0 + \pi_1 \mathbb{E}_t[K_{i,t+1}] + v_{jit}. \quad (845)$$

If expectations are correlated with unobserved accumulation shocks, then

$$\text{Cov}(x_{jit}, u_{it}) \neq 0. \quad (846)$$

To address reverse causality, empirical designs require predetermined features, valid instruments, lag structures, natural experiments, or policy variation that shifts the feature without directly shifting unobserved capital accumulation shocks.

12.4 Omitted Variable Bias

Omitted variable bias arises when an unobserved determinant of capital accumulation is correlated with the included features. Suppose the true model is

$$K_{i,t+1} = \mathbf{x}_{it}^\top \boldsymbol{\beta} + \delta w_{it} + u_{it}, \quad \tilde{\boldsymbol{\beta}} = \boldsymbol{\beta} + \delta \left(\mathbb{E}[\mathbf{x}\mathbf{x}^\top] \right)^{-1} \mathbb{E}[\mathbf{x}w]. \quad (847)$$

Here, w_{it} is an omitted determinant of future capital. If w_{it} is correlated with $\mathbf{x}_{it}$ and $\delta \neq 0$, then the estimated coefficient vector is biased.

The bias term is

$$\text{Bias}(\tilde{\boldsymbol{\beta}}) = \delta \left(\mathbb{E}[\mathbf{x}\mathbf{x}^\top] \right)^{-1} \mathbb{E}[\mathbf{x}w]. \quad (848)$$

Thus, omitted variable bias disappears if either

$$\delta = 0 \quad (849)$$

or

$$\mathbb{E}[\mathbf{x}w] = \mathbf{0}. \quad (850)$$

The first condition means the omitted variable has no effect on capital. The second means the omitted variable is uncorrelated with the included features.

For a scalar feature x and scalar omitted variable w , the bias has the familiar form

$$\tilde{\beta} - \beta = \delta \frac{\text{Cov}(x, w)}{\text{Var}(x)}. \quad (851)$$

The direction of the bias is determined by the sign of δ and the sign of $\text{Cov}(x, w)$.

In capital accumulation studies, omitted variables may include institutional capacity, sectoral bargaining power, monopoly position, unobserved technological quality, credit access, state support, global demand shocks, or political conditions. If these factors are correlated with observed features, then the estimated accumulation effects partly absorb their influence.

Fixed effects mitigate omitted variable bias only for omitted factors that are time invariant at the unit level or common across units at a given time:

$$w_{it} = \mu_i + \lambda_t \implies \text{controlled by fixed effects.} \quad (852)$$

They do not solve bias from time-varying unit-specific omitted factors:

$$w_{it} \neq \mu_i + \lambda_t. \quad (853)$$

12.5 Instrumental Variables

When features are endogenous, instrumental variables provide one route to identification. Let $\mathbf{z}_{it}$ denote a vector of instruments. Valid instruments satisfy the moment, relevance, and exclusion conditions:

$$\mathbb{E}[\mathbf{z}_{it}u_{it}] = \mathbf{0}, \quad \text{rank} \left(\mathbb{E}[\mathbf{z}_{it}\mathbf{x}_{it}^\top] \right) = d, \quad \mathbf{z}_{it} \perp K_{i,t+1}(\mathbf{x}) \forall \mathbf{x}, \quad \hat{\boldsymbol{\beta}}_{2SLS} = (X^\top P_Z X)^{-1} X^\top P_Z Y. \quad (854)$$

The first condition is instrument exogeneity. The second is instrument relevance. The third is an exclusion or independence condition stated in potential-outcome notation. The final expression is the two-stage least squares estimator.

The projection matrix onto the instrument space is

$$P_Z = Z(Z^\top Z)^{-1}Z^\top. \quad (855)$$

The first stage projects the endogenous features onto the instruments:

$$X = Z\Pi + V, \quad \hat{X} = P_Z X. \quad (856)$$

The second stage regresses the outcome on the fitted values:

$$Y = \hat{X}\beta + u. \quad (857)$$

In population form, the IV estimand satisfies

$$\mathbb{E}[\mathbf{z}_{it}(Y_{it} - \mathbf{x}_{it}^\top \beta)] = \mathbf{0}. \quad (858)$$

If the number of instruments equals the number of endogenous regressors, the identified coefficient is

$$\beta_{IV} = \left(\mathbb{E}[\mathbf{z}_{it}\mathbf{x}_{it}^\top] \right)^{-1} \mathbb{E}[\mathbf{z}_{it}Y_{it}]. \quad (859)$$

In overidentified models, where the number of instruments exceeds the number of endogenous regressors, the GMM estimator solves

$$\hat{\beta}_{GMM} = \arg \min_{\beta} \left[\frac{1}{n} \sum_i \mathbf{z}_i(Y_i - \mathbf{x}_i^\top \beta) \right]^\top W \left[\frac{1}{n} \sum_i \mathbf{z}_i(Y_i - \mathbf{x}_i^\top \beta) \right]. \quad (860)$$

Instrumental variables are credible only when the instruments affect capital accumulation through the endogenous features and not through other channels. In this framework, a valid instrument shifts the feature vector $\mathbf{x}_{it}$ while remaining orthogonal to the unobserved determinants of $K_{i,t+1}$.

12.6 Difference-in-Differences Form

Difference-in-differences identifies the effect of a treatment or policy change by comparing treated and control units before and after an intervention:

$$Y_{it} = \alpha_i + \tau_t + \theta(D_i \times Post_t) + \mathbf{w}_{it}^\top \gamma + u_{it}, \quad \theta = \Delta \bar{Y}_T - \Delta \bar{Y}_C, \quad \mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 1] = \mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 0] \quad (861)$$

Here, D_i indicates treated units, $Post_t$ indicates the post-treatment period, and θ is the treatment effect under the parallel-trends assumption.

The two changes in the difference-in-differences contrast are

$$\Delta \bar{Y}_T = \bar{Y}_{T,post} - \bar{Y}_{T,pre}, \quad \Delta \bar{Y}_C = \bar{Y}_{C,post} - \bar{Y}_{C,pre}. \quad (862)$$

The estimator subtracts the control-group change from the treated-group change:

$$\hat{\theta}_{DID} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre}). \quad (863)$$

In the capital accumulation setting, the outcome may be capital, the capital increment, or the growth rate:

$$Y_{it} \in \{K_{i,t+1}, \Delta K_{it}, g_{it}\}. \quad (864)$$

The treatment may be a policy change, credit reform, environmental regulation, industrial subsidy, tax change, labor-market reform, or trade shock.

The identifying assumption is parallel trends in untreated potential outcomes:

$$\mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 1] = \mathbb{E}[Y_{it}(0) - Y_{is}(0) \mid D_i = 0]. \quad (865)$$

This condition states that, absent treatment, treated and control units would have experienced the same average change in the outcome.

If the treatment changes one or more accumulation features, then DID can be interpreted as identifying the causal effect of a policy-induced shift in the accumulation environment:

$$\theta = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid D_i = 1]. \quad (866)$$

This provides a quasi-experimental route to identifying part of the accumulation function.

12.7 Regression Discontinuity Form

Regression discontinuity identifies causal effects when treatment changes discontinuously at a cutoff of a running variable. The basic specification is

$$Y_i = \alpha + \tau \mathbf{1}\{r_i \geq c\} + f_-(r_i - c) \mathbf{1}\{r_i < c\} + f_+(r_i - c) \mathbf{1}\{r_i \geq c\} + u_i, \quad \lim_{r \downarrow c} \mathbb{E}[Y_i(0) \mid r_i = r] = \lim_{r \uparrow c} \mathbb{E}[Y_i(0) \mid r_i = r]. \quad (867)$$

Here, r_i is the running variable, c is the cutoff, and τ is the discontinuity in the outcome at the cutoff.

The treatment indicator is

$$D_i = \mathbf{1}\{r_i \geq c\}. \quad (868)$$

The treatment effect at the cutoff is

$$\tau = \lim_{r \downarrow c} \mathbb{E}[Y_i \mid r_i = r] - \lim_{r \uparrow c} \mathbb{E}[Y_i \mid r_i = r]. \quad (869)$$

The key identifying assumption is continuity of untreated potential outcomes at the cutoff:

$$\lim_{r \downarrow c} \mathbb{E}[Y_i(0) \mid r_i = r] = \lim_{r \uparrow c} \mathbb{E}[Y_i(0) \mid r_i = r]. \quad (870)$$

This means that units just below and just above the cutoff are comparable except for treatment assignment.

In the accumulation framework, the running variable may be firm size, leverage, pollution level, eligibility score, asset value, employment level, or policy threshold. The outcome may again be capital, capital increment, or capital growth:

$$Y_i \in \{K_{i,t+1}, \Delta K_i, g_i\}. \quad (871)$$

Regression discontinuity is especially relevant when institutional rules create sharp thresholds in access to subsidies, credit, regulation, taxation, or environmental obligations. In such settings, the RD design can identify the local effect of crossing a policy threshold on capital accumulation.

12.8 Moment Conditions

A general identification framework can be written in terms of moment conditions. Let W_i denote the observed data, let θ_0 be the true parameter vector, and let $\mathbf{g}$ be a vector of moments:

$$\mathbb{E}[\mathbf{g}(W_i, \theta_0)] = \mathbf{0}, \quad \mathbf{g}(W_i, \theta) = \mathbf{z}_i(Y_i - h(X_i; \theta)), \quad \text{rank} \left(\mathbb{E} \left[\frac{\partial \mathbf{g}(W_i, \theta_0)}{\partial \theta^\top} \right] \right) = \dim(\theta). \quad (872)$$

Here, $h(X_i; \theta)$ is the structural or conditional-mean model, and $\mathbf{z}_i$ is a vector of instruments or conditioning variables.

The moment restriction states that the residual is orthogonal to the instrument vector:

$$\mathbb{E}[\mathbf{z}_i(Y_i - h(X_i; \theta_0))] = \mathbf{0}. \quad (873)$$

This nests ordinary least squares, fixed effects, instrumental variables, and generalized method of moments.

Identification requires that the moments vary sufficiently with the parameters. The rank condition

$$\text{rank} \left(\mathbb{E} \left[\frac{\partial \mathbf{g}(W_i, \theta_0)}{\partial \theta^\top} \right] \right) = \dim(\theta) \quad (874)$$

ensures local identification. If this matrix is rank deficient, multiple parameter values may satisfy the same moment restrictions.

The sample analogue of the moment condition is

$$\bar{\mathbf{g}}_n(\theta) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(W_i, \theta). \quad (875)$$

A GMM estimator solves

$$\hat{\theta}_{GMM} = \arg \min_{\theta} \bar{\mathbf{g}}_n(\theta)^\top W_n \bar{\mathbf{g}}_n(\theta), \quad (876)$$

where W_n is a positive semidefinite weighting matrix.

In the accumulation model, one may set

$$Y_i = \Delta K_i, \quad h(X_i; \theta) = \mathcal{A}(K_i, \mathbf{x}_i; \theta), \quad (877)$$

so that

$$\mathbf{g}(W_i, \theta) = \mathbf{z}_i [\Delta K_i - \mathcal{A}(K_i, \mathbf{x}_i; \theta)]. \quad (878)$$

The identifying restriction is then

$$\mathbb{E} \{ \mathbf{z}_i [\Delta K_i - \mathcal{A}(K_i, \mathbf{x}_i; \theta_0)] \} = \mathbf{0}. \quad (879)$$

The moment-condition framework makes explicit the logic of identification: the theoretical accumulation function is identified only when the residual variation left after imposing the model is orthogonal to valid conditioning variables or instruments. When this orthogonality is credible, the estimated accumulation function can be interpreted structurally. When it is not, the model should be treated as descriptive unless supported by a stronger research design:

$$\text{identification} = \text{functional structure} + \text{valid orthogonality} + \text{rank condition}. \quad (880)$$

13 Infinite-Dimensional Capital Formula

The finite-dimensional framework represents the accumulation environment by a vector of features. This is appropriate when the determinants of accumulation can be summarized by a finite list of variables. In many settings, however, the relevant state is more naturally represented as a function: a spatial distribution of production, a continuum of technologies, a sectoral density of capital, a skill distribution, a network of tasks, or an ecological field. This section extends the capital accumulation formula from finite-dimensional vectors to infinite-dimensional function spaces.

13.1 Measure Space

Let the underlying feature domain be a measure space:

$$(\Omega, \mathcal{F}, \mu), \quad x_t : \Omega \rightarrow \mathbb{R}, \quad x_t \in L_2(\Omega, \mu). \quad (881)$$

Here, Ω denotes the domain over which accumulation-relevant features are distributed. The sigma-algebra $\mathcal{F}$ specifies the measurable subsets of Ω , and μ is a measure assigning size or weight to those subsets.

The element $\omega \in \Omega$ may represent a sector, location, technology, task, input type, skill category, ecological state, or institutional dimension. The function $x_t(\omega)$ gives the intensity or value of the feature at point ω in period t . Thus, instead of observing a finite vector

$$\mathbf{x}_t = (x_{1t}, \dots, x_{dt})^\top, \quad (882)$$

we observe a feature function

$$x_t = \{x_t(\omega) : \omega \in \Omega\}. \quad (883)$$

The square-integrability condition

$$x_t \in L_2(\Omega, \mu) \quad (884)$$

requires

$$\int_{\Omega} |x_t(\omega)|^2 d\mu(\omega) < \infty. \quad (885)$$

This condition ensures that the feature function has finite energy or finite second moment over the domain. Economically, it rules out feature distributions with unbounded total intensity.

The finite-dimensional model is nested as a special case. If

$$\Omega = \{1, \dots, d\}, \quad \mu(\{j\}) = 1, \quad (886)$$

then

$$L_2(\Omega, \mu) \simeq \mathbb{R}^d, \quad (887)$$

and the function $x_t(\omega)$ reduces to the vector component x_{jt} . Thus, the infinite-dimensional formulation is a genuine extension rather than a separate model.

The measure μ controls the economic weight assigned to each part of the domain. For example, if Ω is a set of sectors, then μ may weight sectors by size, employment, output, or strategic importance. If Ω is geographical space, then μ may correspond to area, population, or regional economic mass. If Ω is a technology space, then μ may weight technologies by prevalence or productive relevance.

13.2 Feature Function

The feature function can be denoted by

$$\varphi_t(\omega) = x_t(\omega), \quad \boldsymbol{\varphi}_t = (\varphi_{1t}, \varphi_{2t}, \dots). \quad (888)$$

The notation φ_t emphasizes that the state of accumulation is now a function over the domain Ω . The sequence $\boldsymbol{\varphi}_t = (\varphi_{1t}, \varphi_{2t}, \dots)$ may represent an expansion of the feature function in a basis or a countable collection of functional components.

For example, if $\{\phi_m\}_{m=1}^\infty$ is an orthonormal basis of $L_2(\Omega, \mu)$, the feature function can be expanded as

$$x_t(\omega) = \sum_{m=1}^{\infty} \xi_{mt} \phi_m(\omega), \quad (889)$$

where the coefficient sequence is

$$\xi_{mt} = \langle x_t, \phi_m \rangle_{\mathcal{H}}. \quad (890)$$

The vector

$$\boldsymbol{\varphi}_t = (\xi_{1t}, \xi_{2t}, \dots) \quad (891)$$

then provides a countably infinite representation of the feature state.

This notation is useful because it connects the functional formulation to finite-dimensional approximations. A truncated representation uses only the first M basis coefficients:

$$x_t^{(M)}(\omega) = \sum_{m=1}^M \xi_{mt} \phi_m(\omega). \quad (892)$$

As M increases, the approximation becomes richer and converges to the original feature function in the Hilbert-space norm under standard conditions:

$$\|x_t - x_t^{(M)}\|_{\mathcal{H}} \rightarrow 0 \quad \text{as} \quad M \rightarrow \infty. \quad (893)$$

Economically, the feature function allows the model to represent heterogeneity that cannot be adequately summarized by a small number of variables. For instance, two economies may have the same aggregate level of technology but very different distributions of technology across sectors. The function $x_t(\omega)$ preserves such distributional information.

13.3 Hilbert Space Representation

The natural space for the feature function is the Hilbert space $L_2(\Omega, \mu)$:

$$\mathcal{H} = L_2(\Omega, \mu), \quad \langle f, g \rangle_{\mathcal{H}} = \int_{\Omega} f(\omega)g(\omega)d\mu(\omega), \quad \|f\|_{\mathcal{H}}^2 = \langle f, f \rangle_{\mathcal{H}}. \quad (894)$$

The inner product measures the weighted overlap between two functions, while the norm measures the magnitude of a function over the domain.

The Hilbert-space norm is explicitly

$$\|f\|_{\mathcal{H}}^2 = \int_{\Omega} f(\omega)^2 d\mu(\omega). \quad (895)$$

The distance between two feature functions is

$$\|f - g\|_{\mathcal{H}} = \left(\int_{\Omega} [f(\omega) - g(\omega)]^2 d\mu(\omega) \right)^{1/2}. \quad (896)$$

This distance quantifies how far two accumulation environments are from one another in the functional state space.

The inner product permits linear functionals of the feature state. If $b \in \mathcal{H}$, then

$$\langle b, x_t \rangle_{\mathcal{H}} = \int_{\Omega} b(\omega) x_t(\omega) d\mu(\omega). \quad (897)$$

By the Cauchy–Schwarz inequality,

$$|\langle b, x_t \rangle_{\mathcal{H}}| \leq \|b\|_{\mathcal{H}} \|x_t\|_{\mathcal{H}}. \quad (898)$$

Thus, the linear contribution of the feature function to accumulation is finite whenever both b and x_t are square-integrable.

The Hilbert-space representation gives the model a geometric structure. The direction b identifies the functional pattern most directly associated with accumulation. The inner product $\langle b, x_t \rangle_{\mathcal{H}}$ measures how strongly the current feature distribution aligns with that accumulation-relevant direction.

13.4 Linear Infinite-Dimensional Formula

The simplest infinite-dimensional accumulation law is linear in the feature function:

$$K_{t+1} = K_t + a_0 + \langle b, x_t \rangle_{\mathcal{H}}, \quad b \in \mathcal{H}. \quad (899)$$

Here, a_0 is a baseline accumulation term and b is the functional coefficient determining the contribution of each point in the domain Ω .

The corresponding capital increment is

$$\Delta K_t = K_{t+1} - K_t = a_0 + \langle b, x_t \rangle_{\mathcal{H}}. \quad (900)$$

In integral form, this is

$$\Delta K_t = a_0 + \int_{\Omega} b(\omega) x_t(\omega) d\mu(\omega). \quad (901)$$

The coefficient function $b(\omega)$ has a marginal interpretation. A local perturbation of the feature function at point ω changes accumulation according to

$$\frac{\delta \mathcal{A}[x_t]}{\delta x_t(\omega)} = b(\omega). \quad (902)$$

If $b(\omega) > 0$, increasing the feature intensity near ω raises capital accumulation. If $b(\omega) < 0$, increasing the feature intensity near ω reduces accumulation.

The finite-dimensional linear model is again a special case. If $\Omega = \{1, \dots, d\}$, then

$$\langle b, x_t \rangle_{\mathcal{H}} = \sum_{j=1}^d b_j x_{jt}, \quad (903)$$

and Equation (899) becomes

$$K_{t+1} = K_t + a_0 + \sum_{j=1}^d b_j x_{jt}. \quad (904)$$

Thus, the linear infinite-dimensional formula generalizes the finite multi-feature formula by replacing a coefficient vector with a coefficient function.

If the feature function is expanded in an orthonormal basis,

$$x_t = \sum_{m=1}^{\infty} \xi_{mt} \phi_m, \quad b = \sum_{m=1}^{\infty} b_m \phi_m, \quad (905)$$

then

$$\langle b, x_t \rangle_{\mathcal{H}} = \sum_{m=1}^{\infty} b_m \xi_{mt}. \quad (906)$$

The linear infinite-dimensional formula can therefore be approximated by

$$K_{t+1} \approx K_t + a_0 + \sum_{m=1}^M b_m \xi_{mt}. \quad (907)$$

This provides a practical route to estimation through finite-dimensional sieve approximation.

13.5 Quadratic Kernel Formula

To allow nonlinear interaction across the functional domain, we introduce a quadratic kernel term:

$$K_{t+1} = K_t + a_0 + \langle b, x_t \rangle + \iint_{\Omega \times \Omega} x_t(\omega) \kappa(\omega, \omega') x_t(\omega') d\mu(\omega) d\mu(\omega'). \quad (908)$$

The kernel $\kappa(\omega, \omega')$ measures the interaction between the feature value at ω and the feature value at ω' .

The corresponding accumulation function is

$$\mathcal{A}[x_t] = a_0 + \langle b, x_t \rangle_{\mathcal{H}} + \iint_{\Omega \times \Omega} x_t(\omega) \kappa(\omega, \omega') x_t(\omega') d\mu(\omega) d\mu(\omega'). \quad (909)$$

The square brackets emphasize that $\mathcal{A}$ is now a functional: it maps an entire function x_t into a scalar capital increment.

The kernel term is the infinite-dimensional analogue of the finite quadratic form:

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} = \sum_{i=1}^d \sum_{j=1}^d x_i h_{ij} x_j. \quad (910)$$

In the functional case, the matrix entries h_{ij} are replaced by the kernel values $\kappa(\omega, \omega')$, and the double sum is replaced by a double integral:

$$\sum_{i=1}^d \sum_{j=1}^d x_i h_{ij} x_j \quad \longrightarrow \quad \iint_{\Omega \times \Omega} x(\omega) \kappa(\omega, \omega') x(\omega') d\mu(\omega) d\mu(\omega'). \quad (911)$$

If $\kappa(\omega, \omega') > 0$, the features at ω and ω' are locally complementary in the quadratic interaction. If $\kappa(\omega, \omega') < 0$, they are locally substitutive:

$$\kappa(\omega, \omega') > 0 \Rightarrow \text{functional complementarity}, \quad \kappa(\omega, \omega') < 0 \Rightarrow \text{functional substitution}. \quad (912)$$

A sufficient condition for the quadratic term to be finite is

$$\int_{\Omega} \int_{\Omega} |\kappa(\omega, \omega')|^2 d\mu(\omega) d\mu(\omega') < \infty. \quad (913)$$

Under this condition, the kernel defines a Hilbert–Schmidt operator and the quadratic interaction is well defined for $x_t \in L_2(\Omega, \mu)$.

The functional derivative of the quadratic kernel formula is

$$\frac{\delta \mathcal{A}[x]}{\delta x(\omega)} = b(\omega) + \int_{\Omega} \kappa(\omega, \omega') x(\omega') d\mu(\omega') + \int_{\Omega} \kappa(\omega', \omega) x(\omega') d\mu(\omega'). \quad (914)$$

If the kernel is symmetric, this simplifies to

$$\frac{\delta \mathcal{A}[x]}{\delta x(\omega)} = b(\omega) + 2 \int_{\Omega} \kappa(\omega, \omega') x(\omega') d\mu(\omega'). \quad (915)$$

13.6 Integral Operator

The quadratic kernel formula can be written compactly using an integral operator. Define

$$(\mathcal{K}f)(\omega) = \int_{\Omega} \kappa(\omega, \omega') f(\omega') d\mu(\omega'), \quad \mathcal{K} : \mathcal{H} \rightarrow \mathcal{H}. \quad (916)$$

The operator $\mathcal{K}$ maps a feature function f into another function whose value at ω is a weighted integral of f over the domain.

Using this operator, the quadratic interaction becomes

$$\iint_{\Omega \times \Omega} x(\omega) \kappa(\omega, \omega') x(\omega') d\mu(\omega) d\mu(\omega') = \langle x, \mathcal{K}x \rangle_{\mathcal{H}}. \quad (917)$$

Therefore, the quadratic infinite-dimensional accumulation formula can be written as

$$K_{t+1} = K_t + a_0 + \langle b, x_t \rangle_{\mathcal{H}} + \langle x_t, \mathcal{K}x_t \rangle_{\mathcal{H}}. \quad (918)$$

Equivalently,

$$\mathcal{A}[x_t] = a_0 + \langle b, x_t \rangle_{\mathcal{H}} + \langle x_t, \mathcal{K}x_t \rangle_{\mathcal{H}}. \quad (919)$$

The boundedness of $\mathcal{K}$ requires

$$\|\mathcal{K}f\|_{\mathcal{H}} \leq C_K \|f\|_{\mathcal{H}}, \quad \forall f \in \mathcal{H}, \quad (920)$$

for some finite constant $C_K < \infty$. Equivalently,

$$\|\mathcal{K}\|_{\text{op}} = \sup_{\|f\|_{\mathcal{H}} \leq 1} \|\mathcal{K}f\|_{\mathcal{H}} < \infty. \quad (921)$$

Boundedness ensures that the interaction term is finite and stable:

$$|\langle x, \mathcal{K}x \rangle_{\mathcal{H}}| \leq \|x\|_{\mathcal{H}} \|\mathcal{K}x\|_{\mathcal{H}} \leq \|\mathcal{K}\|_{\text{op}} \|x\|_{\mathcal{H}}^2. \quad (922)$$

If the kernel is symmetric,

$$\kappa(\omega, \omega') = \kappa(\omega', \omega), \quad (923)$$

then the operator is self-adjoint:

$$\langle f, \mathcal{K}g \rangle_{\mathcal{H}} = \langle \mathcal{K}f, g \rangle_{\mathcal{H}}, \quad f, g \in \mathcal{H}. \quad (924)$$

Self-adjointness makes the operator the exact infinite-dimensional analogue of a symmetric interaction matrix.

The integral-operator representation is therefore the central mathematical device that carries the finite-dimensional quadratic formula into functional space:

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} \longrightarrow \langle x, \mathcal{K}x \rangle_{\mathcal{H}}. \quad (925)$$

It permits the model to capture distributed nonlinear interactions across an entire continuum of accumulation-relevant features.

13.7 Kernel Symmetry

The quadratic kernel formula is analytically most transparent when the interaction kernel is symmetric. We impose

$$\kappa(\omega, \omega') = \kappa(\omega', \omega), \quad \langle f, \mathcal{K}g \rangle_{\mathcal{H}} = \langle \mathcal{K}f, g \rangle_{\mathcal{H}}. \quad (926)$$

The first condition states that the interaction between feature locations ω and ω' is reciprocal. The second condition states that the integral operator $\mathcal{K}$ induced by the kernel is self-adjoint on the Hilbert space $\mathcal{H}$.

To see the connection, recall that

$$(\mathcal{K}g)(\omega) = \int_{\Omega} \kappa(\omega, \omega') g(\omega') d\mu(\omega'). \quad (927)$$

Then

$$\langle f, \mathcal{K}g \rangle_{\mathcal{H}} = \int_{\Omega} f(\omega) \left[\int_{\Omega} \kappa(\omega, \omega') g(\omega') d\mu(\omega') \right] d\mu(\omega). \quad (928)$$

By kernel symmetry and interchange of integration,

$$\langle f, \mathcal{K}g \rangle_{\mathcal{H}} = \int_{\Omega} g(\omega') \left[\int_{\Omega} \kappa(\omega', \omega) f(\omega) d\mu(\omega) \right] d\mu(\omega') = \langle \mathcal{K}f, g \rangle_{\mathcal{H}}. \quad (929)$$

Thus, a symmetric kernel generates a self-adjoint operator.

Kernel symmetry is the infinite-dimensional analogue of matrix symmetry in the finite-dimensional quadratic form. In the finite-dimensional model,

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} = \mathbf{x}^\top \left(\frac{\mathbf{H} + \mathbf{H}^\top}{2} \right) \mathbf{x}, \quad (930)$$

so only the symmetric part of $\mathbf{H}$ affects the scalar accumulation function. In the functional model, only the symmetric part of the kernel contributes to the quadratic interaction:

$$\kappa_S(\omega, \omega') = \frac{1}{2} [\kappa(\omega, \omega') + \kappa(\omega', \omega)]. \quad (931)$$

The antisymmetric part integrates to zero in the quadratic scalar functional:

$$\iint x(\omega) \kappa_A(\omega, \omega') x(\omega') d\mu(\omega) d\mu(\omega') = 0, \quad (932)$$

where

$$\kappa_A(\omega, \omega') = \frac{1}{2} [\kappa(\omega, \omega') - \kappa(\omega', \omega)]. \quad (933)$$

Therefore, imposing kernel symmetry entails no loss of generality for a scalar quadratic accumulation functional. It ensures that the interaction structure is economically interpretable as reciprocal complementarity or substitution across the feature domain.

13.8 Kernel Boundedness

For the infinite-dimensional accumulation formula to be well defined, the operator induced by the kernel must be bounded. We impose

$$\sup_{\|f\|_{\mathcal{H}} \leq 1} \|\mathcal{K}f\|_{\mathcal{H}} < \infty, \quad \int_{\Omega} \int_{\Omega} |\kappa(\omega, \omega')|^2 d\mu(\omega) d\mu(\omega') < \infty. \quad (934)$$

The first condition states that $\mathcal{K}$ is a bounded linear operator on $\mathcal{H}$. The second is a Hilbert–Schmidt condition on the kernel.

The operator norm is defined as

$$\|\mathcal{K}\|_{\text{op}} \equiv \sup_{\|f\|_{\mathcal{H}} \leq 1} \|\mathcal{K}f\|_{\mathcal{H}}. \quad (935)$$

Boundedness means that there exists a finite constant C_K such that

$$\|\mathcal{K}f\|_{\mathcal{H}} \leq C_K \|f\|_{\mathcal{H}}, \quad \forall f \in \mathcal{H}. \quad (936)$$

This condition rules out infinite amplification of small functional perturbations.

The Hilbert–Schmidt norm of the kernel is

$$\|\mathcal{K}\|_{HS}^2 = \int_{\Omega} \int_{\Omega} |\kappa(\omega, \omega')|^2 d\mu(\omega) d\mu(\omega'). \quad (937)$$

If this quantity is finite, then $\mathcal{K}$ is bounded, and

$$\|\mathcal{K}\|_{\text{op}} \leq \|\mathcal{K}\|_{HS} < \infty. \quad (938)$$

Thus, Hilbert–Schmidt integrability is a sufficient condition for operator boundedness.

Boundedness ensures that the quadratic interaction term is finite for every $x \in \mathcal{H}$:

$$|\langle x, \mathcal{K}x \rangle_{\mathcal{H}}| \leq \|x\|_{\mathcal{H}} \|\mathcal{K}x\|_{\mathcal{H}} \leq \|\mathcal{K}\|_{\text{op}} \|x\|_{\mathcal{H}}^2. \quad (939)$$

Economically, this means that distributed feature interactions cannot produce an infinite one-period accumulation increment when the underlying feature function has finite magnitude.

The boundedness condition also stabilizes marginal effects. Since

$$\|\mathcal{K}x\|_{\mathcal{H}} \leq \|\mathcal{K}\|_{\text{op}} \|x\|_{\mathcal{H}}, \quad (940)$$

the interaction component of the functional gradient remains controlled by the size of the feature function. Hence, the infinite-dimensional formula preserves the regularity properties needed for local analysis, optimization, and approximation.

13.9 Existence of Integrals

The linear and quadratic terms in the infinite-dimensional accumulation formula must be finite. We therefore require

$$\int_{\Omega} |b(\omega)x_t(\omega)| d\mu(\omega) < \infty, \quad \iint |x_t(\omega)\kappa(\omega, \omega')x_t(\omega')| d\mu(\omega)d\mu(\omega') < \infty. \quad (941)$$

These conditions guarantee that both the linear functional and the quadratic kernel interaction are well defined.

The linear term is finite if $b \in \mathcal{H}$ and $x_t \in \mathcal{H}$. Indeed, by the Cauchy–Schwarz inequality,

$$\int_{\Omega} |b(\omega)x_t(\omega)| d\mu(\omega) \leq \|b\|_{\mathcal{H}} \|x_t\|_{\mathcal{H}} < \infty. \quad (942)$$

Thus, square-integrability of both the coefficient function and the feature function is sufficient for the existence of the linear component.

For the quadratic term, if the kernel is Hilbert–Schmidt, then

$$\left| \iint x_t(\omega)\kappa(\omega, \omega')x_t(\omega') d\mu(\omega)d\mu(\omega') \right| = |\langle x_t, \mathcal{K}x_t \rangle_{\mathcal{H}}| \leq \|\mathcal{K}\|_{\text{op}} \|x_t\|_{\mathcal{H}}^2 < \infty. \quad (943)$$

Hence, the quadratic kernel term is finite whenever $x_t \in \mathcal{H}$ and $\mathcal{K}$ is bounded.

The accumulation functional is therefore well defined as

$$\mathcal{A}[x_t] = a_0 + \langle b, x_t \rangle_{\mathcal{H}} + \langle x_t, \mathcal{K}x_t \rangle_{\mathcal{H}}. \quad (944)$$

The existence conditions ensure that

$$\mathcal{A}[x_t] \in \mathbb{R}. \quad (945)$$

This is essential because the functional is intended to represent a finite capital increment.

These integrability restrictions also clarify the economic scope of the model. The feature distribution may be infinite-dimensional, but its aggregate intensity and interaction effects must remain finite. The model therefore allows rich distributed heterogeneity without permitting mathematically or economically explosive accumulation from nonintegrable features.

13.10 Infinite-Dimensional Marginal Effect

The infinite-dimensional marginal effect is the functional derivative of the accumulation functional with respect to the feature function. For the quadratic kernel formula with symmetric kernel, it is

$$\frac{\delta \mathcal{A}[x]}{\delta x(\omega)} = b(\omega) + 2 \int_{\Omega} \kappa(\omega, \omega') x(\omega') d\mu(\omega'). \quad (946)$$

This expression gives the local effect of increasing the feature intensity at point ω on the capital accumulation increment.

To derive this expression, consider a perturbation $x + \epsilon h$, where $h \in \mathcal{H}$ and ϵ is scalar. The first variation is

$$\left. \frac{d}{d\epsilon} \mathcal{A}[x + \epsilon h] \right|_{\epsilon=0} = \langle b, h \rangle_{\mathcal{H}} + \langle h, \mathcal{K}x \rangle_{\mathcal{H}} + \langle x, \mathcal{K}h \rangle_{\mathcal{H}}. \quad (947)$$

If $\mathcal{K}$ is self-adjoint, then

$$\langle x, \mathcal{K}h \rangle_{\mathcal{H}} = \langle \mathcal{K}x, h \rangle_{\mathcal{H}} = \langle h, \mathcal{K}x \rangle_{\mathcal{H}}. \quad (948)$$

Therefore,

$$D\mathcal{A}[x][h] = \langle b + 2\mathcal{K}x, h \rangle_{\mathcal{H}}. \quad (949)$$

The Hilbert-space gradient is

$$\nabla_{\mathcal{H}} \mathcal{A}[x] = b + 2\mathcal{K}x. \quad (950)$$

Writing this gradient pointwise gives Equation (946).

The term $b(\omega)$ is the direct marginal contribution of the feature at ω . The integral term is the indirect interaction contribution:

$$2 \int_{\Omega} \kappa(\omega, \omega') x(\omega') d\mu(\omega'). \quad (951)$$

It captures how the marginal value of feature location ω depends on the entire distribution of features across Ω .

If

$$\frac{\delta \mathcal{A}[x]}{\delta x(\omega)} > 0, \quad (952)$$

then increasing $x(\omega)$ locally raises accumulation. If

$$\frac{\delta \mathcal{A}[x]}{\delta x(\omega)} < 0, \quad (953)$$

then increasing $x(\omega)$ locally reduces accumulation. If the derivative equals zero, the feature intensity at ω is locally neutral.

The infinite-dimensional marginal effect therefore generalizes the finite gradient

$$\nabla_{\mathbf{x}} \mathcal{A}(\mathbf{x}) = \mathbf{a} + \mathbf{H}\mathbf{x} \quad (954)$$

to the functional gradient

$$\nabla_{\mathcal{H}} \mathcal{A}[x] = b + 2\mathcal{K}x. \quad (955)$$

13.11 Infinite-Dimensional Critical Point

An interior critical point of the infinite-dimensional accumulation functional is a feature function $x^* \in \mathcal{H}$ for which the functional marginal effect vanishes almost everywhere:

$$\frac{\delta \mathcal{A}[x^*]}{\delta x(\omega)} = 0 \quad \text{a.e. } \omega, \quad b + 2\mathcal{K}x^* = 0. \quad (956)$$

The first condition states the pointwise vanishing of the functional derivative. The second expresses the same condition as an operator equation in $\mathcal{H}$.

The critical point therefore solves

$$\mathcal{K}x^\star = -\frac{1}{2}b. \quad (957)$$

This is the infinite-dimensional analogue of the finite-dimensional stationary condition

$$\mathbf{H}x^\star = -\mathbf{a}. \quad (958)$$

If $\mathcal{K}^{-1}$ exists, then the critical point is unique and given by

$$x^\star = -\frac{1}{2}\mathcal{K}^{-1}b. \quad (959)$$

In this case, the operator equation is well posed: every linear component $b \in \mathcal{H}$ corresponds to a unique critical feature function.

If $\mathcal{K}^{-1}$ fails to exist but $b \in \text{Range}(\mathcal{K})$, then the equation remains solvable. When the Moore–Penrose generalized inverse $\mathcal{K}^\dagger$ exists, the minimum-norm solution is

$$x^\star = -\frac{1}{2}\mathcal{K}^\dagger b. \quad (960)$$

This solution is characterized by

$$x^\star \in \arg \min \left\{ \|x\|_{\mathcal{H}} : \mathcal{K}x = -\frac{1}{2}b \right\}. \quad (961)$$

If

$$b \notin \text{Range}(\mathcal{K}), \quad (962)$$

then the critical-point equation has no exact solution in $\mathcal{H}$:

$$\nexists x^\star \in \mathcal{H} \quad \text{such that} \quad b + 2\mathcal{K}x^\star = 0. \quad (963)$$

In that case, the accumulation functional has no interior stationary point in the unconstrained functional space. One may instead consider a least-squares or regularized approximate solution:

$$x_\lambda^\star = \arg \min_{x \in \mathcal{H}} \left\{ \left\| \mathcal{K}x + \frac{1}{2}b \right\|_{\mathcal{H}}^2 + \lambda \|x\|_{\mathcal{H}}^2 \right\}. \quad (964)$$

The classification of the critical point depends on the curvature operator. The second variation is

$$D^2\mathcal{A}[x][h, h] = 2\langle h, \mathcal{K}h \rangle_{\mathcal{H}}. \quad (965)$$

If

$$\langle h, \mathcal{K}h \rangle_{\mathcal{H}} < 0 \quad \forall h \neq 0, \quad (966)$$

then $x^\star$ is a strict local maximum of the accumulation functional. If

$$\langle h, \mathcal{K}h \rangle_{\mathcal{H}} > 0 \quad \forall h \neq 0, \quad (967)$$

then $x^\star$ is a strict local minimum. If the quadratic form takes both positive and negative values, then $x^\star$ is a saddle point:

$$\exists h_1, h_2 \in \mathcal{H} \quad \text{such that} \quad \langle h_1, \mathcal{K}h_1 \rangle_{\mathcal{H}} > 0, \quad \langle h_2, \mathcal{K}h_2 \rangle_{\mathcal{H}} < 0. \quad (968)$$

The infinite-dimensional critical-point condition therefore extends the finite-dimensional theory of stationary points to functional capital accumulation. The vector equation

$$\mathbf{a} + \mathbf{H}\mathbf{x}^* = \mathbf{0} \quad (969)$$

becomes the operator equation

$$b + 2\mathcal{K}\mathbf{x}^* = \mathbf{0}. \quad (970)$$

The inverse or generalized inverse of $\mathcal{K}$ then determines whether the optimal functional configuration is uniquely defined, minimally selected, or not exactly attainable within the Hilbert space.

14 Finite-Dimensional Approximation

The infinite-dimensional capital formula allows the accumulation environment to be represented by a function. For empirical implementation, however, one must approximate this functional object by a finite number of coefficients. This section develops the finite-dimensional approximation that connects the Hilbert-space formula to estimable models.

14.1 Basis Expansion

Let $\{\phi_m\}_{m=1}^{\infty}$ be an orthonormal basis of the Hilbert space $\mathcal{H} = L_2(\Omega, \mu)$. Any feature function $x \in \mathcal{H}$ can be represented as

$$x(\omega) = \sum_{m=1}^{\infty} \xi_m \phi_m(\omega), \quad \langle \phi_m, \phi_n \rangle_{\mathcal{H}} = \mathbf{1}\{m = n\}. \quad (971)$$

Here, ξ_m is the coordinate of the feature function along the basis direction ϕ_m . The orthonormality condition ensures that each basis component captures an independent direction of functional variation.

The coefficient associated with basis element ϕ_m is obtained by inner projection:

$$\xi_m = \langle x, \phi_m \rangle_{\mathcal{H}} = \int_{\Omega} x(\omega) \phi_m(\omega) d\mu(\omega). \quad (972)$$

Thus, the infinite-dimensional function $x(\omega)$ is equivalently represented by the infinite sequence of coefficients

$$\boldsymbol{\xi} = (\xi_1, \xi_2, \dots). \quad (973)$$

The Hilbert-space norm can be expressed in terms of these coefficients:

$$\|x\|_{\mathcal{H}}^2 = \sum_{m=1}^{\infty} \xi_m^2. \quad (974)$$

This identity follows from Parseval's theorem. It implies that $x \in L_2(\Omega, \mu)$ if and only if the coefficient sequence is square-summable:

$$\sum_{m=1}^{\infty} \xi_m^2 < \infty. \quad (975)$$

The choice of basis depends on the economic structure of the feature domain. If Ω is a spatial domain, the basis may consist of Fourier functions, wavelets, splines, or spatial eigenfunctions. If Ω indexes sectors or technologies, the basis may be constructed from principal components, eigenvectors of an input-output network, or other economically meaningful functional directions.

The basis expansion is not merely a technical device. It decomposes the feature function into interpretable modes of variation. Low-frequency or low-index components may capture broad structural features, while high-frequency components may capture localized or fine-scale heterogeneity:

$$x(\omega) = \underbrace{\sum_{m=1}^M \xi_m \phi_m(\omega)}_{\text{coarse structure}} + \underbrace{\sum_{m=M+1}^{\infty} \xi_m \phi_m(\omega)}_{\text{residual heterogeneity}}. \quad (976)$$

14.2 Orthogonal Projection

A finite-dimensional approximation is obtained by projecting the feature function onto the span of the first M basis functions:

$$\Pi_M x = \sum_{m=1}^M \xi_m \phi_m, \quad \xi_m = \langle x, \phi_m \rangle_{\mathcal{H}}. \quad (977)$$

The operator Π_M is the orthogonal projection from $\mathcal{H}$ onto the finite-dimensional subspace

$$\mathcal{H}_M = \text{span}\{\phi_1, \dots, \phi_M\}. \quad (978)$$

The projected function $\Pi_M x$ is the best approximation to x within $\mathcal{H}_M$ in the Hilbert norm:

$$\Pi_M x = \arg \min_{h \in \mathcal{H}_M} \|x - h\|_{\mathcal{H}}. \quad (979)$$

This optimality property follows from orthogonality of the residual:

$$x - \Pi_M x \perp \mathcal{H}_M. \quad (980)$$

Equivalently,

$$\langle x - \Pi_M x, \phi_m \rangle_{\mathcal{H}} = 0, \quad m = 1, \dots, M. \quad (981)$$

The approximation error is

$$R_M(x) \equiv x - \Pi_M x = \sum_{m=M+1}^{\infty} \xi_m \phi_m. \quad (982)$$

Its squared norm is

$$\|R_M(x)\|_{\mathcal{H}}^2 = \sum_{m=M+1}^{\infty} \xi_m^2. \quad (983)$$

Because $x \in \mathcal{H}$, this residual converges to zero:

$$\|x - \Pi_M x\|_{\mathcal{H}} \rightarrow 0 \quad \text{as} \quad M \rightarrow \infty. \quad (984)$$

The projection converts the functional state into a finite vector of coordinates:

$$\boldsymbol{\xi}_M = (\xi_1, \dots, \xi_M)^\top \in \mathbb{R}^M. \quad (985)$$

Thus, the infinite-dimensional accumulation model can be estimated through the finite-dimensional state vector $\boldsymbol{\xi}_M$.

14.3 Truncated Approximation

Substituting the projected feature function into the infinite-dimensional capital formula yields the finite-dimensional truncated approximation:

$$K_{t+1} \approx K_t + a_0 + \sum_{m=1}^M b_m \xi_{mt} + \sum_{m=1}^M \sum_{n=1}^M \kappa_{mn} \xi_{mt} \xi_{nt}. \quad (986)$$

Here, ξ_{mt} is the coefficient of the feature function x_t along basis direction ϕ_m at time t . The coefficient b_m is the projection of the linear functional coefficient b onto the basis:

$$b_m = \langle b, \phi_m \rangle_{\mathcal{H}}. \quad (987)$$

The interaction coefficient κ_{mn} is the basis representation of the operator $\mathcal{K}$:

$$\kappa_{mn} = \langle \phi_m, \mathcal{K} \phi_n \rangle_{\mathcal{H}}. \quad (988)$$

Equivalently, in kernel form,

$$\kappa_{mn} = \int_{\Omega} \int_{\Omega} \phi_m(\omega) \kappa(\omega, \omega') \phi_n(\omega') d\mu(\omega) d\mu(\omega'). \quad (989)$$

The truncated accumulation function can be written compactly as

$$\mathcal{A}_M(\boldsymbol{\xi}_{Mt}) = a_0 + \mathbf{b}_M^\top \boldsymbol{\xi}_{Mt} + \boldsymbol{\xi}_{Mt}^\top \mathbf{K}_M \boldsymbol{\xi}_{Mt}, \quad (990)$$

where

$$\mathbf{b}_M = (b_1, \dots, b_M)^\top, \quad \mathbf{K}_M = (\kappa_{mn})_{m,n=1}^M. \quad (991)$$

Therefore,

$$K_{t+1} \approx K_t + \mathcal{A}_M(\boldsymbol{\xi}_{Mt}). \quad (992)$$

If the operator $\mathcal{K}$ is self-adjoint, then the truncated interaction matrix is symmetric:

$$\kappa_{mn} = \kappa_{nm}, \quad \mathbf{K}_M = \mathbf{K}_M^\top. \quad (993)$$

Thus, the finite-dimensional approximation preserves the same structure as the finite quadratic model:

$$\langle x_t, \mathcal{K} x_t \rangle_{\mathcal{H}} \longrightarrow \boldsymbol{\xi}_{Mt}^\top \mathbf{K}_M \boldsymbol{\xi}_{Mt}. \quad (994)$$

The approximation error in the accumulation function is

$$\mathcal{R}_M(x_t) = \mathcal{A}[x_t] - \mathcal{A}_M(\boldsymbol{\xi}_{Mt}). \quad (995)$$

Under square-integrability of b , boundedness of $\mathcal{K}$, and convergence of $\Pi_M x_t$ to x_t , this error converges to zero:

$$|\mathcal{R}_M(x_t)| \rightarrow 0 \quad \text{as} \quad M \rightarrow \infty. \quad (996)$$

For empirical implementation, the truncated model can be written with an error term:

$$K_{i,t+1} = K_{it} + a_0 + \sum_{m=1}^M b_m \xi_{mit} + \sum_{m=1}^M \sum_{n=1}^M \kappa_{mn} \xi_{mit} \xi_{nit} + u_{it}. \quad (997)$$

This specification is finite-dimensional and can be estimated using the same tools developed for linear and quadratic multi-feature capital formulas.

The truncated approximation therefore provides the empirical bridge from functional capital accumulation to finite-dimensional econometrics:

$$x_t(\omega) \in \mathcal{H} \longrightarrow \boldsymbol{\xi}_{Mt} \in \mathbb{R}^M \longrightarrow K_{t+1} \approx K_t + \mathcal{A}_M(\boldsymbol{\xi}_{Mt}). \quad (998)$$

It preserves the theoretical content of the infinite-dimensional model while making estimation and computation feasible.

14.4 Approximation Error

The finite-dimensional approximation replaces the full feature function $x \in \mathcal{H}$ with its projection onto the first M basis functions. The approximation error is

$$R_M(x) = x - \Pi_M x, \quad \|R_M(x)\|_{\mathcal{H}}^2 = \sum_{m>M} \xi_m^2 \rightarrow 0. \quad (999)$$

Here, $R_M(x)$ is the residual functional component omitted by the M -dimensional projection.

Using the basis expansion

$$x = \sum_{m=1}^{\infty} \xi_m \phi_m, \quad (1000)$$

and the projection

$$\Pi_M x = \sum_{m=1}^M \xi_m \phi_m, \quad (1001)$$

the residual is

$$R_M(x) = \sum_{m=M+1}^{\infty} \xi_m \phi_m. \quad (1002)$$

By orthonormality of the basis,

$$\|R_M(x)\|_{\mathcal{H}}^2 = \left\| \sum_{m=M+1}^{\infty} \xi_m \phi_m \right\|_{\mathcal{H}}^2 = \sum_{m=M+1}^{\infty} \xi_m^2. \quad (1003)$$

Since $x \in \mathcal{H}$ implies

$$\sum_{m=1}^{\infty} \xi_m^2 < \infty, \quad (1004)$$

the tail sum converges to zero as $M \rightarrow \infty$.

The approximation error in the capital accumulation functional can also be bounded. For the linear functional,

$$|\langle b, x \rangle_{\mathcal{H}} - \langle b, \Pi_M x \rangle_{\mathcal{H}}| = |\langle b, R_M(x) \rangle_{\mathcal{H}}| \leq \|b\|_{\mathcal{H}} \|R_M(x)\|_{\mathcal{H}}. \quad (1005)$$

Thus, the linear approximation error vanishes when $\|R_M(x)\|_{\mathcal{H}} \rightarrow 0$.

For the quadratic operator term,

$$|\langle x, \mathcal{K}x \rangle_{\mathcal{H}} - \langle \Pi_M x, \mathcal{K} \Pi_M x \rangle_{\mathcal{H}}| \quad (1006)$$

can be bounded using operator boundedness. Since

$$x = \Pi_M x + R_M(x), \quad (1007)$$

we obtain

$$\langle x, \mathcal{K}x \rangle_{\mathcal{H}} - \langle \Pi_M x, \mathcal{K} \Pi_M x \rangle_{\mathcal{H}} = 2\langle \Pi_M x, \mathcal{K} R_M(x) \rangle_{\mathcal{H}} + \langle R_M(x), \mathcal{K} R_M(x) \rangle_{\mathcal{H}}, \quad (1008)$$

when $\mathcal{K}$ is self-adjoint. Therefore,

$$|\langle x, \mathcal{K}x \rangle_{\mathcal{H}} - \langle \Pi_M x, \mathcal{K} \Pi_M x \rangle_{\mathcal{H}}| \leq 2\|\mathcal{K}\|_{\text{op}} \|\Pi_M x\|_{\mathcal{H}} \|R_M(x)\|_{\mathcal{H}} + \|\mathcal{K}\|_{\text{op}} \|R_M(x)\|_{\mathcal{H}}^2. \quad (1009)$$

Hence, if $\|x\|_{\mathcal{H}} < \infty$ and $\|R_M(x)\|_{\mathcal{H}} \rightarrow 0$, the quadratic approximation error also vanishes:

$$|\langle x, \mathcal{K}x \rangle_{\mathcal{H}} - \langle \Pi_M x, \mathcal{K} \Pi_M x \rangle_{\mathcal{H}}| \rightarrow 0. \quad (1010)$$

The total functional approximation error is therefore controlled by the projection residual:

$$|\mathcal{A}[x] - \mathcal{A}[\Pi_M x]| \rightarrow 0 \quad \text{as} \quad M \rightarrow \infty. \quad (1011)$$

This justifies using finite-dimensional approximations to estimate and interpret the infinite-dimensional capital accumulation formula.

14.5 Sieve Approximation

The same logic applies to a general nonparametric accumulation function. Let $g_0(\mathbf{x})$ denote the true finite-dimensional accumulation function. A sieve approximation represents g_0 by a sequence of finite-dimensional function spaces:

$$g(\mathbf{x}) \approx g_M(\mathbf{x}) = \sum_{m=1}^M \theta_m \psi_m(\mathbf{x}), \quad \mathcal{S}_M = \text{span}\{\psi_1, \dots, \psi_M\}. \quad (1012)$$

Here, $\{\psi_m\}_{m=1}^M$ are basis functions and $\boldsymbol{\theta}_M = (\theta_1, \dots, \theta_M)^\top$ is the coefficient vector.

The sieve space $\mathcal{S}_M$ increases in flexibility as M grows:

$$\mathcal{S}_1 \subset \mathcal{S}_2 \subset \dots \subset \mathcal{S}_M \subset \dots. \quad (1013)$$

The union of these spaces should be rich enough to approximate the true accumulation function:

$$g_0 \in \overline{\bigcup_{M=1}^{\infty} \mathcal{S}_M}. \quad (1014)$$

The approximation error of the sieve is

$$r_M(\mathbf{x}) = g_0(\mathbf{x}) - g_M(\mathbf{x}). \quad (1015)$$

A valid sieve sequence satisfies

$$\inf_{g \in \mathcal{S}_M} \|g - g_0\|_{L_2} \rightarrow 0 \quad \text{as} \quad M \rightarrow \infty. \quad (1016)$$

Common choices of sieve basis functions include polynomials, splines, wavelets, Fourier bases, radial basis functions, and neural-network features. For example, a polynomial sieve may use

$$\psi_m(\mathbf{x}) \in \{\mathbf{x}^\nu : |\nu| \leq p_M\}, \quad (1017)$$

where the polynomial degree p_M increases with M . A spline sieve may use piecewise polynomial functions with an increasing number of knots:

$$\psi_m(\mathbf{x}) = \text{spline basis function}. \quad (1018)$$

The sieve approximation provides a disciplined compromise between parametric structure and non-parametric flexibility. For fixed M , the model is finite-dimensional and estimable. As M increases, the model can approximate increasingly complex accumulation functions:

$$g_M(\mathbf{x}) \rightarrow g_0(\mathbf{x}) \quad \text{in } L_2. \quad (1019)$$

14.6 Sieve Estimation

Given observations $\{K_{it}, K_{i,t+1}, \mathbf{x}_{it}\}$, the sieve estimator chooses coefficients to minimize the squared difference between the observed capital increment and the sieve approximation:

$$\hat{\boldsymbol{\theta}}_M = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^M} \sum_{i,t} \left(K_{i,t+1} - K_{it} - \sum_{m=1}^M \theta_m \psi_m(\mathbf{x}_{it}) \right)^2. \quad (1020)$$

The estimated accumulation function is

$$\hat{g}_M(\mathbf{x}) = \sum_{m=1}^M \hat{\theta}_m \psi_m(\mathbf{x}). \quad (1021)$$

Let

$$Y_{it} \equiv K_{i,t+1} - K_{it} \quad (1022)$$

denote the observed capital increment, and define the basis vector

$$\boldsymbol{\psi}_M(\mathbf{x}_{it}) = (\psi_1(\mathbf{x}_{it}), \dots, \psi_M(\mathbf{x}_{it}))^\top. \quad (1023)$$

Then the sieve model is

$$Y_{it} = \boldsymbol{\psi}_M(\mathbf{x}_{it})^\top \boldsymbol{\theta}_M + u_{it}. \quad (1024)$$

Although the model is nonlinear in the original features, it is linear in the sieve coefficients.

Stacking observations gives

$$\mathbf{Y} = \boldsymbol{\Psi}_M \boldsymbol{\theta}_M + \mathbf{u}, \quad (1025)$$

where $\boldsymbol{\Psi}_M$ is the matrix of evaluated basis functions. The least squares estimator is

$$\hat{\boldsymbol{\theta}}_M = (\boldsymbol{\Psi}_M^\top \boldsymbol{\Psi}_M)^{-1} \boldsymbol{\Psi}_M^\top \mathbf{Y}, \quad (1026)$$

provided $\boldsymbol{\Psi}_M^\top \boldsymbol{\Psi}_M$ is nonsingular.

When M is large, regularization may be introduced:

$$\hat{\boldsymbol{\theta}}_{M,\lambda} = \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^M} \left\{ \sum_{i,t} \left(Y_{it} - \boldsymbol{\psi}_M(\mathbf{x}_{it})^\top \boldsymbol{\theta} \right)^2 + \lambda \mathcal{P}(\boldsymbol{\theta}) \right\}. \quad (1027)$$

An ℓ_2 penalty stabilizes estimation:

$$\mathcal{P}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_2^2, \quad (1028)$$

while an ℓ_1 penalty selects a sparse subset of basis functions:

$$\mathcal{P}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_1. \quad (1029)$$

The sieve estimator can also incorporate fixed effects:

$$K_{i,t+1} - K_{it} = \alpha_i + \tau_t + \sum_{m=1}^M \theta_m \psi_m(\mathbf{x}_{it}) + u_{it}. \quad (1030)$$

In this case, the basis functions are estimated after controlling for time-invariant unit heterogeneity and common time shocks.

14.7 Convergence Conditions

For sieve estimation to be consistent, the dimension of the sieve space must increase with the sample size, but not too quickly. A standard set of conditions is

$$M \rightarrow \infty, \quad \frac{M}{n} \rightarrow 0, \quad \|\hat{g}_M - g_0\|_{L_2} \xrightarrow{p} 0. \quad (1031)$$

Here, n denotes the effective sample size and g_0 is the true accumulation function.

The condition

$$M \rightarrow \infty \quad (1032)$$

ensures that the sieve space becomes rich enough to approximate the true function. If M were fixed, the estimator would generally converge only to the best approximation within a fixed finite-dimensional space rather than to g_0 itself.

The condition

$$\frac{M}{n} \rightarrow 0 \quad (1033)$$

controls estimation variance. It requires the number of estimated basis coefficients to grow more slowly than the sample size. Without this restriction, the estimator may overfit the sample and fail to converge to the true accumulation function.

The convergence statement

$$\|\hat{g}_M - g_0\|_{L_2} \xrightarrow{p} 0 \quad (1034)$$

means that the estimated accumulation function converges in probability to the true function under the L_2 norm. Equivalently,

$$\int [\hat{g}_M(\mathbf{x}) - g_0(\mathbf{x})]^2 dP(\mathbf{x}) \xrightarrow{p} 0. \quad (1035)$$

The total estimation error can be decomposed into approximation error and statistical error:

$$\|\hat{g}_M - g_0\|_{L_2} \leq \|\hat{g}_M - g_M^*\|_{L_2} + \|g_M^* - g_0\|_{L_2}, \quad (1036)$$

where

$$g_M^* = \arg \min_{g \in \mathcal{S}_M} \|g - g_0\|_{L_2}. \quad (1037)$$

The first term is estimation error, and the second is approximation error. As M increases, approximation error decreases:

$$\|g_M^* - g_0\|_{L_2} \rightarrow 0. \quad (1038)$$

At the same time, estimation error generally increases with M :

$$\|\hat{g}_M - g_M^*\|_{L_2} = O_p\left(\sqrt{\frac{M}{n}}\right) \quad (1039)$$

under standard regularity conditions. The choice of M therefore balances bias and variance.

The convergence conditions justify treating finite-dimensional approximations as consistent representations of the underlying accumulation law. They complete the bridge from the infinite-dimensional theory to empirical implementation:

$$x \in \mathcal{H} \quad \longrightarrow \quad \Pi_M x \in \mathcal{H}_M \quad \longrightarrow \quad \hat{g}_M \xrightarrow{p} g_0. \quad (1040)$$

15 Statistical Estimation

The preceding sections specify the capital accumulation function in linear, regularized, high-dimensional, and nonparametric forms. This section gives the corresponding statistical estimators. Throughout, let

$$\mathbf{Y} = (Y_1, \dots, Y_n)^\top \quad (1041)$$

denote the observed outcome vector, where in the accumulation setting one may take

$$Y_i = K_{i,t+1} - K_{it} \quad (1042)$$

or, equivalently, use $K_{i,t+1}$ as the dependent variable after including K_{it} among the regressors. Let $X \in \mathbb{R}^{n \times p}$ denote the design matrix of observed features, transformations, interactions, or basis functions.

15.1 Least Squares Estimator

The baseline estimator is ordinary least squares:

$$\hat{\beta}_{OLS} = \arg \min_{\beta} \|\mathbf{Y} - X\beta\|_2^2 = (X^\top X)^{-1} X^\top \mathbf{Y}. \quad (1043)$$

This estimator applies to linear-in-parameters specifications of the accumulation equation:

$$Y_i = \mathbf{x}_i^\top \beta + u_i, \quad i = 1, \dots, n. \quad (1044)$$

In the simplest capital-increment specification,

$$K_{i,t+1} - K_{it} = \alpha + \beta^\top \mathbf{x}_{it} + u_{it}. \quad (1045)$$

The least-squares estimator chooses the coefficient vector that minimizes the sum of squared residuals:

$$SSR(\beta) = \sum_{i=1}^n (Y_i - \mathbf{x}_i^\top \beta)^2. \quad (1046)$$

The first-order condition is

$$X^\top(\mathbf{Y} - X\hat{\boldsymbol{\beta}}_{OLS}) = \mathbf{0}. \quad (1047)$$

Equivalently,

$$X^\top X \hat{\boldsymbol{\beta}}_{OLS} = X^\top \mathbf{Y}. \quad (1048)$$

If $X^\top X$ is nonsingular, the solution is the closed-form expression in Equation (1043).

The fitted values and residuals are

$$\hat{\mathbf{Y}} = X\hat{\boldsymbol{\beta}}_{OLS}, \quad \hat{\mathbf{u}} = \mathbf{Y} - \hat{\mathbf{Y}}. \quad (1049)$$

The projection matrix is

$$P_X = X(X^\top X)^{-1}X^\top, \quad (1050)$$

so that

$$\hat{\mathbf{Y}} = P_X \mathbf{Y}, \quad \hat{\mathbf{u}} = (I - P_X) \mathbf{Y}. \quad (1051)$$

Under the exogeneity condition

$$\mathbb{E}[u_i \mid X] = 0, \quad (1052)$$

and the full-rank condition

$$\text{rank}(X) = p, \quad (1053)$$

the least-squares estimator consistently estimates the population linear projection coefficient. If the model is correctly specified, it estimates the structural marginal accumulation effects.

The estimated residual variance is

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum_{i=1}^n \hat{u}_i^2 = \frac{1}{n-p} \|\mathbf{Y} - X\hat{\boldsymbol{\beta}}_{OLS}\|_2^2. \quad (1054)$$

Under homoskedastic errors, the conventional variance estimator is

$$\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}_{OLS}) = \hat{\sigma}^2 (X^\top X)^{-1}. \quad (1055)$$

In the present framework, OLS is best understood as the benchmark estimator. It is transparent and directly interpretable, but it is sensitive to endogeneity, omitted variables, multicollinearity, and high-dimensional feature spaces. These limitations motivate regularized and nonparametric estimators.

15.2 Regularized Estimator

When the feature matrix contains many correlated variables or when the estimation problem is ill-conditioned, regularization stabilizes estimation. A ridge-type regularized estimator is

$$\hat{\boldsymbol{\beta}}_\lambda = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{n} \|\mathbf{Y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right\}. \quad (1056)$$

Here, $\lambda \geq 0$ is the regularization parameter. Larger values of λ impose stronger shrinkage toward zero.

The objective function balances empirical fit and coefficient magnitude:

$$Q_\lambda(\boldsymbol{\beta}) = \frac{1}{n} \|\mathbf{Y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2. \quad (1057)$$

The first term rewards fit to the observed capital increments, while the second term penalizes large coefficients.

The first-order condition is

$$-\frac{2}{n}X^\top(\mathbf{Y} - X\hat{\boldsymbol{\beta}}_\lambda) + 2\lambda\hat{\boldsymbol{\beta}}_\lambda = \mathbf{0}. \quad (1058)$$

Therefore,

$$\left(X^\top X + n\lambda I_p\right)\hat{\boldsymbol{\beta}}_\lambda = X^\top \mathbf{Y}. \quad (1059)$$

The closed-form solution is

$$\hat{\boldsymbol{\beta}}_\lambda = \left(X^\top X + n\lambda I_p\right)^{-1} X^\top \mathbf{Y}. \quad (1060)$$

If the intercept should not be penalized, the penalty can be modified as

$$\lambda \sum_{j=1}^p \beta_j^2 \quad (1061)$$

after separating the intercept from the remaining coefficients. This is often appropriate because the intercept represents baseline accumulation rather than a feature effect.

The regularization parameter controls the bias–variance trade-off:

$$\lambda \uparrow \implies \text{lower variance but greater shrinkage bias.} \quad (1062)$$

When $\lambda = 0$, ridge reduces to OLS:

$$\hat{\boldsymbol{\beta}}_{\lambda=0} = \hat{\boldsymbol{\beta}}_{OLS}. \quad (1063)$$

When $\lambda \rightarrow \infty$, the penalized coefficients shrink toward zero:

$$\hat{\boldsymbol{\beta}}_\lambda \rightarrow \mathbf{0}. \quad (1064)$$

In the capital accumulation framework, ridge estimation is useful when $\mathbf{x}_{it}$ contains many related indicators, such as technology, finance, labor, demand, and ecological variables that move together. It reduces instability in coefficient estimates while retaining all features in the model.

The regularized estimator can also be applied to polynomial and interaction models. If X contains terms such as $x_j x_\ell$ or basis functions $\psi_m(\mathbf{x})$, ridge stabilizes estimation of nonlinear accumulation effects:

$$Y_i = \sum_{m=1}^M \theta_m \psi_m(\mathbf{x}_i) + u_i. \quad (1065)$$

In this case, the ridge penalty controls the complexity of the estimated accumulation surface.

15.3 Penalized High-Dimensional Estimator

When the number of candidate features is large, it may be desirable to select a sparse subset of relevant predictors. The LASSO estimator solves

$$\hat{\boldsymbol{\beta}}_{LASSO} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\mathbf{Y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \right\}. \quad (1066)$$

The ℓ_1 penalty is

$$\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|. \quad (1067)$$

Unlike ridge regression, the LASSO can set some coefficients exactly equal to zero.

The LASSO objective is convex but not differentiable at zero. Its optimality conditions can be written using subgradients:

$$-\frac{1}{n}X^\top(\mathbf{Y} - X\hat{\boldsymbol{\beta}}_{LASSO}) + \lambda \mathbf{s} = \mathbf{0}, \quad (1068)$$

where

$$s_j = \begin{cases} \text{sign}(\hat{\beta}_j), & \hat{\beta}_j \neq 0, \\ \in [-1, 1], & \hat{\beta}_j = 0. \end{cases} \quad (1069)$$

A coefficient is set to zero when the corresponding feature has insufficient partial correlation with the residual:

$$\left| \frac{1}{n}X_j^\top(\mathbf{Y} - X\hat{\boldsymbol{\beta}}_{LASSO}) \right| \leq \lambda \implies \hat{\beta}_j = 0 \text{ is admissible.} \quad (1070)$$

Thus, the LASSO performs estimation and feature selection simultaneously.

The selected feature set is

$$\hat{S} = \{j : \hat{\beta}_{j,LASSO} \neq 0\}. \quad (1071)$$

In the accumulation framework, this set identifies the features that are most predictive of capital changes after penalization. These may include individual variables, interaction terms, polynomial terms, or sieve basis functions.

The LASSO is especially useful when the candidate accumulation formula is high-dimensional:

$$\Delta K_t = \alpha + \sum_{j=1}^d \beta_j x_{jt} + \sum_{j < \ell} \gamma_{j\ell} x_{jt} x_{\ell t} + \sum_{|\nu| \leq p} a_\nu \mathbf{x}_t^\nu + u_t. \quad (1072)$$

The number of possible nonlinear terms can grow rapidly, and sparse penalization helps isolate the most relevant channels.

A common extension is the elastic net estimator:

$$\hat{\boldsymbol{\beta}}_{EN} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\mathbf{Y} - X\boldsymbol{\beta}\|_2^2 + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2 \right\}. \quad (1073)$$

This combines sparsity with ridge-type stabilization and is useful when features are highly correlated.

For causal interpretation, variable selection alone is not sufficient. If features are endogenous, penalized estimation may improve prediction but does not by itself solve identification. The selected coefficients should therefore be interpreted structurally only under valid orthogonality or with appropriate instruments, designs, or controls.

15.4 Kernel Estimator

A fully nonparametric alternative estimates the conditional mean function directly. Let X_i denote the state vector and Y_i the observed capital outcome or increment. The Nadaraya–Watson kernel estimator is

$$\hat{m}(x) = \frac{\sum_{i=1}^n \mathcal{K}_h(X_i - x) Y_i}{\sum_{i=1}^n \mathcal{K}_h(X_i - x)}, \quad \mathcal{K}_h(u) = h^{-d} \mathcal{K}_0(u/h). \quad (1074)$$

Here, $\mathcal{K}_0$ is a kernel function, $h > 0$ is a bandwidth, and d is the dimension of the state vector.

The estimator can be written as a weighted average:

$$\hat{m}(x) = \sum_{i=1}^n w_i(x) Y_i, \quad (1075)$$

where

$$w_i(x) = \frac{\mathcal{K}_h(X_i - x)}{\sum_{\ell=1}^n \mathcal{K}_h(X_\ell - x)}. \quad (1076)$$

The weights satisfy

$$w_i(x) \geq 0, \quad \sum_{i=1}^n w_i(x) = 1 \quad (1077)$$

when the kernel is nonnegative.

In the capital accumulation setting, one may set

$$Y_i = K_{i,t+1} - K_{it}, \quad X_i = (K_{it}, \mathbf{x}_{it}^\top)^\top. \quad (1078)$$

Then the kernel estimator estimates the conditional accumulation function:

$$\hat{\mathcal{A}}(K, \mathbf{x}) = \hat{m}(K, \mathbf{x}) \approx \mathbb{E}[K_{i,t+1} - K_{it} \mid K_{it} = K, \mathbf{x}_{it} = \mathbf{x}]. \quad (1079)$$

If instead $Y_i = K_{i,t+1}$, then the estimated accumulation function is

$$\hat{\mathcal{A}}(K, \mathbf{x}) = \hat{m}(K, \mathbf{x}) - K. \quad (1080)$$

The bandwidth h controls the degree of smoothing. A small bandwidth uses only observations close to x and produces a flexible but variable estimate. A large bandwidth uses many observations and produces a smoother but more biased estimate:

$$h \downarrow \implies \text{lower bias and higher variance}, \quad h \uparrow \implies \text{higher bias and lower variance}. \quad (1081)$$

Under standard regularity conditions, consistency requires

$$h \rightarrow 0, \quad nh^d \rightarrow \infty. \quad (1082)$$

The first condition ensures that the local neighborhood shrinks, reducing bias. The second ensures that enough observations remain within the local neighborhood, reducing variance.

The kernel estimator is useful because it imposes minimal functional-form restrictions. It can reveal nonlinearities, turning points, and local heterogeneity in capital accumulation. Its main limitation is the curse of dimensionality: as d grows, the required sample size increases rapidly. For high-dimensional feature vectors, kernel estimation may therefore require dimension reduction, additive structure, or sieve methods.

A local-linear alternative reduces boundary bias by solving

$$(\hat{a}(x), \hat{\mathbf{b}}(x)) = \arg \min_{a, \mathbf{b}} \sum_{i=1}^n \mathcal{K}_h(X_i - x) \left[Y_i - a - \mathbf{b}^\top (X_i - x) \right]^2. \quad (1083)$$

The local-linear estimate is

$$\hat{m}_{LL}(x) = \hat{a}(x). \quad (1084)$$

This version provides both a local estimate of the accumulation function and a local estimate of its gradient through $\widehat{\mathbf{b}}(x)$.

Together, OLS, ridge, LASSO, and kernel estimation provide a hierarchy of statistical tools for the capital accumulation model:

$$\text{linear estimation} \longrightarrow \text{regularized estimation} \longrightarrow \text{high-dimensional selection} \longrightarrow \text{nonparametric estimation} \quad (1085)$$

Each estimator corresponds to a different balance among interpretability, stability, flexibility, and data requirements.

15.5 Generalized Method of Moments

The generalized method of moments provides a unified framework for estimating accumulation models defined by orthogonality restrictions. Let W_i denote the observed data for unit i , let $\boldsymbol{\theta}$ denote the parameter vector, and let $\mathbf{g}(W_i, \boldsymbol{\theta})$ be a vector of moment functions. The GMM estimator is

$$\widehat{\boldsymbol{\theta}}_{GMM} = \arg \min_{\boldsymbol{\theta}} \bar{\mathbf{g}}_n(\boldsymbol{\theta})^\top W_n \bar{\mathbf{g}}_n(\boldsymbol{\theta}), \quad \bar{\mathbf{g}}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(W_i, \boldsymbol{\theta}). \quad (1086)$$

Here, W_n is a positive semidefinite weighting matrix. The estimator chooses the parameter value that makes the sample moments as close to zero as possible.

In the capital accumulation framework, a natural moment condition is

$$\mathbb{E}[\mathbf{z}_i (\Delta K_i - \mathcal{A}(K_i, \mathbf{x}_i; \boldsymbol{\theta}_0))] = \mathbf{0}, \quad (1087)$$

where $\mathbf{z}_i$ is a vector of instruments or conditioning variables. The corresponding sample moment is

$$\bar{\mathbf{g}}_n(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^n \mathbf{z}_i [\Delta K_i - \mathcal{A}(K_i, \mathbf{x}_i; \boldsymbol{\theta})]. \quad (1088)$$

For a linear accumulation model,

$$\Delta K_i = \mathbf{x}_i^\top \boldsymbol{\beta} + u_i, \quad (1089)$$

the moment function becomes

$$\mathbf{g}(W_i, \boldsymbol{\beta}) = \mathbf{z}_i (\Delta K_i - \mathbf{x}_i^\top \boldsymbol{\beta}). \quad (1090)$$

If $\mathbf{z}_i = \mathbf{x}_i$, GMM reduces to ordinary least squares under the identity weighting matrix. If $\mathbf{z}_i$ contains external instruments, the same framework yields instrumental-variables estimation.

The population moment condition is

$$\mathbb{E}[\mathbf{g}(W_i, \boldsymbol{\theta}_0)] = \mathbf{0}. \quad (1091)$$

Identification requires that this condition uniquely determines $\boldsymbol{\theta}_0$:

$$\mathbb{E}[\mathbf{g}(W_i, \boldsymbol{\theta})] = \mathbf{0} \iff \boldsymbol{\theta} = \boldsymbol{\theta}_0. \quad (1092)$$

When the number of moments equals the number of parameters, the model is exactly identified. In that case, the weighting matrix does not affect the asymptotic point estimate:

$$\dim(\mathbf{g}) = \dim(\boldsymbol{\theta}) \implies \bar{\mathbf{g}}_n(\widehat{\boldsymbol{\theta}}) = \mathbf{0}. \quad (1093)$$

When the number of moments exceeds the number of parameters, the model is overidentified:

$$\dim(\mathbf{g}) > \dim(\boldsymbol{\theta}). \quad (1094)$$

In this case, the estimator minimizes a weighted distance because the sample moments cannot generally all be set exactly equal to zero.

The optimal weighting matrix is the inverse of the long-run covariance matrix of the moments:

$$\mathbf{W}^* = \Omega^{-1}, \quad \Omega = \mathbb{E}[\mathbf{g}(W_i, \boldsymbol{\theta}_0)\mathbf{g}(W_i, \boldsymbol{\theta}_0)^\top]. \quad (1095)$$

With serial correlation or clustering, Ω should be replaced by the appropriate long-run or cluster-robust covariance matrix.

A common two-step GMM procedure first estimates $\boldsymbol{\theta}$ using an initial weighting matrix $\mathbf{W}_n^{(1)}$, then estimates the moment covariance matrix at the first-step estimate:

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(W_i, \hat{\boldsymbol{\theta}}^{(1)})\mathbf{g}(W_i, \hat{\boldsymbol{\theta}}^{(1)})^\top. \quad (1096)$$

The second-step estimator then uses

$$\mathbf{W}_n^{(2)} = \hat{\Omega}^{-1}. \quad (1097)$$

GMM is particularly useful for the present framework because it accommodates endogenous features, instrumental variables, nonlinear accumulation functions, dynamic panel moments, and overidentifying restrictions within one estimation structure.

15.6 Asymptotic Normality

Under standard regularity conditions, the estimator is asymptotically normal:

$$\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, V), \quad V = (G^\top \mathbf{W} G)^{-1} G^\top \mathbf{W} \Omega \mathbf{W} G (G^\top \mathbf{W} G)^{-1}. \quad (1098)$$

Here,

$$G = \mathbb{E} \left[\frac{\partial \mathbf{g}(W_i, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}^\top} \right] \quad (1099)$$

is the Jacobian matrix of the moment conditions, and

$$\Omega = \mathbb{E}[\mathbf{g}(W_i, \boldsymbol{\theta}_0)\mathbf{g}(W_i, \boldsymbol{\theta}_0)^\top] \quad (1100)$$

is the covariance matrix of the moments.

The matrix V is the asymptotic variance of the estimator. It captures three objects: the sensitivity of the moment conditions to the parameters, the weighting matrix used in estimation, and the variability of the empirical moments. If the optimal weighting matrix is used,

$$\mathbf{W} = \Omega^{-1}, \quad (1101)$$

then the asymptotic variance simplifies to

$$\mathbf{V}_{\text{opt}} = (G^\top \Omega^{-1} G)^{-1}. \quad (1102)$$

For a linear least-squares model with homoskedastic errors, the familiar asymptotic variance is obtained as a special case:

$$V_{OLS} = \sigma^2 \left(\mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top] \right)^{-1}. \quad (1103)$$

With heteroskedastic errors, the robust sandwich form is

$$V_{HC} = Q^{-1} \Omega_x Q^{-1}, \quad (1104)$$

where

$$Q = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top], \quad \Omega_x = \mathbb{E}[u_i^2 \mathbf{x}_i \mathbf{x}_i^\top]. \quad (1105)$$

The asymptotic normality result provides the basis for confidence intervals and hypothesis tests. For the j th component of $\boldsymbol{\theta}$, an asymptotic confidence interval is

$$\hat{\theta}_j \pm z_{1-\alpha/2} \sqrt{\frac{\hat{V}_{jj}}{n}}. \quad (1106)$$

Here, $\hat{V}_{jj}$ is the j th diagonal element of a consistent estimate of V .

The Wald statistic for testing linear restrictions

$$H_0 : R\boldsymbol{\theta} = r \quad (1107)$$

is

$$Wald = n(R\hat{\boldsymbol{\theta}} - r)^\top (R\hat{V}R^\top)^{-1} (R\hat{\boldsymbol{\theta}} - r). \quad (1108)$$

Under the null hypothesis,

$$Wald \xrightarrow{d} \chi_q^2, \quad (1109)$$

where q is the number of restrictions.

In the accumulation setting, such tests can be used to evaluate whether specific features affect capital accumulation, whether interaction terms are jointly significant, whether threshold regimes differ, or whether classical value-theoretic restrictions are consistent with the data.

15.7 Variance Estimation

Valid inference requires consistent estimation of the variance of the estimator. Under heteroskedasticity, the heteroskedasticity-consistent variance estimator is

$$\hat{V}_{HC} = (X^\top X)^{-1} \left(\sum_{i=1}^n \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i^\top \right) (X^\top X)^{-1}, \quad \hat{V}_{cluster} = (X^\top X)^{-1} \left(\sum_{g=1}^G X_g^\top \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g^\top X_g \right) (X^\top X)^{-1}. \quad (1110)$$

The first estimator is robust to arbitrary heteroskedasticity across observations. The second is robust to arbitrary correlation within clusters.

The heteroskedasticity-consistent estimator can be written in matrix form as

$$\hat{V}_{HC} = (X^\top X)^{-1} X^\top \hat{\Omega} X (X^\top X)^{-1}, \quad (1111)$$

where

$$\hat{\Omega} = \text{diag}(\hat{u}_1^2, \dots, \hat{u}_n^2). \quad (1112)$$

This estimator allows the variance of u_i to vary with the feature vector:

$$\text{Var}(u_i \mid \mathbf{x}_i) = \sigma_i^2. \quad (1113)$$

Cluster-robust variance estimation is appropriate when observations are independent across clusters but may be correlated within clusters. Let $g = 1, \dots, G$ index clusters, and let X_g and $\hat{\mathbf{u}}_g$ collect the observations and residuals in cluster g . The cluster score is

$$\mathbf{s}_g = X_g^\top \hat{\mathbf{u}}_g. \quad (1114)$$

Then the middle matrix in the cluster-robust estimator is

$$\sum_{g=1}^G \mathbf{s}_g \mathbf{s}_g^\top = \sum_{g=1}^G X_g^\top \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g^\top X_g. \quad (1115)$$

In panel applications, clustering by unit allows arbitrary serial correlation within each unit:

$$\mathbb{E}[u_{it} u_{is} \mid i] \neq 0 \quad \text{for } t \neq s. \quad (1116)$$

Clustering by time allows arbitrary cross-sectional dependence within a period:

$$\mathbb{E}[u_{it} u_{jt} \mid t] \neq 0 \quad \text{for } i \neq j. \quad (1117)$$

When both forms are important, two-way clustering can be used:

$$\hat{V}_{\text{two-way}} = \hat{V}_{\text{unit}} + \hat{V}_{\text{time}} - \hat{V}_{\text{intersection}}. \quad (1118)$$

For GMM estimation, the general sandwich variance estimator is

$$\hat{V}_{GMM} = (\hat{G}^\top W_n \hat{G})^{-1} \hat{G}^\top W_n \hat{\Omega} W_n \hat{G} (\hat{G}^\top W_n \hat{G})^{-1}, \quad (1119)$$

where

$$\hat{G} = \frac{1}{n} \sum_{i=1}^n \frac{\partial \mathbf{g}(W_i, \hat{\boldsymbol{\theta}})}{\partial \boldsymbol{\theta}^\top}, \quad (1120)$$

and

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \mathbf{g}(W_i, \hat{\boldsymbol{\theta}}) \mathbf{g}(W_i, \hat{\boldsymbol{\theta}})^\top. \quad (1121)$$

With the optimal weighting matrix, this simplifies to

$$\hat{V}_{GMM, \text{opt}} = (\hat{G}^\top \hat{\Omega}^{-1} \hat{G})^{-1}. \quad (1122)$$

Variance estimation is essential for determining whether estimated accumulation effects are statistically distinguishable from zero. In the multi-feature capital framework, robust inference is especially important because heteroskedasticity, serial correlation, cross-sectional dependence, and clustered shocks are likely to arise in firm, sectoral, regional, and macroeconomic data.

The statistical estimation section can therefore be summarized as

$$\text{estimation+orthogonality+robust variance} \implies \text{interpretable inference on capital accumulation.} \quad (1123)$$

16 Hypothesis Testing

The preceding sections introduce a hierarchy of accumulation models, ranging from linear specifications to nonlinear coupling, threshold regimes, and infinite-dimensional representations. Hypothesis testing provides a formal way to evaluate whether the data support these additional structures. This section presents tests for linearity, nonlinear coupling, and threshold effects.

16.1 Linearity Test

A first empirical question is whether the accumulation function is adequately described by a linear multi-feature formula, or whether interaction terms are needed. Consider the unrestricted interaction model

$$\Delta K_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t + \sum_{j < \ell} \gamma_{j\ell} x_{jt} x_{\ell t} + u_t. \quad (1124)$$

The restricted linear model imposes that all interaction coefficients are zero:

$$\Delta K_t = \alpha + \boldsymbol{\beta}^\top \mathbf{x}_t + u_t. \quad (1125)$$

The null hypothesis is

$$H_0 : \gamma_{j\ell} = 0 \quad \forall j < \ell, \quad F = \frac{(SSR_r - SSR_u)/q}{SSR_u/(n - k_u)}. \quad (1126)$$

Here, SSR_r is the sum of squared residuals from the restricted linear model, SSR_u is the sum of squared residuals from the unrestricted interaction model, q is the number of restrictions, and k_u is the number of parameters in the unrestricted model.

The number of excluded pairwise interaction terms is

$$q = \binom{d}{2} = \frac{d(d-1)}{2}, \quad (1127)$$

when all pairwise interactions among d features are tested jointly. Under the null hypothesis and standard homoskedastic linear-model assumptions,

$$F \sim F_{q, n-k_u}. \quad (1128)$$

The test compares the fit of the restricted and unrestricted models. If the interaction terms do not improve fit sufficiently, then

$$SSR_r - SSR_u \approx 0, \quad (1129)$$

and the evidence favors the linear accumulation formula. If the reduction in residual variation is large relative to the unrestricted residual variance, then the null hypothesis is rejected:

$$F > F_{q, n-k_u}(1 - \alpha) \implies \text{reject } H_0. \quad (1130)$$

Rejection of the linearity null indicates that the marginal effect of at least one feature depends on the level of another feature. In economic terms, the determinants of capital accumulation are not additively separable:

$$\exists j, \ell \quad \text{such that} \quad \frac{\partial^2 \mathcal{A}(\mathbf{x})}{\partial x_j \partial x_\ell} \neq 0. \quad (1131)$$

This provides empirical support for the nonlinear coupling framework.

In panel settings, the same test can be applied after including unit and time fixed effects:

$$\Delta K_{it} = \alpha_i + \tau_t + \boldsymbol{\beta}^\top \mathbf{x}_{it} + \sum_{j < \ell} \gamma_{j\ell} x_{jit} x_{lit} + u_{it}. \quad (1132)$$

When errors are heteroskedastic or clustered, the classical F statistic should be replaced by a robust Wald test:

$$W = (R\hat{\boldsymbol{\theta}} - r)^\top \left[R\widehat{\text{Var}}(\hat{\boldsymbol{\theta}})R^\top \right]^{-1} (R\hat{\boldsymbol{\theta}} - r), \quad (1133)$$

where $R\boldsymbol{\theta} = r$ encodes the restrictions $\gamma_{j\ell} = 0$ for all $j < \ell$.

16.2 Coupling Test

The coupling test evaluates whether the second-order interaction matrix is statistically different from zero. In the quadratic accumulation model,

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad (1134)$$

the matrix $\mathbf{H}$ captures curvature, complementarity, substitution, and higher-order local structure. The null hypothesis is

$$H_0 : \mathbf{H} = \mathbf{0}, \quad W = n \hat{\mathbf{h}}^\top \left[\widehat{\text{Var}}(\hat{\mathbf{h}}) \right]^{-1} \hat{\mathbf{h}} \xrightarrow{d} \chi_q^2. \quad (1135)$$

Here, $\hat{\mathbf{h}}$ stacks the estimated distinct elements of $\mathbf{H}$, and q is the number of tested second-order coefficients.

If the full symmetric Hessian is tested, then

$$q = \frac{d(d+1)}{2}. \quad (1136)$$

If only off-diagonal interactions are tested, excluding own-curvature terms, then

$$q = \frac{d(d-1)}{2}. \quad (1137)$$

The null hypothesis states that the accumulation function is locally linear in the feature vector:

$$H_0 : \mathbf{H} = \mathbf{0} \implies \nabla_{\mathbf{x}}^2 \mathcal{A}(\mathbf{x}) = \mathbf{0}. \quad (1138)$$

Under the alternative, at least one second-order term is nonzero:

$$H_1 : \mathbf{H} \neq \mathbf{0}. \quad (1139)$$

The Wald statistic in Equation (1135) measures whether the estimated second-order coefficients are jointly large relative to their sampling uncertainty. The rejection rule is

$$W > \chi_q^2(1 - \alpha) \implies \text{reject } H_0. \quad (1140)$$

Rejection indicates that the data support nonlinear accumulation effects.

The coupling test can also be decomposed into economically interpretable components. A test of own curvature imposes

$$H_0 : h_{jj} = 0 \quad \forall j, \quad (1141)$$

while a test of cross-feature coupling imposes

$$H_0 : h_{j\ell} = 0 \quad \forall j \neq \ell. \quad (1142)$$

The first evaluates diminishing or increasing marginal returns. The second evaluates complementarity and substitution among different features.

For a particular pair (j, ℓ) , the individual test is

$$H_0 : h_{j\ell} = 0, \quad t_{j\ell} = \frac{\hat{h}_{j\ell}}{\widehat{\text{se}}(\hat{h}_{j\ell})}. \quad (1143)$$

A positive and significant estimate implies complementarity:

$$\hat{h}_{j\ell} > 0 \implies \text{evidence of complementarity.} \quad (1144)$$

A negative and significant estimate implies substitution:

$$\hat{h}_{j\ell} < 0 \implies \text{evidence of substitution.} \quad (1145)$$

Thus, the coupling test provides a formal bridge between the theoretical interaction matrix and empirical evidence on whether capital accumulation is governed by independent features or by nonlinear feature relations.

16.3 Threshold Test

The threshold test evaluates whether the accumulation law differs across regimes defined by a threshold variable. Consider the threshold regression model

$$\Delta K_t = \mathbf{x}_t^\top \beta_1 \mathbf{1}\{q_t \leq \tau\} + \mathbf{x}_t^\top \beta_2 \mathbf{1}\{q_t > \tau\} + u_t. \quad (1146)$$

The null hypothesis of no threshold effect is

$$H_0 : \beta_1 = \beta_2, \quad LR(\tau) = \frac{SSR_0 - SSR_1(\tau)}{\hat{\sigma}^2}, \quad \sup_{\tau \in \mathcal{T}} LR(\tau). \quad (1147)$$

Here, SSR_0 is the residual sum of squares under the no-threshold model, $SSR_1(\tau)$ is the residual sum of squares under the threshold model for a candidate threshold τ , and $\hat{\sigma}^2$ is an estimate of the error variance.

Under the null, the model reduces to a single-regime specification:

$$\Delta K_t = \mathbf{x}_t^\top \beta + u_t. \quad (1148)$$

Under the alternative, the slope vector changes across regimes:

$$\beta_1 \neq \beta_2. \quad (1149)$$

The threshold variable q_t may represent leverage, ecological burden, network scale, capacity utilization, market stress, or any state variable that may alter the accumulation mechanism.

For each candidate threshold $\tau \in \mathcal{T}$, the threshold model is estimated and the likelihood-ratio-type statistic is computed:

$$LR(\tau) = \frac{SSR_0 - SSR_1(\tau)}{\hat{\sigma}^2}. \quad (1150)$$

The test statistic is the supremum over the admissible threshold set:

$$SupLR = \sup_{\tau \in \mathcal{T}} LR(\tau). \quad (1151)$$

The estimated threshold is the value that minimizes the unrestricted residual sum of squares:

$$\hat{\tau} = \arg \min_{\tau \in \mathcal{T}} SSR_1(\tau). \quad (1152)$$

Equivalently, it maximizes the improvement in fit relative to the restricted model:

$$\hat{\tau} = \arg \max_{\tau \in \mathcal{T}} LR(\tau). \quad (1153)$$

The admissible set $\mathcal{T}$ is typically trimmed to avoid regimes with too few observations:

$$\mathcal{T} = \{\tau : \pi_0 \leq \widehat{\Pr}(q_t \leq \tau) \leq 1 - \pi_0\}, \quad (1154)$$

where $\pi_0 \in (0, 1/2)$ is a trimming parameter. This ensures that both regimes are empirically meaningful.

The distribution of the threshold test statistic is generally nonstandard because the threshold parameter is not identified under the null hypothesis. Therefore, inference is commonly based on simulation or bootstrap procedures. A bootstrap version draws residuals under the null model, constructs bootstrap outcomes, and recomputes

$$SupLR^* = \sup_{\tau \in \mathcal{T}} LR^*(\tau). \quad (1155)$$

The bootstrap p -value is

$$p^* = \widehat{\Pr}^*(SupLR^* \geq SupLR). \quad (1156)$$

Rejecting the null hypothesis indicates that the accumulation function is regime dependent:

$$\mathcal{A}_1(\mathbf{x}) \neq \mathcal{A}_2(\mathbf{x}) \quad \text{across} \quad q_t \leq \tau \quad \text{and} \quad q_t > \tau. \quad (1157)$$

This provides empirical support for financial, ecological, network, or historical threshold mechanisms in the capital accumulation process.

Together, the linearity, coupling, and threshold tests provide a sequence of increasingly rich empirical diagnostics:

$$\text{linear accumulation} \quad \longrightarrow \quad \text{nonlinear coupling} \quad \longrightarrow \quad \text{regime-dependent accumulation.} \quad (1158)$$

They allow the researcher to determine whether a simple linear formula is sufficient, whether second-order feature interactions matter, and whether the accumulation law changes across economically meaningful regimes.

16.4 Monotonicity Test

A monotonicity test evaluates whether a feature has the theoretically expected directional effect on capital accumulation over the entire admissible state space. For features expected to increase accumulation, the null hypothesis is

$$H_0 : \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}, \quad j \in \mathcal{J}_+. \quad (1159)$$

Here, $\mathcal{J}_+$ denotes the set of features whose marginal effect is expected to be nonnegative.

Given an estimator $\hat{\mathcal{A}}$, define the standardized minimum marginal-effect statistic as

$$T_n = \inf_{\mathbf{x} \in \mathcal{X}, j \in \mathcal{J}_+} \frac{\partial \hat{\mathcal{A}}(\mathbf{x}) / \partial x_j}{\hat{s}_j(\mathbf{x})}. \quad (1160)$$

The denominator $\hat{s}_j(\mathbf{x})$ is the estimated standard error of the marginal-effect estimator $\partial \hat{\mathcal{A}}(\mathbf{x}) / \partial x_j$. The intuition is straightforward. Under the null hypothesis, the estimated marginal effects should not be substantially negative anywhere in the state space. Large negative values of T_n provide evidence against global monotonicity:

$$T_n \ll 0 \implies \text{evidence against } H_0. \quad (1161)$$

The alternative hypothesis is

$$H_1 : \exists \mathbf{x} \in \mathcal{X}, j \in \mathcal{J}_+ \text{ such that } \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} < 0. \quad (1162)$$

Thus, rejection means that at least one feature expected to support accumulation has a negative marginal effect somewhere in the admissible domain.

For features expected to reduce accumulation, one may analogously test

$$H_0 : \frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} \leq 0, \quad \forall \mathbf{x} \in \mathcal{X}, \quad j \in \mathcal{J}_-. \quad (1163)$$

The corresponding statistic is

$$T_n^- = \sup_{\mathbf{x} \in \mathcal{X}, j \in \mathcal{J}_-} \frac{\partial \hat{\mathcal{A}}(\mathbf{x}) / \partial x_j}{\hat{s}_j(\mathbf{x})}. \quad (1164)$$

Large positive values of T_n^- indicate violations of the expected negative monotonicity restriction.

In parametric models, the monotonicity condition can often be translated into restrictions on coefficients. For the linear model

$$\mathcal{A}(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}, \quad (1165)$$

the positive monotonicity null becomes

$$H_0 : \beta_j \geq 0, \quad j \in \mathcal{J}_+. \quad (1166)$$

For the quadratic model

$$\mathcal{A}(\mathbf{x}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}, \quad (1167)$$

the marginal effect is

$$\frac{\partial \mathcal{A}(\mathbf{x})}{\partial x_j} = a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell, \quad (1168)$$

so monotonicity requires

$$a_j + \sum_{\ell=1}^d h_{j\ell} x_\ell \geq 0, \quad \forall \mathbf{x} \in \mathcal{X}, \quad j \in \mathcal{J}_+. \quad (1169)$$

In nonparametric models, the test is naturally conducted over a grid or sample support. Let $\mathcal{X}_n$ denote a finite evaluation set. Then the sample analogue is

$$T_{n, \mathcal{X}_n} = \min_{\mathbf{x} \in \mathcal{X}_n, j \in \mathcal{J}_+} \frac{\partial \hat{\mathcal{A}}(\mathbf{x}) / \partial x_j}{\hat{s}_j(\mathbf{x})}. \quad (1170)$$

Critical values may be obtained using bootstrap methods that account for estimation error in the derivative process.

The monotonicity test is useful because economic theory often predicts the direction, but not necessarily the magnitude, of a feature's effect. It therefore provides a weaker and more robust test than imposing a fully parametric accumulation formula:

$$\text{monotonicity} \neq \text{linearity}. \quad (1171)$$

A model may be nonlinear and still satisfy the expected sign restrictions over the relevant domain.

16.5 Infinite-Dimensional Relevance Test

The infinite-dimensional linear component is governed by the coefficient function $b(\omega)$. A natural relevance test asks whether this functional coefficient is identically zero:

$$H_0 : b(\omega) = 0 \text{ a.e.}, \quad T_b = n \sum_{m=1}^{M_n} \frac{\hat{b}_m^2}{\hat{\sigma}_m^2}. \quad (1172)$$

Here, $\hat{b}_m$ is the estimated coefficient of b along basis function ϕ_m , $\hat{\sigma}_m^2$ is its estimated variance, and M_n is the number of basis terms used in the test.

Using the basis expansion

$$b(\omega) = \sum_{m=1}^{\infty} b_m \phi_m(\omega), \quad (1173)$$

the null hypothesis is equivalent to

$$H_0 : b_m = 0 \quad \forall m \geq 1. \quad (1174)$$

Since only a finite number of coefficients can be tested in finite samples, the operational null is

$$H_{0, M_n} : b_m = 0, \quad m = 1, \dots, M_n. \quad (1175)$$

The statistic T_b aggregates standardized squared coefficient estimates. Large values indicate that the feature function has a statistically detectable linear contribution to capital accumulation:

$$T_b \text{ large} \implies \text{reject functional irrelevance of } b. \quad (1176)$$

If the coefficient estimates are asymptotically normal and approximately independent after orthogonalization, then under the truncated null,

$$T_b \xrightarrow{d} \chi_{M_n}^2 \quad (1177)$$

for fixed M_n . When M_n grows with the sample size, the limiting distribution may require normalization or simulation:

$$M_n \rightarrow \infty, \quad \frac{M_n}{n} \rightarrow 0. \quad (1178)$$

The economic interpretation is direct. If the null is not rejected, the data do not provide evidence that the distributed feature function has a direct linear effect on capital accumulation. If the null is rejected, then at least some functional directions are relevant:

$$\exists m \leq M_n \quad \text{such that} \quad b_m \neq 0. \quad (1179)$$

In functional terms, there exists a region or mode of the feature domain whose intensity contributes linearly to accumulation.

One may also test relevance over a subset of basis directions. For example, if $\mathcal{M}_0 \subset \{1, \dots, M_n\}$ denotes a group of economically meaningful functional modes, the group relevance statistic is

$$T_{b, \mathcal{M}_0} = n \sum_{m \in \mathcal{M}_0} \frac{\hat{b}_m^2}{\hat{\sigma}_m^2}. \quad (1180)$$

This allows the researcher to test whether particular sectors, spatial patterns, technological modes, or ecological components matter for accumulation.

16.6 Kernel Interaction Test

The quadratic infinite-dimensional component is governed by the kernel $\kappa(\omega, \omega')$ or, equivalently, by the integral operator $\mathcal{K}$. A natural test asks whether the kernel interaction is absent:

$$H_0 : \kappa(\omega, \omega') = 0 \text{ a.e.}, \quad T_\kappa = n \|\hat{\mathcal{K}}\|_{HS}^2, \quad \|\mathcal{K}\|_{HS}^2 = \iint \kappa^2 d\mu d\mu. \quad (1181)$$

The null hypothesis states that there is no quadratic functional coupling among the distributed features.

Using a basis expansion, the kernel can be represented by coefficients

$$\kappa_{mn} = \langle \phi_m, \mathcal{K} \phi_n \rangle_{\mathcal{H}}. \quad (1182)$$

The Hilbert–Schmidt norm is then

$$\|\mathcal{K}\|_{HS}^2 = \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \kappa_{mn}^2. \quad (1183)$$

The finite-sample test typically uses a truncated version:

$$T_{\kappa, M_n} = n \sum_{m=1}^{M_n} \sum_{n=1}^{M_n} \hat{\kappa}_{mn}^2. \quad (1184)$$

If the kernel is symmetric, only the distinct elements need to be counted:

$$\kappa_{mn} = \kappa_{nm}, \quad 1 \leq m \leq n \leq M_n. \quad (1185)$$

The number of distinct tested coefficients is then

$$q_\kappa = \frac{M_n(M_n + 1)}{2}. \quad (1186)$$

The null hypothesis can be written as

$$H_0 : \mathcal{K} = 0, \quad (1187)$$

which implies

$$\langle x, \mathcal{K}x \rangle_{\mathcal{H}} = 0 \quad \forall x \in \mathcal{H}. \quad (1188)$$

Under the alternative,

$$H_1 : \|\mathcal{K}\|_{HS}^2 > 0. \quad (1189)$$

Rejection of the null implies that distributed features interact in the accumulation process:

$$\exists(\omega, \omega') \text{ such that } \kappa(\omega, \omega') \neq 0. \quad (1190)$$

If the estimated kernel is positive over a region, the corresponding functional components are complementary. If it is negative, they are substitutive:

$$\hat{\kappa}(\omega, \omega') > 0 \Rightarrow \text{functional complementarity}, \quad \hat{\kappa}(\omega, \omega') < 0 \Rightarrow \text{functional substitution}. \quad (1191)$$

Because the kernel test may involve many coefficients, regularization or dimension reduction is often needed. One may test a low-rank approximation

$$\mathcal{K} \approx \sum_{r=1}^R \lambda_r \phi_r \otimes \phi_r, \quad (1192)$$

and evaluate

$$H_0 : \lambda_1 = \dots = \lambda_R = 0. \quad (1193)$$

This focuses the test on the dominant nonlinear directions of the functional accumulation process.

16.7 Overidentification Test

In GMM estimation, overidentification arises when the number of moment conditions exceeds the number of parameters. The overidentification test evaluates whether the full set of moment restrictions is jointly consistent with the data:

$$J = n \bar{\mathbf{g}}_n(\hat{\boldsymbol{\theta}})^\top \widehat{W} \bar{\mathbf{g}}_n(\hat{\boldsymbol{\theta}}), \quad H_0 : \mathbb{E}[\mathbf{g}(W_i, \boldsymbol{\theta}_0)] = \mathbf{0}, \quad J \xrightarrow{d} \chi_{L-K}^2. \quad (1194)$$

Here, L is the number of moment conditions and K is the number of estimated parameters. The difference $L - K$ is the number of overidentifying restrictions.

The test statistic is the minimized GMM objective scaled by the sample size:

$$J = n \cdot Q_n(\hat{\boldsymbol{\theta}}), \quad Q_n(\boldsymbol{\theta}) = \bar{\mathbf{g}}_n(\boldsymbol{\theta})^\top \widehat{W} \bar{\mathbf{g}}_n(\boldsymbol{\theta}). \quad (1195)$$

If the model is exactly identified, then $L = K$ and there are no overidentifying restrictions:

$$L = K \implies \text{no overidentification test.} \quad (1196)$$

Under the null hypothesis, the moment conditions are valid:

$$\mathbb{E}[\mathbf{g}(W_i, \boldsymbol{\theta}_0)] = \mathbf{0}. \quad (1197)$$

A large value of J indicates that the sample moments remain far from zero even at the estimated parameter value:

$$J > \chi_{L-K}^2(1 - \alpha) \implies \text{reject overidentifying restrictions.} \quad (1198)$$

In an instrumental-variables accumulation model, the moment function may be

$$\mathbf{g}(W_i, \boldsymbol{\theta}) = \mathbf{z}_i [\Delta K_i - \mathcal{A}(K_i, \mathbf{x}_i; \boldsymbol{\theta})]. \quad (1199)$$

The overidentification test then evaluates whether the instruments are jointly consistent with the maintained accumulation equation:

$$\mathbb{E} \{ \mathbf{z}_i [\Delta K_i - \mathcal{A}(K_i, \mathbf{x}_i; \boldsymbol{\theta}_0)] \} = \mathbf{0}. \quad (1200)$$

Failure of the overidentification test may indicate invalid instruments, misspecified functional form, omitted dynamics, incorrect exogeneity restrictions, or heterogeneous effects not captured by the model:

$$J \text{ large} \implies \text{some maintained moment restrictions are inconsistent with the data.} \quad (1201)$$

The overidentification test does not reveal which specific moment is invalid. For diagnosis, one may examine individual sample moments:

$$\bar{g}_{\ell,n}(\hat{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{i=1}^n g_{\ell}(W_i, \hat{\boldsymbol{\theta}}), \quad \ell = 1, \dots, L. \quad (1202)$$

Large standardized individual moments can indicate which instruments or restrictions are driving rejection.

Together, the monotonicity, infinite-dimensional relevance, kernel interaction, and overidentification tests extend the hypothesis-testing framework beyond simple parametric restrictions. They allow the researcher to test directional economic theory, functional relevance, nonlinear distributed coupling, and the validity of moment-based identification:

$$\text{hypothesis testing} = \text{shape restrictions} + \text{functional relevance} + \text{kernel interaction} + \text{moment validity.} \quad (1203)$$

17 Unified General Formula

This section consolidates the preceding frameworks into a single unified representation of capital accumulation. It encompasses linear, nonlinear, dynamic, constrained, and infinite-dimensional formulations. The unified formulas provide a consistent notation for theoretical analysis, empirical implementation, and hypothesis testing.

17.1 General Static Formula

The general static accumulation function maps current capital and features to the next-period increment:

$$K^+ = K + \mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta}), \quad \mathcal{A}(K, \mathbf{x}; \boldsymbol{\theta}) = a_0 + \mathbf{a}^\top \mathbf{x} + \frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x} + g(\mathbf{x}). \quad (1204)$$

Here, a_0 is a baseline accumulation term, $\mathbf{a}^\top \mathbf{x}$ captures linear contributions, $\frac{1}{2} \mathbf{x}^\top \mathbf{H} \mathbf{x}$ represents quadratic coupling effects, and $g(\mathbf{x})$ denotes higher-order or nonparametric contributions. The static formulation is appropriate when features are fixed or exogenous over the horizon of interest.

17.2 General Dynamic Formula

The dynamic formula incorporates time evolution, feedback, and stochastic shocks:

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t; \boldsymbol{\theta}) + \varepsilon_t, \quad \mathbf{x}_t = \Gamma(\mathbf{x}_{t-1}, K_t, \boldsymbol{\eta}_t). \quad (1205)$$

The function Γ represents the law of motion for the features, potentially depending on past capital and stochastic innovation $\boldsymbol{\eta}_t$. The error term ε_t captures unobserved shocks or measurement noise. This general dynamic framework includes recursive growth, thresholds, and regime-switching specifications as special cases.

17.3 General Constrained Formula

Capital accumulation is often subject to resource, ecological, market, or policy constraints. Let $\mathcal{F}_t$ denote the feasible set of allocations at time t . The general constrained problem is

$$\max_{\{\mathbf{x}_t\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \delta^t \mathcal{A}(K_t, \mathbf{x}_t) \quad \text{s.t.} \quad K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t), \quad \mathbf{x}_t \in \mathcal{F}_t, \quad (1206)$$

where $\delta \in (0, 1]$ is a discount factor. This formulation generalizes budget-constrained, ecological-constrained, and market-constrained accumulation problems and allows for optimal control, dynamic programming, or Lagrangian approaches.

17.4 General Infinite-Dimensional Formula

For distributed or functional features, the accumulation law can be written in a Hilbert space H :

$$K_{t+1} = K_t + a_0 + \langle b, x_t \rangle_H + \langle x_t, \mathcal{K} x_t \rangle_H + R(x_t), \quad x_t \in H. \quad (1207)$$

Here, $\langle b, x_t \rangle_H$ represents the linear infinite-dimensional effect, $\langle x_t, \mathcal{K} x_t \rangle_H$ is the quadratic kernel-induced interaction, and $R(x_t)$ collects higher-order nonlinear residuals, including nonparametric contributions or sieve approximations. This formula unifies finite-dimensional linear and quadratic accumulation, functional coefficients, kernel interactions, and remainder terms in a single framework suitable for both theoretical and empirical analysis.

The unified formulas highlight that all preceding models—linear, nonlinear, dynamic, constrained, and infinite-dimensional—can be expressed as special cases of the general accumulation law. They provide a canonical representation that facilitates estimation, inference, and policy analysis:

linear $\subset$ quadratic/coupled $\subset$ nonparametric/sieve $\subset$ dynamic/constrained $\subset$ infinite-dimensional. (1208)

17.5 Final Compact Representation (Detailed and Rigorous)

We present a fully general and unified formulation of capital accumulation that explicitly incorporates finite-dimensional, infinite-dimensional, and regime-dependent components, together with stochastic shocks and constraints. This representation is rigorous, suitable for theoretical analysis, econometric implementation, and simulation studies.

$$K_{t+1} = K_t + F(K_t, x_t, x_t(\cdot), S_t; \Theta) + \varepsilon_t, \quad x_t \in \mathcal{F}_t, \quad \mathbb{E}[z_t \varepsilon_t] = 0. \quad (1209)$$

Here:

- $K_t \in \mathbb{R}_+$ is the aggregate or firm-specific capital at time t , with K_0 given as initial condition.
- $x_t \in \mathbb{R}^d$ is a finite-dimensional feature vector including observed inputs, intermediate indicators, and control variables.
- $x_t(\cdot) \in H$ is a functional or infinite-dimensional feature, defined on a Hilbert space H with inner product $\langle \cdot, \cdot \rangle_H$.
- $S_t \in \{1, \dots, R\}$ denotes the realized regime at time t , possibly determined by threshold rules, structural breaks, or stochastic switches.
- Θ denotes the full parameter set, including $\alpha, \beta, H, b, \mathcal{K}$, and regime-specific functions $\{F_r\}_{r=1}^R$.
- ε_t is a mean-zero stochastic disturbance or shock, potentially serially correlated or heteroskedastic, satisfying instrument exogeneity conditions $\mathbb{E}[z_t \varepsilon_t] = 0$.
- $\mathcal{F}_t \subset \mathbb{R}^d$ is the feasible set of finite-dimensional inputs, encoding budget, ecological, or technological constraints.

The mapping F can be decomposed as

$$F(K_t, x_t, x_t(\cdot), S_t; \Theta) = \underbrace{\alpha + \beta^\top x_t + \frac{1}{2} x_t^\top H x_t}_{\text{finite-dimensional block}} + \underbrace{\langle b, x_t \rangle_H + \langle x_t, \mathcal{K} x_t \rangle_H}_{\text{infinite-dimensional block}} + \underbrace{\sum_{r=1}^R \mathbf{1}\{S_t = r\} F_r(\cdot)}_{\text{regime-dependent block}}. \quad (1210)$$

Finite-dimensional block: captures standard linear and quadratic effects, where

- $\alpha \in \mathbb{R}$ is a baseline accumulation term,
- $\beta \in \mathbb{R}^d$ are linear coefficients,
- $H \in \mathbb{R}^{d \times d}$ is a symmetric matrix capturing interactions and curvature.

Infinite-dimensional block: models distributed, functional, or continuous features:

- $b \in H$ is a linear functional coefficient,
- $\mathcal{K} : H \rightarrow H$ is a self-adjoint, bounded integral operator encoding quadratic or kernel interactions,

- $\langle b, x_t \rangle_H$ is the linear contribution, and
- $\langle x_t, \mathcal{K}x_t \rangle_H$ is the quadratic kernel interaction contribution.

Regime-dependent block: incorporates threshold, structural break, or stochastic regime effects. For R regimes, let $F_r(\cdot)$ denote the accumulation function specific to regime r , and

$$\mathbf{1}\{S_t = r\} = \begin{cases} 1, & \text{if regime } r \text{ is realized at time } t, \\ 0, & \text{otherwise.} \end{cases} \quad (1211)$$

The unified representation satisfies the following properties:

1. **Consistency with classical formulas:** setting $b = 0$, $\mathcal{K} = 0$, $F_r = 0$ reduces F to the standard finite-dimensional formula.
2. **Dynamic generalization:** including time-varying x_t , $x_t(\cdot)$, and S_t recovers the dynamic accumulation model $K_{t+1} = K_t + \mathcal{A}(K_t, x_t) + \varepsilon_t$.
3. **Constraint inclusion:** restricting $x_t \in \mathcal{F}_t$ and imposing $\mathbb{E}[z_t \varepsilon_t] = 0$ allows for constrained optimization and valid econometric identification.
4. **Infinite-dimensional convergence:** approximating $x_t(\cdot)$ by a finite basis $\{\phi_m\}_{m=1}^{M_n}$ with sieve estimation yields consistent estimation of b and $\mathcal{K}$ as $M_n \rightarrow \infty$.
5. **Regime flexibility:** arbitrary number R of regimes allows modeling of thresholds, structural breaks, or stochastic Markov switches.

The final representation therefore provides a rigorous, fully unified framework:

$$K_{t+1} = K_t + \underbrace{\mathcal{F}^{(d)}(K_t, \mathbf{x}_t)}_{\text{finite block}} + \underbrace{\mathcal{F}^{(\infty)}(x_t(\cdot))}_{\text{functional block}} + \underbrace{\mathcal{R}(S_t, \mathbf{x}_t)}_{\text{regime block}} + \varepsilon_t. \quad (1212)$$

In Equation (1212), $\mathcal{G}_t$ denotes the total net capital-accumulation increment from period t to period $t+1$. The term $\mathcal{F}^{(d)}(K_t, \mathbf{x}_t)$ represents the finite-dimensional block, including linear effects, quadratic coupling, and observed feature interactions. The term $\mathcal{F}^{(\infty)}(x_t(\cdot))$ denotes the infinite-dimensional or functional block, which captures distributed features through Hilbert-space inner products and kernel operators. The term $\mathcal{R}(S_t, \mathbf{x}_t)$ denotes the regime-dependent block, allowing the accumulation law to vary across thresholds, structural breaks, or latent states. Finally, ε_t is the stochastic disturbance, representing unobserved shocks, measurement error, and residual variation not captured by the systematic accumulation components. This compact form serves as a canonical reference for theory, estimation, and empirical inference across all previously discussed models.

18 Empirical Illustration and Computational Validation

This section provides an empirical and computational validation of the proposed multi-feature capital accumulation formula. The purpose of the exercise is not to claim that the available aggregate macroeconomic data directly measure Marxian capital stock, constant capital, variable capital, or surplus value. Rather, the objective is to show that the theoretical framework developed in

the previous sections can be translated into a reproducible empirical pipeline, estimated from observed data, evaluated out of sample, and interpreted through prediction errors, marginal effects, regime-specific diagnostics, and extrema of the predicted capital-state trajectory.

Throughout this section, K_t should be understood as a *capital-state variable* or *capital-motion proxy*, not as a literal measure of the total capital stock. The empirical target is the one-period change in this observable capital-state proxy:

$$\Delta K_t \equiv K_{t+1} - K_t. \quad (1213)$$

The estimated model therefore evaluates whether the proposed formula can represent the movement of a capital-state variable over time.

The empirical transition equation is

$$K_{t+1} = K_t + \mathcal{A}(K_t, \mathbf{x}_t) + u_t, \quad (1214)$$

or, equivalently,

$$\Delta K_t = \mathcal{A}(K_t, \mathbf{x}_t) + u_t. \quad (1215)$$

The estimated object is the conditional accumulation function

$$\mathcal{A}(K_t, \mathbf{x}_t) = \mathbb{E}[\Delta K_t \mid K_t, \mathbf{x}_t]. \quad (1216)$$

Thus, the empirical exercise is best interpreted as a test of whether the multi-feature formula can model changes in a capital-state variable rather than as a direct structural estimate of total capital stock.

18.1 Empirical Design

The empirical exercise uses quarterly United States macroeconomic data. Since the available dataset does not directly contain Marxian capital categories, we use real aggregate output as an observable capital-state proxy:

$$K_t = \text{real GDP}_t. \quad (1217)$$

The feature vector contains real consumption, real investment, and a real interest-rate variable:

$$\mathbf{x}_t = (x_{\text{cons},t}, x_{\text{inv},t}, x_{\text{rate},t})^\top. \quad (1218)$$

The outcome of interest is the one-period variation of the capital-state proxy:

$$Y_t = \Delta K_t = K_{t+1} - K_t. \quad (1219)$$

This proxy choice is intentionally modest. Real GDP is not identical to capital stock in either the Marxian or national-accounting sense. It is used as a macroeconomic indicator of the movement of the aggregate capitalist economy, allowing the proposed transition formula to be evaluated in a computationally transparent way. The empirical findings should therefore be read as evidence on the model's operational value for capital-state dynamics, not as a complete measurement of value-theoretic capital accumulation.

Table 1: Main empirical performance for predicting the movement of the capital-state variable.

Model	Specification	Training R^2	Test MAE	Test RMSE
Proposed accumulation model	Ridge, $\lambda = 0.8$	0.2373	121.1827	143.1279

18.2 Estimated Specification

The implemented model is a finite-dimensional approximation to the general capital accumulation function. It includes standardized contemporaneous features, lagged features, and second-order terms:

$$\hat{\mathcal{A}}(K_t, \mathbf{x}_t) = \hat{\alpha} + \hat{\boldsymbol{\beta}}^\top \mathbf{x}_t + \hat{\boldsymbol{\lambda}}^\top \mathbf{x}_{t-1} + \frac{1}{2} \mathbf{x}_t^\top \hat{\mathbf{H}} \mathbf{x}_t. \quad (1220)$$

The predicted next-period capital-state variable is therefore

$$\hat{K}_{t+1} = K_t + \hat{\mathcal{A}}(K_t, \mathbf{x}_t). \quad (1221)$$

The feature vector used in estimation can be written as

$$\boldsymbol{\phi}_t = \left(1, \mathbf{x}_t^\top, \mathbf{x}_{t-1}^\top, \text{vech}(\mathbf{x}_t \mathbf{x}_t^\top)^\top \right)^\top. \quad (1222)$$

This construction operationalizes the central theoretical claim of the paper: the movement of the capital-state variable may depend not only on individual macroeconomic features, but also on lagged conditions and second-order feature couplings.

The estimator is a ridge-regularized least-squares estimator:

$$\hat{\boldsymbol{\theta}}_\lambda = \arg \min_{\boldsymbol{\theta}} \left\{ \frac{1}{n} \sum_{t=1}^n \left(\Delta K_t - \boldsymbol{\theta}^\top \boldsymbol{\phi}_t \right)^2 + \lambda \|\boldsymbol{\theta}\|_2^2 \right\}, \quad \lambda = 0.8. \quad (1223)$$

The regularization term stabilizes estimation in the presence of lagged and quadratic features, which can be highly correlated in aggregate macroeconomic data.

18.3 Overall Performance

Table 1 reports the main predictive performance of the proposed model. The model achieves a training R^2 of 0.2373 for the capital-state increment equation. The test-period mean absolute error is 121.1827, and the test-period root mean squared error is 143.1279.

The explanatory power is moderate. This is theoretically reasonable because aggregate capital-state movement is affected by institutional changes, policy shifts, financial disturbances, demand shocks, and crisis dynamics that cannot be fully captured by a small feature vector. The result should therefore be interpreted as evidence that the proposed formula is operational and informative, rather than as a claim of complete predictive dominance.

18.4 Predicted and Observed Capital-State Dynamics

Figure 1 compares the predicted and observed next-period capital-state variable over the test period. The model tracks the broad trajectory of the observed series, including the expansionary movement before the crisis and the subsequent weakening around the 2007–2008 period. The predicted series

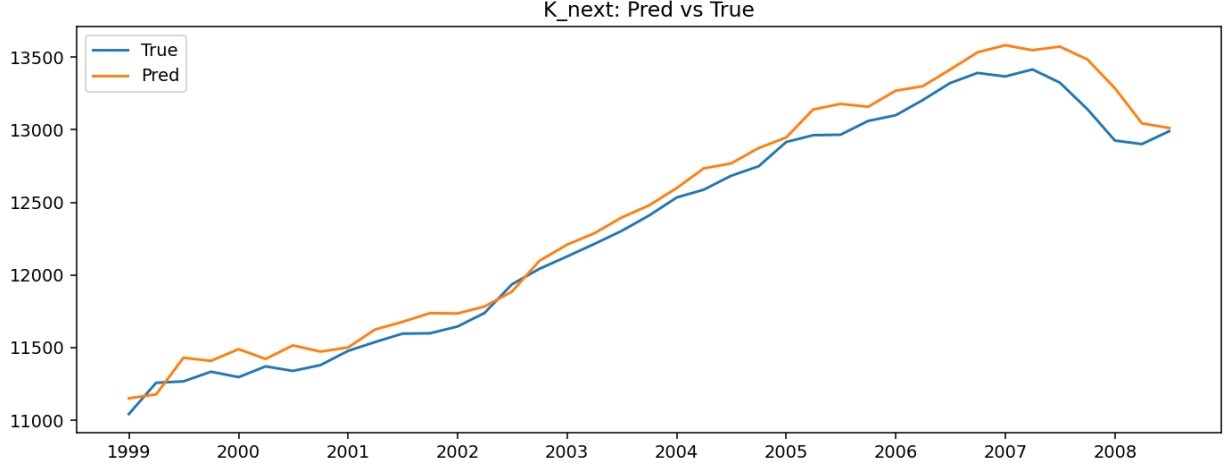


Figure 1: Predicted and observed next-period capital-state variable over the test period. The model tracks the broad trajectory of the aggregate capital-state proxy, while deviations around the crisis period indicate unmodeled regime-specific disturbances.

Table 2: Extrema of predicted and observed next-period capital-state variable in the test period.

Series	Minimum	Maximum
Predicted K_{t+1}	11151.1408	13581.9913
Observed K_{t+1}	11043.0440	13415.2660

is smoother than the observed series, which is consistent with the use of a regularized finite-dimensional approximation.

This figure is important because it evaluates the model as a dynamic transition system:

$$K_t \mapsto \hat{K}_{t+1} = K_t + \hat{\mathcal{A}}(K_t, \mathbf{x}_t). \quad (1224)$$

The predicted path shows that the proposed formula can be used to generate a numerical trajectory for the capital-state proxy, rather than merely estimating a static association.

18.5 Extrema and Range Properties

Beyond average prediction error, it is useful to examine whether the model captures the range and extrema of the observed capital-state variable. Table 2 reports the minimum and maximum values of the predicted and observed next-period capital-state variable in the test period.

The predicted minimum exceeds the observed minimum by

$$11151.1408 - 11043.0440 = 108.0968. \quad (1225)$$

The predicted maximum exceeds the observed maximum by

$$13581.9913 - 13415.2660 = 166.7253. \quad (1226)$$

Thus, the model slightly overstates both the lower and upper extrema of the test-period capital-state variable.

Table 3: Prediction errors by macroeconomic regime. The crisis regime corresponds to observations from 2007 to 2009.

Regime	MAE	RMSE	Observations
Normal period	101.1761	112.7025	29
Crisis period	179.2020	207.5054	10

The predicted range is

$$13581.9913 - 11151.1408 = 2430.8505, \quad (1227)$$

whereas the observed range is

$$13415.2660 - 11043.0440 = 2372.2220. \quad (1228)$$

The ratio of predicted to observed range is therefore

$$\frac{2430.8505}{2372.2220} \approx 1.0247. \quad (1229)$$

This indicates that the model captures the overall amplitude of the capital-state movement quite closely, with the predicted range only about 2.47% larger than the observed range.

This range analysis is important because it shows that the model does not collapse the dynamics into an overly flat trajectory. Although ridge regularization smooths short-run fluctuations, the predicted series still reproduces the broad scale of movement in the observed capital-state proxy:

$$\text{Range}(\hat{K}_{t+1}) \approx \text{Range}(K_{t+1}). \quad (1230)$$

At the same time, the overstatement of extrema suggests that the model does not fully capture the turning-point dynamics of the crisis period. This finding is consistent with the argument that threshold or regime-switching extensions may be necessary for crisis-sensitive accumulation modeling.

18.6 Regime-Specific Prediction Errors

The theoretical framework developed in this paper allows the accumulation law to vary across regimes. The empirical results provide evidence that such regime dependence is relevant. Table 3 divides the test period into normal observations and crisis observations, where the crisis period corresponds to 2007–2009.

The crisis-period MAE is substantially larger than the normal-period MAE:

$$179.2020 - 101.1761 = 78.0259. \quad (1231)$$

The crisis-period RMSE is also much larger:

$$207.5054 - 112.7025 = 94.8029. \quad (1232)$$

The crisis RMSE is approximately

$$\frac{207.5054}{112.7025} \approx 1.84 \quad (1233)$$

times the normal-period RMSE.

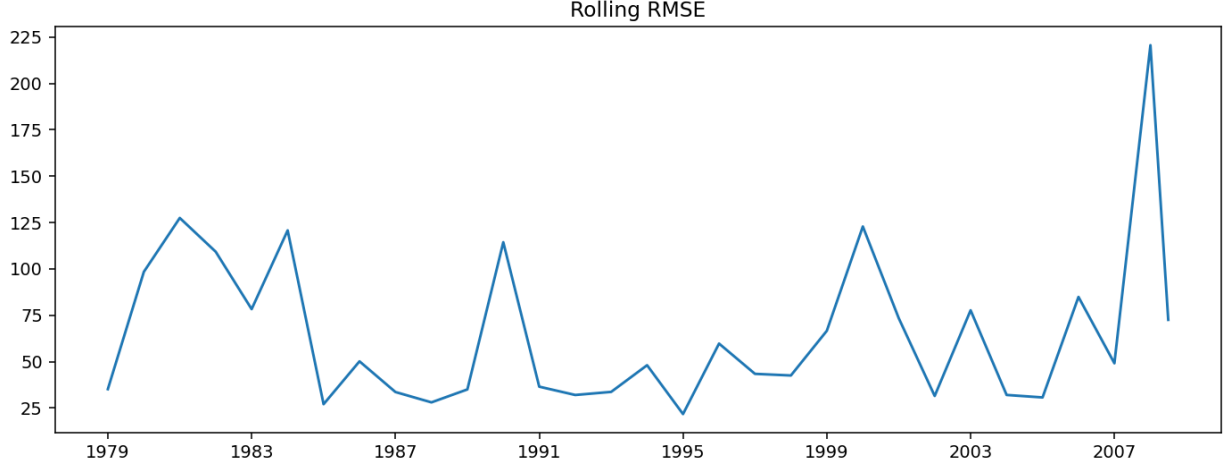


Figure 2: Rolling RMSE of the proposed accumulation model. Prediction errors are relatively moderate in most normal periods but rise sharply around the 2007–2008 crisis episode. This pattern provides empirical motivation for regime-aware extensions of the capital accumulation formula.

This result has a direct theoretical interpretation. The proposed finite-dimensional model describes normal-period capital-state movement more accurately than crisis-period movement. During crisis episodes, the relationship between features and the capital-state variable appears to shift. This motivates a regime-dependent accumulation law of the form

$$K_{t+1} = K_t + \mathcal{A}_{S_t}(K_t, \mathbf{x}_t) + u_t, \quad S_t \in \{\text{normal}, \text{crisis}\}. \quad (1234)$$

Hence, the regime-specific error analysis supports the paper’s earlier threshold and regime-shift formulation.

18.7 Rolling RMSE and Crisis Diagnostics

Figure 2 reports the rolling RMSE of the proposed model. The rolling error path is relatively moderate in many normal periods but rises sharply around the 2007–2008 crisis episode. This confirms that the largest prediction failures are not uniformly distributed across the sample; they are concentrated in specific historical regimes.

The rolling RMSE is defined as

$$RMSE_\ell = \left[\frac{1}{|\mathcal{W}_\ell|} \sum_{t \in \mathcal{W}_\ell} \left(K_{t+1} - \hat{K}_{t+1} \right)^2 \right]^{1/2}, \quad (1235)$$

where $\mathcal{W}_\ell$ is a moving evaluation window. Large values of $RMSE_\ell$ indicate periods in which the estimated accumulation law fails to fully capture realized capital-state movement.

The spike around the crisis period suggests that the accumulation mapping is not time-invariant in a simple sense. In normal periods, the finite-dimensional ridge approximation captures a meaningful part of the transition process. During crisis periods, however, the mapping appears to require additional state variables, nonlinearities, or threshold rules:

$$F_{\text{normal}}(K_t, \mathbf{x}_t) \neq F_{\text{crisis}}(K_t, \mathbf{x}_t). \quad (1236)$$

Table 4: Estimated local marginal effects at the latest test-period observation.

Feature	Estimated marginal effect
Real consumption, x_{cons}	0.016488
Real investment, x_{inv}	0.059808
Real interest rate, x_{rate}	7.047134

This finding supports one of the main theoretical claims of the paper: capital accumulation should not be modeled only as a smooth global function. Crisis episodes may involve discontinuities, structural breaks, or changes in the effective interaction among financial, productive, and demand-side features.

18.8 Estimated Marginal Effects

The proposed model also provides interpretable marginal effects. These effects measure the local response of the estimated capital-state increment to changes in each feature:

$$\widehat{ME}_{j,t} = \frac{\partial \widehat{\mathcal{A}}(K_t, \mathbf{x}_t)}{\partial x_{jt}}. \quad (1237)$$

Table 4 reports the estimated marginal effects at the latest test-period observation.

The estimated marginal effect of real investment is positive, consistent with its interpretation as a productive accumulation channel. The positive marginal effect of real consumption suggests that aggregate demand conditions are also associated with increases in the capital-state increment. The estimated interest-rate marginal effect is positive and numerically large. This coefficient should be interpreted cautiously. In the present reduced-form aggregate model, the interest-rate variable may proxy for broader macro-financial or cyclical conditions rather than representing a pure cost-of-capital mechanism.

The marginal-effect decomposition can be written as

$$d\widehat{\mathcal{A}}_t \approx \sum_{j=1}^d \widehat{ME}_{j,t} dx_{jt}. \quad (1238)$$

This is one of the main advantages of the proposed framework. A purely predictive model would only produce fitted values. The multi-feature accumulation formula also produces economically interpretable local effects.

18.9 Model Value and Theoretical Implications

The empirical exercise supports the value of the proposed model in four respects. First, it demonstrates that the theoretical capital accumulation formula can be implemented as a reproducible statistical pipeline:

$$\text{theoretic} \rightarrow \text{feature construction} \rightarrow \text{estimation} \rightarrow \text{prediction and diagnostics}. \quad (1239)$$

Second, the model captures the broad scale and trajectory of the capital-state proxy. The extrema analysis shows that the predicted range is close to the observed range:

$$\frac{\text{Range}(\widehat{K}_{t+1})}{\text{Range}(K_{t+1})} \approx 1.0247. \quad (1240)$$

This suggests that the model preserves the amplitude of capital-state movement even under ridge regularization.

Third, the regime-specific error analysis shows that crisis periods are much harder to predict than normal periods. This supports the theoretical need for thresholds, structural breaks, and regime-switching accumulation laws:

$$\mathcal{A} \rightarrow \mathcal{A}_{S_t}. \quad (1241)$$

Fourth, the estimated marginal effects provide an interpretable local decomposition of the accumulation process. The model therefore contributes not only as a predictive device but also as a framework for studying how different features contribute to changes in the capital-state variable.

18.10 Summary

This section has shown that the proposed multi-feature capital accumulation formula can be operationalized as an empirical model of capital-state movement. Using real GDP as a capital-state proxy, the model estimates the one-period variation

$$\Delta K_t = K_{t+1} - K_t \quad (1242)$$

through a ridge-regularized finite-dimensional approximation with standardized features, lagged terms, and quadratic terms.

The model obtains a training R^2 of 0.2373, a test MAE of 121.1827, and a test RMSE of 143.1279. It captures the broad trajectory and amplitude of the observed capital-state proxy, with the predicted range only about 2.47% larger than the observed range. At the same time, the model's errors increase substantially during the 2007–2009 crisis period, where the RMSE is approximately 1.84 times the normal-period RMSE. This pattern supports the theoretical claim that capital accumulation is regime dependent and that crisis episodes may require threshold or switching formulations. The empirical validation therefore has a precise interpretation. It does not prove that the selected macroeconomic proxy is the true capital stock. Instead, it demonstrates that the proposed formula can be used as an estimable framework for the dynamics of a capital-state variable:

$$\text{classical capital circuit} \rightarrow \text{multi-feature} \rightarrow \text{empirical capital-state transition model}. \quad (1243)$$

In this sense, the computational project validates the operational value of the theoretical framework: it transforms the classical idea of capital as self-expanding value into a flexible, estimable, and diagnostically rich model of capital-state movement.

References

- [1] Laibman, D. (1977). Toward a Marxian model of economic growth. *The American Economic Review*, 67(1), 387–392.
- [2] Lange, O. (1945). Marxian economics in the Soviet Union. *The American Economic Review*, 35(1), 127–133.
- [3] Roemer, J. E. (1981). *Analytical Foundations of Marxian Economic Theory*. Cambridge: Cambridge University Press.

- [4] Itoh, M. (2020). *Value and Crisis: Essays on Marxian Economics in Japan*. New York: Monthly Review Press.
- [5] Bowles, S. (1985). Post-Marxian economics: labour, learning and history. *Social Science Information*, 24(3), 507–528.
- [6] Samuelson, P. A. (1957). Wages and interest: a modern dissection of Marxian economic models. *The American Economic Review*, 47(6), 884–912.
- [7] Morishima, M. (1973). *Marx's Economics: A Dual Theory of Value and Growth*. Cambridge: Cambridge University Press.
- [8] Brenner, R. (1982). The origins of capitalist development: a critique of neo-Smithian Marxism. In *Introduction to the Sociology of "Developing Societies"* (pp. 54–71). Berlin: Springer.
- [9] Howard, M. C., and King, J. E. (1992). *A History of Marxian Economics: Volume II: 1929–1990*. London: Bloomsbury Publishing.
- [10] Lange, O. (1935). Marxian economics and modern economic theory. *The Review of Economic Studies*, 2(3), 189–201.
- [11] Zhou, S., Cao, Y., Nie, J., Fu, Y., Zhao, Z., Lu, X., and Wang, S. (2026). Comptrack: Information bottleneck-guided low-rank dynamic token compression for point cloud tracking. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(16), 13773–13781.
- [12] Gao, Q., Ning, L., Huang, L., Liu, C., Chen, X., and Zhou, F. (2026). Aligning authentic location shares with mobility information bottleneck. *IEEE Transactions on Big Data*.
- [13] Wang, J., Yang, Z., Ding, Y., Hu, Y., and Shikh-Bahaei, M. (2026). TOIB: Task-oriented orthogonalised information bottleneck for distributed semantic communication. *arXiv preprint arXiv:2604.11053*.
- [14] Yu, Y., Guo, Y., Liu, J., Sun, Y., and Du, J. (2026). Interpretable smart contract vulnerability detection with LLM-augmented Hilbert-Schmidt information bottleneck. *Computers, Materials, & Continua*, 87(2).
- [15] Gupta, V., Jain, N., Anand, A., Jain, D., Yadav, M. P., and Singh, Y. P. (2026). LogHyper-IB: Spatio-temporal hypergraph representation learning via information bottleneck for anomaly detection in distributed systems. *IEEE Access*.
- [16] Robinson, J. (1967). *Essay on Marxian Economics*. London: Springer.
- [17] Perraton, J. (2007). Evaluating Marxian contributions to development economics. *Journal of Economic Methodology*, 14(1), 27–46.
- [18] Bardhan, P. (1985). Marxist ideas in development economics: a brief evaluation. *Economic and Political Weekly*, 550–555.
- [19] Lazonick, W. (1986). Theory and history in Marxian economics. In *The Future of Economic History* (pp. 255–312). Berlin: Springer.
- [20] Dunayevskaya, R. (1944). A new revision of Marxian economics. *The American Economic Review*, 531–537.

- [21] Chowdhury, A. R., Kong, Z., and Chaudhuri, K. (2025). On the reliability of membership inference attacks. In *2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)* (pp. 534–549). IEEE.
- [22] Ran, D., He, X., Cong, T., Wang, A., Li, Q., and Wang, X. (2025). Lora-leak: Membership inference attacks against LoRA fine-tuned language models. *arXiv preprint arXiv:2507.18302*.
- [23] Wu, Z., Lin, M., Zhang, Z., Wang, F., Zhang, X., Zhang, X., and Wang, S. (2026). Image corruption-inspired membership inference attacks against large vision-language models. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 7945–7957).
- [24] Du, Y., Li, J., Chen, Y., Zhang, K., Yuan, Z., Xiao, H., Ribeiro, B., and Li, N. (2025). Cascading and proxy membership inference attacks. *arXiv preprint arXiv:2507.21412*.
- [25] Cui, Z., Zhang, M., and Pei, J. (2025). On membership inference attacks in knowledge distillation. *arXiv preprint arXiv:2505.11837*.
- [26] Mao, W., Wu, J., Liu, H., Sui, Y., and Wang, X. (2026). Invariant graph learning meets information bottleneck for out-of-distribution generalization. *Frontiers of Computer Science*, 20(1), 1–16.
- [27] Du, H., Ben, X., Feng, X., Xu, Z., Chen, L., and Wu, Q. (2026). GaitADIB: Adversarial disentangled information bottleneck network for unseen-view gait recognition. *Pattern Recognition*, 113243.
- [28] Li, M., Zhao, K., Zhu, J., Lu, X., Zhang, Y., and Du, Y. (2026). TUL-IB: Enhancing explainability in trajectory user linking with information bottleneck. *ACM Transactions on Knowledge Discovery from Data*.
- [29] Martini, K. M., Abdelaleem, E., Gulati, P., and Nemenman, I. (2026). Information bottleneck for learning the phase space of dynamics from high-dimensional experimental data. *arXiv preprint arXiv:2604.24662*.
- [30] Bhatt, M. (2026). Predictive coding and information bottleneck for hallucination detection in large language models. *arXiv preprint arXiv:2601.15652*.